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(54) **AUTOMATED EXCHANGE OF
HEALTHCARE INFORMATION FOR
FULFILLMENT OF MEDICATION DOSES**

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(58) **Field of Classification Search**

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See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

641,748 A 1/1900 Smith
819,339 A 5/1906 Cleland
(Continued)

FOREIGN PATENT DOCUMENTS

CN 1516257 5/1999
CN 2440518 8/2001
(Continued)

OTHER PUBLICATIONS

Texas Administrative Code, Title 22, Part 15, Ch 291, Rules 20, 36,
and 71-74. Feb. 10, 2004.

(Continued)

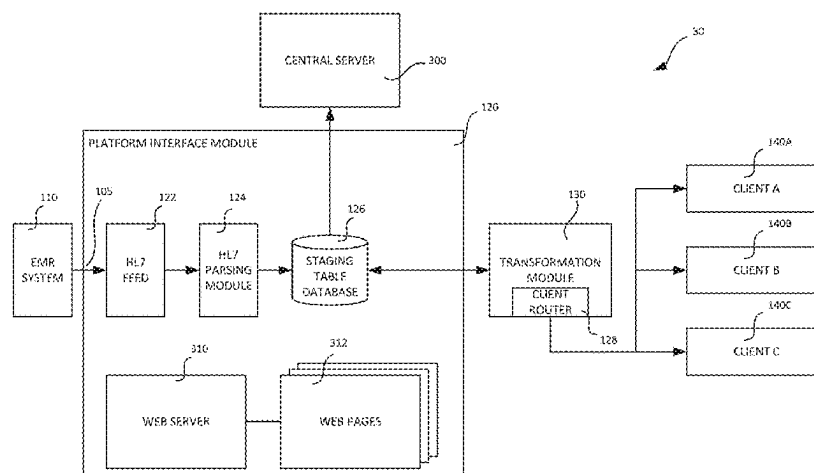
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(57) **ABSTRACT**

Automated healthcare information exchange. The healthcare information may comprise one or more dose orders corresponding to dose medications to be administered to a patient. The healthcare information may be received (e.g., from an EMR system such as a hospital information system (HIS) or the like) in the form of a healthcare information data stream. The information may then be standardized in to a standardized intermediate form. For instance, data may be parsed from the data stream and used to populate a staging table. In turn, data from the staging table may be transformed into an input format specific to a given dose fulfillment client to which the dose order is provided for fulfillment of the dose. Additionally, exception processing, logistical processing, and management functionality may be applied to the dose orders.

25 Claims, 18 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

3,426,150	A	2/1969	Tygart	4,622,979	A	11/1986	Katchis et al.
3,739,943	A	6/1973	Williamsen et al.	4,624,661	A	11/1986	Arimond
3,742,938	A	7/1973	Stern	4,636,950	A	1/1987	Caswell et al.
3,756,752	A	9/1973	Stenner	4,637,817	A	1/1987	Archibald et al.
3,774,762	A	11/1973	Lichtenstein	4,650,469	A	3/1987	Berg et al.
3,786,190	A	1/1974	Pori	4,652,262	A	3/1987	Veracchi
3,809,871	A	5/1974	Howard et al.	4,653,010	A	3/1987	Figler et al.
3,810,102	A	5/1974	Parks, III et al.	4,676,776	A	6/1987	Howson
3,848,112	A	11/1974	Weichselbaum et al.	4,681,563	A	7/1987	Deckert et al.
3,858,574	A	1/1975	Page	4,688,167	A	8/1987	Agarwal
3,878,967	A	4/1975	Joslin et al.	4,691,580	A	9/1987	Fosslien
3,910,257	A	10/1975	Fletcher et al.	4,695,954	A	9/1987	Rose et al.
3,910,260	A	10/1975	Sarnoff et al.	4,696,671	A	9/1987	Epstein et al.
3,921,196	A	11/1975	Patterson	4,697,928	A	10/1987	Csongor
3,971,000	A	7/1976	Cromwell	4,702,595	A	10/1987	Mutschler et al.
3,995,630	A	12/1976	Verrdonk	4,705,506	A	11/1987	Archibald
3,998,103	A	12/1976	Bjorklund et al.	D293,135	S	12/1987	Medema et al.
4,032,908	A	6/1977	Rice et al.	4,714,462	A	12/1987	DiComenico
4,078,562	A	3/1978	Friedman	4,717,042	A	1/1988	McLaughlin
4,144,496	A	3/1979	Cunningham et al.	4,718,576	A	1/1988	Tamura et al.
4,151,407	A	4/1979	McBride et al.	4,722,224	A	2/1988	Scheller et al.
4,156,867	A	5/1979	Bench et al.	4,722,349	A	2/1988	Baumberg
4,164,320	A	8/1979	Irazoqui et al.	4,722,734	A	2/1988	Kolln
4,173,971	A	11/1979	Karz	4,730,849	A	3/1988	Siegel
4,199,307	A	4/1980	Jassawalla	4,731,058	A	3/1988	Doan
4,270,532	A	6/1981	Franetzki et al.	4,732,411	A	3/1988	Siegel
4,273,121	A	6/1981	Jassawalla	4,741,732	A	5/1988	Crankshaw et al.
4,282,872	A	8/1981	Franetzki et al.	4,741,736	A	5/1988	Brown
4,308,866	A	1/1982	Jelliffe et al.	4,756,706	A	7/1988	Kerns et al.
4,319,338	A	3/1982	Grudowski et al.	4,759,756	A	7/1988	Forman
4,320,757	A	3/1982	Whitney et al.	4,770,184	A	9/1988	Greene et al.
4,354,252	A	10/1982	Lamb et al.	4,778,449	A	10/1988	Weber et al.
4,369,780	A	1/1983	Sakai	4,779,626	A	10/1988	Peel et al.
4,370,983	A	2/1983	Lichtenstein	4,784,645	A	11/1988	Fischell
4,373,527	A	2/1983	Fischell	4,785,799	A	11/1988	Schoon et al.
4,381,776	A	5/1983	Latham, Jr.	4,785,969	A	11/1988	McLaughlin
4,385,630	A	5/1983	Gilcher et al.	4,796,644	A	1/1989	Polaschegg
4,398,289	A	8/1983	Schoate	4,797,840	A	1/1989	Fraden
4,398,908	A	8/1983	Siposs	4,803,625	A	2/1989	Fu et al.
4,414,566	A	11/1983	Peyton et al.	4,810,090	A	3/1989	Boucher
4,416,654	A	11/1983	Schoendorfer et al.	4,810,243	A	3/1989	Howson
4,425,114	A	1/1984	Schoendorfer et al.	4,811,844	A	3/1989	Moulding, Jr. et al.
4,428,381	A	1/1984	Hepp	4,816,208	A	3/1989	Woods et al.
4,443,216	A	4/1984	Chappell	4,817,044	A	3/1989	Ogren
4,447,224	A	5/1984	DeCant, Jr. et al.	4,828,545	A	5/1989	Epstein et al.
4,449,538	A	5/1984	Corbitt et al.	4,829,524	A	5/1989	Yoshida
4,451,255	A	5/1984	Bujan et al.	4,830,018	A	5/1989	Treach
4,457,750	A	7/1984	Hill	4,831,562	A	5/1989	Mcintosh et al.
4,458,693	A	7/1984	Badzinski et al.	4,832,033	A	5/1989	Maher et al.
4,460,358	A	7/1984	Somerville et al.	4,835,372	A	5/1989	Gombrich et al.
4,464,172	A	8/1984	Lichtenstein	4,835,521	A	5/1989	Andrejasich et al.
4,469,481	A	9/1984	Kobayashi	4,838,275	A	6/1989	Lee
4,475,901	A	10/1984	Kraegen et al.	4,839,806	A	6/1989	Goldfischer et al.
4,476,381	A	10/1984	Rubin	4,845,644	A	7/1989	Anthias et al.
4,480,751	A	11/1984	Lueptow	4,847,764	A	7/1989	Halvorson
4,481,670	A	11/1984	Freeburg	4,850,009	A	7/1989	Zook et al.
4,487,604	A	12/1984	Iwatschenko et al.	4,850,972	A	7/1989	Schulman et al.
4,490,798	A	12/1984	Franks et al.	4,853,521	A	8/1989	Claeys et al.
4,496,351	A	1/1985	Hillel et al.	4,854,324	A	8/1989	Hirschman et al.
4,511,352	A	4/1985	Theeuwes et al.	4,857,713	A	8/1989	Brown
4,525,861	A	6/1985	Freeburg	4,857,716	A	8/1989	Gombrich et al.
4,526,574	A	7/1985	Pekkarinen	4,865,584	A	9/1989	Epstein et al.
4,529,401	A	7/1985	Leslie et al.	4,871,351	A	10/1989	Feingold
4,531,527	A	7/1985	Reinhold, Jr. et al.	4,878,175	A	10/1989	Norden-Paul et al.
4,538,138	A	8/1985	Harvey et al.	4,880,013	A	11/1989	Chio
4,545,071	A	10/1985	Freeburg	4,889,132	A	12/1989	Hutcheson et al.
4,551,133	A	11/1985	Zegers de Beyl et al.	4,889,134	A	12/1989	Greenwold et al.
4,559,038	A	12/1985	Berg et al.	4,893,270	A	1/1990	Beck et al.
4,560,979	A	12/1985	Roskopf	4,897,777	A	1/1990	Janke et al.
4,561,443	A	12/1985	Hogrefe et al.	4,898,209	A	2/1990	Zbed
4,562,751	A	1/1986	Nason	4,898,576	A	2/1990	Philip
4,564,054	A	1/1986	Gustavsson	4,898,578	A	2/1990	Rubalcaba, Jr.
4,590,473	A	5/1986	Burke et al.	4,901,728	A	2/1990	Hutchison
4,602,249	A	7/1986	Abbott	4,905,163	A	2/1990	Garber et al.
4,619,653	A	10/1986	Fischell	4,908,017	A	3/1990	Howson et al.
				4,912,623	A	3/1990	Rantala et al.
				4,916,441	A	4/1990	Gombrich et al.
				4,922,922	A	5/1990	Pollock et al.
				4,925,444	A	5/1990	Orkin et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

4,933,843	A	6/1990	Scheller et al.	5,157,595	A	10/1992	Lovrenich
4,937,777	A	6/1990	Flood et al.	5,158,091	A	10/1992	Butterfield et al.
4,941,808	A	7/1990	Qureshi et al.	5,159,673	A	10/1992	Sackmann et al.
4,943,279	A	7/1990	Samiotes et al.	5,160,320	A	11/1992	Yum et al.
4,943,987	A	7/1990	Asahina et al.	5,161,211	A	11/1992	Taguchi et al.
4,946,445	A	8/1990	Lynn	5,167,235	A	12/1992	Seacord et al.
4,949,274	A	8/1990	Hollander et al.	5,169,642	A	12/1992	Brinker et al.
4,952,928	A	8/1990	Carroll et al.	5,172,698	A	12/1992	Stanko
4,953,074	A	8/1990	Kametani et al.	5,176,004	A	1/1993	Gaudet
4,960,230	A	10/1990	Marelli	5,179,569	A	1/1993	Sawyer
4,964,847	A	10/1990	Prince	5,179,700	A	1/1993	Aihara et al.
4,966,579	A	10/1990	Polaschegg	5,181,910	A	1/1993	Scanlon
4,967,928	A	11/1990	Carter	5,190,185	A	3/1993	Blechl
4,968,295	A	11/1990	Neumann	5,190,522	A	3/1993	Wojcicki et al.
4,975,647	A	12/1990	Downer et al.	5,191,891	A	3/1993	Righter
4,977,590	A	12/1990	Milovancevic	5,208,762	A	5/1993	Charhut et al.
4,978,335	A	12/1990	Arthur, III	5,208,907	A	5/1993	Shelton et al.
4,991,091	A	2/1991	Allen	5,211,849	A	5/1993	Kitaevich et al.
4,992,926	A	2/1991	Janke et al.	5,213,099	A	5/1993	Tripp, Jr.
4,993,068	A	2/1991	Piosenka et al.	5,213,232	A	5/1993	Kraft et al.
4,993,506	A	2/1991	Angel	5,213,568	A	5/1993	Lattin et al.
4,998,249	A	3/1991	Bennett et al.	5,219,330	A	6/1993	Bollich et al.
5,002,055	A	3/1991	Merki et al.	5,219,331	A	6/1993	Vanderveen
5,003,296	A	3/1991	Lee	5,225,974	A	7/1993	Mathews et al.
5,006,699	A	4/1991	Felkner et al.	5,226,425	A	7/1993	Righter
5,007,429	A	4/1991	Treatch et al.	5,228,450	A	7/1993	Sellers
5,012,402	A	4/1991	Akiyama	5,231,990	A	8/1993	Gauglitz
5,012,411	A	4/1991	PolICASTRO et al.	5,234,404	A	8/1993	Tuttle et al.
5,014,875	A	5/1991	McLaughlin et al.	5,235,510	A	8/1993	Yamada et al.
5,016,172	A	5/1991	Dessertine	5,236,416	A	8/1993	McDaniel et al.
5,023,770	A	6/1991	Siverling	5,238,001	A	8/1993	Gallant et al.
5,025,374	A	6/1991	Roizen et al.	5,240,007	A	8/1993	Pytel et al.
5,036,852	A	8/1991	Leishman	5,244,463	A	9/1993	Cordner, Jr. et al.
5,038,800	A	8/1991	Oba	5,245,704	A	9/1993	Weber et al.
5,041,086	A	8/1991	Koenig et al.	5,254,096	A	10/1993	Rondelet et al.
5,045,048	A	9/1991	Kaleskas et al.	5,256,156	A	10/1993	Kern et al.
5,047,959	A	9/1991	Phillips et al.	5,256,157	A	10/1993	Samiotes et al.
5,053,031	A	10/1991	Borsanyi	5,261,884	A	11/1993	Stern et al.
5,053,990	A	10/1991	Kreifels et al.	5,262,943	A	11/1993	Thibado et al.
5,055,001	A	10/1991	Natwick et al.	5,265,010	A	11/1993	Evans-Paganelli et al.
5,057,076	A	10/1991	Polaschegg	5,265,431	A	11/1993	Gaudet et al.
5,061,243	A	10/1991	Winchell et al.	5,267,174	A	11/1993	Kaufman et al.
5,072,356	A	12/1991	Watt et al.	5,271,405	A	12/1993	Boyer et al.
5,072,383	A	12/1991	Brimm et al.	5,272,318	A	12/1993	Gorman
5,072,412	A	12/1991	Henderson, Jr. et al.	5,272,321	A	12/1993	Otsuka et al.
5,078,683	A	1/1992	Sancoff et al.	5,273,517	A	12/1993	Barone et al.
5,084,828	A	1/1992	Kaufman et al.	5,277,188	A	1/1994	Selker
5,087,245	A	2/1992	Doan	5,283,861	A	2/1994	Dangler et al.
5,088,904	A	2/1992	Okada	5,284,150	A	2/1994	Butterfield et al.
5,088,981	A	2/1992	Howson et al.	5,286,252	A	2/1994	Tuttle et al.
5,088,990	A	2/1992	Hivale et al.	5,292,029	A	3/1994	Pearson
5,096,385	A	3/1992	Georgi et al.	5,297,257	A	3/1994	Struger et al.
5,098,377	A	3/1992	Borsanyi et al.	5,298,021	A	3/1994	Sherer
5,100,380	A	3/1992	Epstein et al.	5,304,126	A	4/1994	Epstein et al.
5,100,394	A	3/1992	Dudar et al.	5,307,263	A	4/1994	Brown
5,103,211	A	4/1992	Daoud et al.	5,307,372	A	4/1994	Sawyer et al.
5,104,374	A	4/1992	Bishko et al.	5,307,463	A	4/1994	Hyatt et al.
5,108,131	A	4/1992	Nassim	5,311,908	A	5/1994	Barone et al.
5,108,363	A	4/1992	Tuttle et al.	5,314,243	A	5/1994	McDonald et al.
5,108,367	A	4/1992	Epstein et al.	5,315,505	A	5/1994	Pratt et al.
5,108,372	A	4/1992	Swenson	5,317,506	A	5/1994	Coutre et al.
5,109,487	A	4/1992	Ohgomori et al.	5,319,363	A	6/1994	Welch et al.
5,109,849	A	5/1992	Goodman et al.	5,319,543	A	6/1994	Wilhelm
5,112,319	A	5/1992	Lai	5,321,618	A	6/1994	Gessman
5,116,203	A	5/1992	Natwick et al.	5,321,829	A	6/1994	Zifferer
5,116,312	A	5/1992	Blankenship et al.	5,324,422	A	6/1994	Colleran et al.
5,131,092	A	7/1992	Sackmann et al.	5,325,478	A	6/1994	Shelton et al.
5,134,574	A	7/1992	Beaverstock et al.	5,327,341	A	7/1994	Whalen et al.
5,135,500	A	8/1992	Zdeb	5,331,549	A	7/1994	Crawford, Jr.
5,137,023	A	8/1992	Mendelson et al.	5,336,245	A	8/1994	Adams et al.
5,151,978	A	9/1992	Bronikowski et al.	5,337,230	A	8/1994	Baumgartner et al.
5,152,296	A	10/1992	Simons	5,337,747	A	8/1994	Nefel
5,153,416	A	10/1992	Neeley	5,337,919	A	8/1994	Spaulding et al.
5,153,827	A	10/1992	Coutre et al.	5,338,157	A	8/1994	Blomquist
5,155,693	A	10/1992	Altmayer et al.	5,339,421	A	8/1994	Housel, III
				5,339,821	A	8/1994	Fujimoto
				5,341,291	A	8/1994	Roizen et al.
				5,341,412	A	8/1994	Ramot et al.
				5,348,008	A	9/1994	Bornn et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,348,539	A	9/1994	Herskowitz	5,514,088	A	5/1996	Zakko
5,349,675	A	9/1994	Fitzgerald et al.	5,514,095	A	5/1996	Brightbill et al.
5,356,378	A	10/1994	Doan	5,515,426	A	5/1996	Yacenda et al.
5,360,410	A	11/1994	Wacks	5,520,450	A	5/1996	Colson, Jr. et al.
5,361,202	A	11/1994	Doue	5,520,637	A	5/1996	Pager et al.
5,361,758	A	11/1994	Hall et al.	5,522,396	A	6/1996	Langer et al.
5,366,896	A	11/1994	Margrey et al.	5,522,798	A	6/1996	Johnson et al.
5,366,904	A	11/1994	Qureshi et al.	5,526,428	A	6/1996	Arnold
5,367,555	A	11/1994	Isoyama	5,528,503	A	6/1996	Moore et al.
5,368,562	A	11/1994	Blomquist et al.	5,529,063	A	6/1996	Hill
5,370,612	A	12/1994	Maeda et al.	5,531,680	A	7/1996	Dumas et al.
5,371,687	A	12/1994	Holmes, II et al.	5,531,697	A	7/1996	Olsen et al.
5,374,251	A	12/1994	Smith	5,531,698	A	7/1996	Olsen
5,374,813	A	12/1994	Shipp	5,533,079	A	7/1996	Colburn et al.
5,374,965	A	12/1994	Kanno	5,533,981	A	7/1996	Mandro et al.
5,375,604	A	12/1994	Kelly et al.	5,534,691	A	7/1996	Holdaway et al.
5,376,070	A	12/1994	Purvis et al.	5,536,084	A	7/1996	Curtis et al.
5,377,864	A	1/1995	Blechl et al.	5,537,313	A	7/1996	Pirelli
5,378,231	A	1/1995	Johnson et al.	5,537,853	A	7/1996	Finburgh et al.
5,379,214	A	1/1995	Arbuckle et al.	5,542,420	A	8/1996	Goldman et al.
5,389,078	A	2/1995	Zalesky et al.	5,544,649	A	8/1996	David et al.
5,390,238	A	2/1995	Kirk et al.	5,544,651	A	8/1996	Wilk
5,392,951	A	2/1995	Gardner et al.	5,544,661	A	8/1996	Davis et al.
5,395,320	A	3/1995	Padda et al.	5,545,140	A	8/1996	Conero et al.
5,395,321	A	3/1995	Kawahara et al.	5,546,580	A	8/1996	Seliger et al.
5,398,336	A	3/1995	Tantry et al.	5,547,470	A	8/1996	Johnson et al.
5,401,059	A	3/1995	Ferrario	5,549,117	A	8/1996	Tacklind et al.
5,404,292	A	4/1995	Hendrickson	5,549,460	A	8/1996	O'Leary
5,404,384	A	4/1995	Colburn et al.	5,553,609	A	9/1996	Chen et al.
5,406,473	A	4/1995	Yoshikura et al.	5,558,638	A	9/1996	Evers et al.
5,412,715	A	5/1995	Volpe	5,558,640	A	9/1996	Pfeiler et al.
5,415,167	A	5/1995	Wilk	5,560,352	A	10/1996	Heim et al.
5,416,695	A	5/1995	Stutman et al.	5,562,232	A	10/1996	Pearson
5,417,222	A	5/1995	Dempsey et al.	5,562,621	A	10/1996	Claude et al.
5,420,977	A	5/1995	Sztipanovits et al.	5,563,347	A	10/1996	Martin et al.
5,421,343	A	6/1995	Feng	5,564,434	A	10/1996	Halperin et al.
5,423,746	A	6/1995	Burkett et al.	5,564,803	A	10/1996	McDonald et al.
5,429,602	A	7/1995	Hauser	5,568,912	A	10/1996	Minami et al.
5,431,201	A	7/1995	Torchia et al.	5,569,186	A	10/1996	Lord et al.
5,431,299	A	7/1995	Brewer et al.	5,569,187	A	10/1996	Kaiser
5,431,627	A	7/1995	Pastrone et al.	5,571,258	A	11/1996	Pearson
5,433,736	A	7/1995	Nilsson	5,573,502	A	11/1996	LeCocq et al.
5,438,607	A	8/1995	Przygoda, Jr. et al.	5,573,506	A	11/1996	Vasko
5,440,699	A	8/1995	Farrand et al.	5,575,632	A	11/1996	Morris et al.
5,441,047	A	8/1995	David et al.	5,576,952	A	11/1996	Stutman et al.
5,445,294	A	8/1995	Gardner et al.	5,579,001	A	11/1996	Dempsey et al.
5,445,621	A	8/1995	Poli et al.	5,579,378	A	11/1996	Arlinghaus, Jr.
5,446,868	A	8/1995	Gardea, II et al.	5,581,369	A	12/1996	Righter et al.
5,453,098	A	9/1995	Botts et al.	5,581,687	A	12/1996	Lyle et al.
5,455,851	A	10/1995	Chaco et al.	5,582,593	A	12/1996	Hultman
5,458,123	A	10/1995	Unger	5,583,758	A	12/1996	Mcilroy et al.
5,460,294	A	10/1995	Williams	5,588,815	A	12/1996	Zaleski, II
5,460,605	A	10/1995	Tuttle et al.	5,589,932	A	12/1996	Garcia-Rubio et al.
5,461,665	A	10/1995	Shur et al.	5,590,648	A	1/1997	Mitchell et al.
5,462,051	A	10/1995	Oka et al.	5,591,344	A	1/1997	Kenley et al.
5,464,392	A	11/1995	Epstein et al.	5,593,267	A	1/1997	McDonald et al.
5,465,286	A	11/1995	Clare et al.	5,594,637	A	1/1997	Eisenberg et al.
5,467,773	A	11/1995	Bergelson et al.	5,594,786	A	1/1997	Chaco et al.
5,468,110	A	11/1995	McDonald et al.	5,597,995	A	1/1997	Williams et al.
5,469,855	A	11/1995	Pompei et al.	5,598,536	A	1/1997	Slaughter, III et al.
5,471,382	A	11/1995	Tallman et al.	5,601,445	A	2/1997	Schipper et al.
5,474,552	A	12/1995	Palti	5,609,575	A	3/1997	Larson et al.
5,482,043	A	1/1996	Zulauf	5,609,576	A	3/1997	Voss et al.
5,482,446	A	1/1996	Williamson et al.	5,613,115	A	3/1997	Gihl et al.
5,485,408	A	1/1996	Blomquist	5,619,428	A	4/1997	Lee et al.
5,490,610	A	2/1996	Pearson	5,619,991	A	4/1997	Sloane
5,494,592	A	2/1996	Latham, Jr. et al.	5,623,652	A	4/1997	Vora et al.
5,496,265	A	3/1996	Langley et al.	5,623,925	A	4/1997	Swenson et al.
5,496,273	A	3/1996	Pastrone et al.	5,626,144	A	5/1997	Tacklind et al.
5,502,944	A	4/1996	Kraft et al.	5,628,619	A	5/1997	Wilson
5,507,412	A	4/1996	Ebert et al.	5,630,710	A	5/1997	Tune et al.
5,508,912	A	4/1996	Schneiderman	5,631,844	A	5/1997	Margrey et al.
5,509,318	A	4/1996	Gomes	5,633,910	A	5/1997	Cohen
5,509,422	A	4/1996	Fukami	D380,260	S	6/1997	Hyman
5,513,957	A	5/1996	O'Leary	5,634,893	A	6/1997	Rishton
				5,637,082	A	6/1997	Pages et al.
				5,637,093	A	6/1997	Hyman et al.
				5,640,301	A	6/1997	Roecher et al.
				5,640,953	A	6/1997	Bishop et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,641,628 A	6/1997	Bianchi	5,795,327 A	8/1998	Wilson et al.
5,643,193 A	7/1997	Papillon et al.	5,797,515 A	8/1998	Liff et al.
5,643,212 A	7/1997	Coutre et al.	5,800,387 A	9/1998	Duffy et al.
5,647,853 A	7/1997	Feldmann et al.	5,801,755 A	9/1998	Echerer
5,647,854 A	7/1997	Olsen et al.	5,803,906 A	9/1998	Pratt et al.
5,651,775 A	7/1997	Walker et al.	5,805,442 A	9/1998	Crater et al.
5,652,566 A	7/1997	Lambert	5,805,454 A	9/1998	Valerino et al.
5,658,240 A	8/1997	Urdahl et al.	5,805,456 A	9/1998	Higham et al.
5,658,250 A	8/1997	Blomquist et al.	5,805,505 A	9/1998	Zheng et al.
5,661,978 A	9/1997	Holmes et al.	5,807,321 A	9/1998	Stoker et al.
5,664,270 A	9/1997	Bell et al.	5,807,322 A	9/1998	Lindsey et al.
5,666,404 A	9/1997	Ciccotelli et al.	5,807,336 A	9/1998	Russo et al.
D385,646 S	10/1997	Chan	5,810,747 A	9/1998	Brudny et al.
5,678,562 A	10/1997	Sellers	5,812,410 A	9/1998	Lion et al.
5,678,568 A	10/1997	Uchikubo et al.	5,814,015 A	9/1998	Gargano et al.
5,681,285 A	10/1997	Ford et al.	5,815,566 A	9/1998	Ramot et al.
5,682,526 A	10/1997	Smokoff et al.	5,818,528 A	10/1998	Roth et al.
5,683,367 A	11/1997	Jordan et al.	5,822,418 A	10/1998	Yacenda et al.
5,685,844 A	11/1997	Marttila	5,822,544 A	10/1998	Chaco et al.
5,687,717 A	11/1997	Halpern	5,823,949 A	10/1998	Goltra
5,687,734 A	11/1997	Dempsey et al.	5,826,237 A	10/1998	Macrae et al.
5,695,473 A	12/1997	Olsen	5,829,438 A	11/1998	Gibbs et al.
5,697,951 A	12/1997	Harpstead	5,832,447 A	11/1998	Rieker et al.
5,700,998 A	12/1997	Palti	5,832,448 A	11/1998	Brown
5,701,894 A	12/1997	Cherry et al.	5,832,450 A	11/1998	Myers et al.
5,704,351 A	1/1998	Mortara et al.	5,833,599 A	11/1998	Schrier et al.
5,704,364 A	1/1998	Saltzstein et al.	5,835,897 A	11/1998	Dang
5,704,366 A	1/1998	Tacklind et al.	5,836,910 A	11/1998	Duffy et al.
5,712,798 A	1/1998	Langley et al.	5,841,975 A	11/1998	Layne et al.
5,712,912 A	1/1998	Tomko et al.	5,842,841 A	12/1998	Danby et al.
5,713,485 A	2/1998	Liff et al.	5,842,976 A	12/1998	Williamson
5,713,856 A	2/1998	Eggers et al.	5,845,253 A	12/1998	Rensimer et al.
5,715,823 A	2/1998	Wood et al.	5,848,593 A	12/1998	McGrady et al.
5,716,114 A	2/1998	Holmes et al.	5,851,186 A	12/1998	Wood et al.
5,716,194 A	2/1998	Butterfield et al.	5,852,590 A	12/1998	De La Huerga
5,718,562 A	2/1998	Lawless et al.	5,853,387 A	12/1998	Clegg et al.
5,719,761 A	2/1998	Gatti et al.	5,855,550 A	1/1999	Lai et al.
RE35,743 E	3/1998	Pearson	5,857,967 A	1/1999	Frid et al.
5,724,025 A	3/1998	Tavori	5,859,972 A	1/1999	Subramaniam et al.
5,724,580 A	3/1998	Levin et al.	5,865,745 A	2/1999	Schmitt et al.
5,732,709 A	3/1998	Tacklind et al.	5,865,786 A	2/1999	Sibalis et al.
5,733,259 A	3/1998	Valcke et al.	5,867,821 A	2/1999	Ballantyne et al.
5,735,887 A	4/1998	Barreras, Sr. et al.	5,871,465 A	2/1999	Vasko
5,737,539 A	4/1998	Edelson et al.	5,876,926 A	3/1999	Beecham
5,740,185 A	4/1998	Bosse	5,880,443 A	3/1999	McDonald et al.
5,740,800 A	4/1998	Hendrickson et al.	5,882,338 A	3/1999	Gray
5,745,366 A	4/1998	Higham et al.	5,883,370 A	3/1999	Walker et al.
5,745,378 A	4/1998	Barker et al.	5,883,576 A	3/1999	De La Huerga
5,752,917 A	5/1998	Fuchs	5,884,273 A	3/1999	Sattizahn et al.
5,752,976 A	5/1998	Duffin et al.	5,884,457 A	3/1999	Ortiz et al.
5,755,563 A	5/1998	Clegg et al.	5,885,245 A	3/1999	Lynch et al.
5,758,095 A	5/1998	Albaum et al.	5,891,035 A	4/1999	Wood et al.
5,764,923 A	6/1998	Tallman et al.	5,891,734 A	4/1999	Gill et al.
5,766,155 A	6/1998	Hyman et al.	5,893,697 A	4/1999	Zimi et al.
5,769,811 A	6/1998	Stacey et al.	5,894,273 A	4/1999	Meador et al.
5,771,657 A	6/1998	Lasher et al.	5,895,371 A	4/1999	Levitas et al.
5,772,585 A	6/1998	Lavin et al.	5,897,493 A	4/1999	Brown
5,772,586 A	6/1998	Heinonen et al.	5,897,530 A	4/1999	Jackson
5,772,635 A	6/1998	Dastur et al.	5,897,989 A	4/1999	Beecham
5,772,637 A	6/1998	Heinzmann et al.	5,899,665 A	5/1999	Makino et al.
5,776,057 A	7/1998	Swenson et al.	5,899,855 A	5/1999	Brown
5,778,345 A	7/1998	McCartney	5,901,150 A	5/1999	Jhuboo et al.
5,778,882 A	7/1998	Raymond et al.	5,904,668 A	5/1999	Hyman et al.
5,781,442 A	7/1998	Engleson et al.	5,905,653 A	5/1999	Higham et al.
5,782,805 A	7/1998	Meinzer et al.	5,907,291 A	5/1999	Chen et al.
5,782,878 A	7/1998	Morgan et al.	5,907,493 A	5/1999	Boyer et al.
5,785,650 A	7/1998	Akasaka et al.	5,908,027 A	6/1999	Butterfield et al.
5,788,669 A	8/1998	Peterson	5,910,107 A	6/1999	Iliff
5,788,851 A	8/1998	Kenley et al.	5,910,252 A	6/1999	Truitt et al.
5,790,409 A	8/1998	Fedor et al.	5,911,132 A	6/1999	Sloane
5,791,342 A	8/1998	Woodard	5,911,687 A	6/1999	Sato et al.
5,791,880 A	8/1998	Wilson	5,912,818 A	6/1999	McGrady et al.
5,793,861 A	8/1998	Haigh	5,913,197 A	6/1999	Kameda
5,793,969 A	8/1998	Kamentsky et al.	5,913,310 A	6/1999	Brown
5,795,317 A	8/1998	Brierton et al.	5,915,089 A	6/1999	Stevens et al.
			5,915,240 A	6/1999	Karpf
			5,919,154 A	7/1999	Toavs et al.
			5,921,938 A	7/1999	Aoyama et al.
			5,923,018 A	7/1999	Kameda et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

5,924,074	A	7/1999	Evans	6,021,392	A	2/2000	Lester et al.
5,924,103	A	7/1999	Ahmed et al.	6,022,315	A	2/2000	Iliff
5,927,540	A	7/1999	Godlewski	6,023,522	A	2/2000	Draganoff et al.
5,931,791	A	8/1999	Saltzstein et al.	6,024,539	A	2/2000	Blomquist
5,935,060	A	8/1999	Iliff	6,024,699	A	2/2000	Surwit et al.
5,935,099	A	8/1999	Peterson et al.	6,027,217	A	2/2000	McClure et al.
5,935,106	A	8/1999	Olsen	6,029,138	A	2/2000	Khorasani et al.
5,938,413	A	8/1999	Makino et al.	6,032,119	A	2/2000	Brown et al.
5,939,326	A	8/1999	Chupp et al.	6,032,155	A	2/2000	de la Huerga
5,939,699	A	8/1999	Perttunen et al.	6,033,076	A	3/2000	Braeuning et al.
5,940,306	A	8/1999	Gardner et al.	6,039,251	A	3/2000	Holowko et al.
5,940,802	A	8/1999	Hildebrand et al.	6,039,467	A	3/2000	Holmes
5,941,829	A	8/1999	Saltzstein et al.	6,047,259	A	4/2000	Campbell et al.
5,941,846	A	8/1999	Duffy et al.	6,048,086	A	4/2000	Valerino
5,942,986	A	8/1999	Shabot et al.	6,050,940	A	4/2000	Braun et al.
5,943,423	A	8/1999	Muftic	6,055,487	A	4/2000	Margery et al.
5,943,633	A	8/1999	Wilson et al.	6,057,758	A	5/2000	Dempsey et al.
5,944,659	A	8/1999	Flach et al.	6,059,736	A	5/2000	Tapper
5,945,651	A	8/1999	Chorosinski et al.	6,061,603	A	5/2000	Papadopoulos et al.
5,946,083	A	8/1999	Melendez et al.	6,065,819	A	5/2000	Holmes et al.
5,946,659	A	8/1999	Lancelot et al.	6,068,153	A	5/2000	Young et al.
D414,578	S	9/1999	Chen et al.	6,068,156	A	5/2000	Liff et al.
5,950,006	A	9/1999	Crater et al.	6,073,046	A	6/2000	Patel et al.
5,951,300	A	9/1999	Brown	6,074,345	A	6/2000	van Oostrom et al.
5,951,510	A	9/1999	Barak	6,079,621	A	6/2000	Vardanyan et al.
5,954,640	A	9/1999	Szabo	6,080,106	A	6/2000	Lloyd et al.
5,954,885	A	9/1999	Bollich et al.	6,081,048	A	6/2000	Bergmann et al.
5,954,971	A	9/1999	Pages et al.	6,081,786	A	6/2000	Barry et al.
5,956,023	A	9/1999	Lyle et al.	6,082,776	A	7/2000	Feinberg
5,957,885	A	9/1999	Bollich et al.	6,083,206	A	7/2000	Molko
5,959,529	A	9/1999	Kail, IV	6,093,146	A	7/2000	Filangeri
5,960,085	A	9/1999	de la Huerga	6,095,985	A	8/2000	Raymond et al.
5,960,403	A	9/1999	Brown	6,096,561	A	8/2000	Tayi
5,960,991	A	10/1999	Ophardt	6,098,892	A	8/2000	Peoples, Jr.
5,961,446	A	10/1999	Beller et al.	6,101,407	A	8/2000	Groezinger
5,961,448	A	10/1999	Swenson et al.	6,101,478	A	8/2000	Brown
5,961,487	A	10/1999	Davis	6,102,856	A	8/2000	Groff et al.
5,961,923	A	10/1999	Nova et al.	6,108,399	A	8/2000	Hernandez-Guerra et al.
5,963,641	A	10/1999	Crandall et al.	6,108,588	A	8/2000	McGrady
5,964,700	A	10/1999	Tallman et al.	6,109,774	A	8/2000	Holmes et al.
5,966,304	A	10/1999	Cook et al.	6,112,224	A	8/2000	Peifer et al.
5,967,975	A	10/1999	Ridgeway	RE36,871	E	9/2000	Epstein et al.
5,970,423	A	10/1999	Langley et al.	6,113,554	A	9/2000	Gilcher et al.
5,971,593	A	10/1999	McGrady	6,116,461	A	9/2000	Broadfield et al.
5,971,921	A	10/1999	Timbel	6,117,940	A	9/2000	Mjalli
5,971,948	A	10/1999	Pages et al.	6,123,524	A	9/2000	Danby et al.
5,974,124	A	10/1999	Schlueter, Jr. et al.	6,125,350	A	9/2000	Dirbas
5,975,737	A	11/1999	Crater et al.	6,129,517	A	10/2000	Danby et al.
5,980,490	A	11/1999	Tsoukalis	6,132,371	A	10/2000	Dempsey et al.
5,983,193	A	11/1999	Heinonen et al.	6,134,504	A	10/2000	Douglas et al.
5,987,519	A	11/1999	Peifer et al.	6,135,949	A	10/2000	Russo et al.
5,991,731	A	11/1999	Colon et al.	6,139,177	A	10/2000	Venkatraman et al.
5,993,046	A	11/1999	McGrady et al.	6,139,495	A	10/2000	De La Huerga
5,993,420	A	11/1999	Hyman et al.	6,141,412	A	10/2000	Smith et al.
5,995,077	A	11/1999	Wilcox et al.	6,144,922	A	11/2000	Douglas et al.
5,995,939	A	11/1999	Berman et al.	6,145,695	A	11/2000	Garrigues
5,995,965	A	11/1999	Experton	6,146,523	A	11/2000	Kenley et al.
5,997,167	A	12/1999	Crater et al.	6,148,297	A	11/2000	Swor et al.
5,997,476	A	12/1999	Brown	6,149,063	A	11/2000	Reynolds et al.
6,003,006	A	12/1999	Colella et al.	6,151,536	A	11/2000	Arnold et al.
6,004,020	A	12/1999	Bartur	6,152,364	A	11/2000	Schoonen et al.
6,004,276	A	12/1999	Wright et al.	6,154,668	A	11/2000	Pedersen et al.
6,006,191	A	12/1999	DeRienzo	6,154,726	A	11/2000	Rensimer et al.
6,006,946	A	12/1999	Williams et al.	6,157,914	A	12/2000	Seto et al.
6,009,333	A	12/1999	Chaco	6,158,965	A	12/2000	Butterfield et al.
6,010,454	A	1/2000	Arieff et al.	6,160,478	A	12/2000	Jacobsen et al.
6,011,858	A	1/2000	Stock et al.	6,161,095	A	12/2000	Brown
6,011,999	A	1/2000	Holmes	6,161,141	A	12/2000	Dillon
6,012,034	A	1/2000	Hamparian et al.	6,163,737	A	12/2000	Fedor et al.
6,013,057	A	1/2000	Danby et al.	6,165,154	A	12/2000	Gray et al.
6,014,631	A	1/2000	Teagarden et al.	6,168,563	B1	1/2001	Brown
6,016,444	A	1/2000	John	6,170,007	B1	1/2001	Venkatraman et al.
6,017,318	A	1/2000	Gauthier et al.	6,170,746	B1	1/2001	Brook et al.
6,018,713	A	1/2000	Coli et al.	6,171,112	B1	1/2001	Clark et al.
6,019,745	A	2/2000	Gray	6,171,237	B1	1/2001	Avitall et al.
				6,171,264	B1	1/2001	Bader
				6,173,198	B1	1/2001	Schulze et al.
				6,175,779	B1	1/2001	Barrett
				6,175,977	B1	1/2001	Schumacher et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,176,392 B1	1/2001	William et al.	6,338,007 B1	1/2002	Broadfield et al.
6,182,047 B1	1/2001	Dirbas	6,339,732 B1	1/2002	Phoon et al.
6,183,417 B1	2/2001	Geheb et al.	6,345,260 B1	2/2002	Cummings, Jr. et al.
6,186,145 B1	2/2001	Brown	6,346,886 B1	2/2002	De La Huerga
6,192,320 B1	2/2001	Margrey et al.	6,347,553 B1	2/2002	Morris et al.
6,193,480 B1	2/2001	Butterfield	6,352,200 B1	3/2002	Schoonen et al.
6,195,887 B1	3/2001	Danby et al.	6,353,817 B1	3/2002	Jacobs et al.
6,198,394 B1	3/2001	Jacobsen et al.	6,358,225 B1	3/2002	Butterfield
6,200,264 B1	3/2001	Satherley et al.	6,358,237 B1	3/2002	Paukovits et al.
6,200,289 B1	3/2001	Hochman et al.	6,361,263 B1	3/2002	Dewey et al.
6,202,923 B1	3/2001	Boyer et al.	6,362,591 B1	3/2002	Moberg
6,203,495 B1	3/2001	Bardy	6,363,282 B1	3/2002	Nichols et al.
6,203,528 B1	3/2001	Deckert et al.	6,363,290 B1	3/2002	Lyle et al.
6,206,238 B1	3/2001	Ophardt	6,364,834 B1	4/2002	Reuss et al.
6,206,829 B1	3/2001	Iliff	6,368,273 B1	4/2002	Brown
6,210,361 B1	4/2001	Kamen et al.	6,370,841 B1	4/2002	Chudy et al.
6,213,391 B1	4/2001	Lewis	6,381,577 B1	4/2002	Brown
6,213,738 B1	4/2001	Danby et al.	6,385,505 B1	5/2002	Lipps
6,213,972 B1	4/2001	Butterfield et al.	6,393,369 B1	5/2002	Carr
6,219,439 B1	4/2001	Burger	6,397,190 B1	5/2002	Goetz
6,219,587 B1	4/2001	Ahlin et al.	6,401,072 B1	6/2002	Haudenschild et al.
6,221,009 B1	4/2001	Doi et al.	6,402,702 B1	6/2002	Gilcher et al.
6,221,011 B1	4/2001	Bardy	6,406,426 B1	6/2002	Reuss et al.
6,221,012 B1	4/2001	Maschke et al.	6,407,335 B1	6/2002	Franklin-Lees et al.
6,222,619 B1	4/2001	Herron et al.	6,408,330 B1	6/2002	DeLaHuerga
6,224,549 B1	5/2001	Drongelen	6,416,471 B1	7/2002	Kumar et al.
6,225,901 B1	5/2001	Kail, IV	6,421,650 B1	7/2002	Goetz et al.
6,226,564 B1	5/2001	Stuart	6,424,996 B1	7/2002	Killcommons et al.
6,226,745 B1	5/2001	Wiederhold	6,427,088 B1	7/2002	Bowman, IV et al.
6,230,142 B1	5/2001	Benigno et al.	6,434,531 B1	8/2002	Lancelot et al.
6,230,927 B1	5/2001	Schoonen et al.	6,434,569 B1	8/2002	Tomlinson et al.
6,234,997 B1	5/2001	Kamen et al.	6,438,451 B1	8/2002	Lion
6,245,013 B1	6/2001	Minoz et al.	RE37,829 E	9/2002	Charhut et al.
6,246,473 B1	6/2001	Smith, Jr. et al.	6,449,927 B2	9/2002	Hebron et al.
6,248,063 B1	6/2001	Barnhill et al.	6,450,956 B1	9/2002	Rappaport et al.
6,248,065 B1	6/2001	Brown	6,458,102 B1	10/2002	Mann et al.
6,255,951 B1	7/2001	De La Huerga	6,461,037 B1	10/2002	O'Leary
6,256,643 B1	7/2001	Cork et al.	6,463,310 B1	10/2002	Swedlow et al.
6,256,967 B1	7/2001	Hebron et al.	6,468,242 B1	10/2002	Wilson et al.
6,259,355 B1	7/2001	Chaco et al.	6,470,234 B1	10/2002	McGrady
6,259,654 B1	7/2001	De La Huerga	6,471,089 B2	10/2002	Liff et al.
6,260,021 B1	7/2001	Wong et al.	6,471,645 B1	10/2002	Warkentin et al.
6,266,645 B1	7/2001	Simpson	6,471,646 B1	10/2002	Thede
6,269,340 B1	7/2001	Ford et al.	6,475,146 B1	11/2002	Frelburger et al.
D446,854 S	8/2001	Cheney, II et al.	6,475,148 B1	11/2002	Jackson et al.
6,270,455 B1	8/2001	Brown	6,475,180 B2	11/2002	Peterson et al.
6,270,457 B1	8/2001	Bardy	6,478,737 B2	11/2002	Bardy
6,272,394 B1	8/2001	Lipps	6,485,465 B2	11/2002	Moberg et al.
6,272,505 B1	8/2001	De La Huerga	6,494,831 B1	12/2002	Koritzinsky
6,277,072 B1	8/2001	Bardy	6,511,138 B1	1/2003	Gardner et al.
6,278,999 B1	8/2001	Knapp	6,519,569 B1	2/2003	White et al.
6,283,322 B1	9/2001	Liff et al.	6,537,244 B2	3/2003	Paukovits
6,283,944 B1	9/2001	McMullen et al.	6,542,902 B2	4/2003	Dulong et al.
6,290,646 B1	9/2001	Cosentino et al.	6,542,910 B2	4/2003	Cork et al.
6,290,650 B1	9/2001	Butterfield et al.	6,544,174 B2	4/2003	West et al.
6,294,999 B1	9/2001	Yarin et al.	6,544,228 B1	4/2003	Heitmeier
6,295,506 B1	9/2001	Heinonen	6,551,243 B2	4/2003	Bocionek et al.
6,304,788 B1	10/2001	Eady et al.	6,551,276 B1	4/2003	Mann et al.
6,306,088 B1	10/2001	Krausman et al.	6,554,791 B1	4/2003	Cartledge et al.
6,307,956 B1	10/2001	Black	6,554,798 B1	4/2003	Mann et al.
6,308,171 B1	10/2001	De La Huerga	6,555,986 B2	4/2003	Moberg
6,311,163 B1	10/2001	Sheehan et al.	6,558,321 B1	5/2003	Burd et al.
6,312,227 B1	11/2001	Davis	6,561,975 B1	5/2003	Pool et al.
6,312,378 B1	11/2001	Bardy	6,562,001 B2	5/2003	Lebel et al.
6,314,384 B1	11/2001	Goetz	6,564,104 B2	5/2003	Nelson et al.
6,317,719 B1	11/2001	Schrier et al.	6,564,105 B2	5/2003	Starkweather et al.
6,319,200 B1	11/2001	Lai et al.	6,564,121 B1	5/2003	Wallace et al.
6,321,203 B1	11/2001	Kameda	6,571,128 B2	5/2003	Lebel et al.
6,322,502 B1	11/2001	Schoenberg et al.	6,575,900 B1	6/2003	Zweig et al.
6,322,504 B1	11/2001	Kirshner	6,577,899 B2	6/2003	Lebel et al.
6,322,515 B1	11/2001	Goor et al.	6,579,232 B2	6/2003	Sakamaki et al.
6,330,491 B1	12/2001	Lion	6,581,069 B1	6/2003	Robinson et al.
6,332,090 B1	12/2001	DeFrank et al.	6,581,798 B2	6/2003	Liff et al.
RE37,531 E	1/2002	Chaco et al.	6,584,336 B1	6/2003	Ali et al.
6,337,631 B1	1/2002	Pai et al.	6,585,157 B2	7/2003	Brandt et al.
			6,585,644 B2	7/2003	Lebel et al.
			6,585,675 B1	7/2003	O'Mahony et al.
			6,592,551 B1	7/2003	Cobb
			6,593,528 B2	7/2003	Franklin-Lees et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

6,602,469	B1	8/2003	Maus et al.	7,277,579	B2	10/2007	Huang
6,607,485	B2	8/2003	Bardy	7,277,757	B2	10/2007	Casavant et al.
6,610,973	B1	8/2003	Davis, III	7,317,967	B2	1/2008	DiGianfilippo et al.
6,613,009	B1	9/2003	Bainbridge et al.	7,321,861	B1	1/2008	Oon
6,616,633	B1	9/2003	Butterfield et al.	7,343,224	B2	3/2008	DiGianfilippo et al.
6,635,014	B2	10/2003	Starkweather et al.	7,403,901	B1	7/2008	Carley et al.
6,648,821	B2	11/2003	Lebel et al.	7,427,002	B2	9/2008	Liff et al.
6,659,948	B2	12/2003	Lebel et al.	7,493,263	B2	2/2009	Helmus et al.
6,664,893	B1	12/2003	Eveland et al.	7,499,581	B2	3/2009	Tribble et al.
6,668,196	B1	12/2003	Villegas et al.	7,509,280	B1	3/2009	Haudenschild
6,669,663	B1	12/2003	Thompson	7,555,557	B2	6/2009	Bradley et al.
6,673,314	B1	1/2004	Burbank et al.	7,561,312	B1	7/2009	Proudfoot et al.
6,687,546	B2	2/2004	Lebel et al.	7,581,953	B2	9/2009	Lehmann et al.
6,689,091	B2	2/2004	Bui et al.	7,599,516	B2	10/2009	Limer et al.
6,694,191	B2	2/2004	Starkweather et al.	7,610,115	B2	10/2009	Rob et al.
6,694,334	B2	2/2004	DuLong et al.	7,630,908	B1	12/2009	Amrien et al.
6,711,460	B1	3/2004	Reese	7,636,718	B1	12/2009	Steen et al.
6,731,324	B2	5/2004	Levy	7,672,859	B1	3/2010	Louie et al.
6,733,447	B2	5/2004	Lai et al.	7,698,019	B2	4/2010	Moncrief et al.
6,735,497	B2	5/2004	Wallace et al.	7,698,154	B2	4/2010	Marchosky
6,740,075	B2	5/2004	Lebel et al.	7,734,478	B2	6/2010	Goodall et al.
6,746,398	B2	6/2004	Hervy et al.	7,753,085	B2	7/2010	Tribble et al.
6,758,810	B2	7/2004	Lebel et al.	7,769,601	B1	8/2010	Bleser et al.
6,768,425	B2	7/2004	Flaherty et al.	7,783,383	B2	8/2010	Eliuk et al.
6,771,369	B2	8/2004	Rzasa et al.	D624,225	S	9/2010	Federico et al.
6,775,602	B2	8/2004	Gordon, Jr. et al.	7,801,642	B2	9/2010	Ansari et al.
6,776,304	B2	8/2004	Liff et al.	7,847,970	B1	12/2010	McGrady
6,790,198	B1	9/2004	White et al.	7,853,621	B2	12/2010	Guo
6,804,656	B1	10/2004	Rosenfeld et al.	7,904,822	B2	3/2011	Monteleone et al.
6,810,290	B2	10/2004	Lebel et al.	7,931,859	B2	4/2011	Mlodzinski et al.
6,811,533	B2	11/2004	Lebel et al.	7,937,290	B2	5/2011	Bahir
6,811,534	B2	11/2004	Bowman, IV et al.	7,986,369	B1	7/2011	Burns
6,811,707	B2	11/2004	Rovatti et al.	7,991,507	B2	8/2011	Liff et al.
6,813,473	B1	11/2004	Braker	8,170,271	B2	5/2012	Chen
6,813,519	B2	11/2004	Lebel et al.	8,191,339	B2	6/2012	Tribble et al.
6,814,255	B2	11/2004	Liff et al.	8,215,557	B1	7/2012	Reno et al.
6,820,093	B2	11/2004	De La Huerga	8,220,503	B2	7/2012	Tribble et al.
6,842,736	B1	1/2005	Brzozowski	8,225,824	B2	7/2012	Eliuk et al.
6,847,861	B2	1/2005	Lunak et al.	D667,961	S	9/2012	Marmier
6,854,088	B2	2/2005	Massengale et al.	8,267,129	B2	9/2012	Doherty et al.
6,871,211	B2	3/2005	Labounty et al.	8,271,138	B2	9/2012	Eliuk et al.
6,873,268	B2	3/2005	Lebel et al.	8,280,549	B2	10/2012	Liff et al.
6,877,530	B2	4/2005	Osborne et al.	8,284,305	B2	10/2012	Newcomb et al.
6,880,034	B2	4/2005	Manke et al.	8,374,887	B1	2/2013	Alexander
6,885,288	B2	4/2005	Pincus	8,386,070	B2	2/2013	Eliuk et al.
6,887,201	B2	5/2005	Bardy	8,548,824	B1	10/2013	daCosta
6,892,941	B2	5/2005	Rosenblum	8,554,579	B2	10/2013	Tribble et al.
6,912,549	B2	6/2005	Rotter et al.	D693,480	S	11/2013	Spiess et al.
6,913,590	B2	7/2005	Sorenson et al.	8,595,206	B1	11/2013	Ansari et al.
6,915,265	B1	7/2005	Johnson	8,666,541	B1	3/2014	Ansari et al.
6,915,823	B2	7/2005	Osborne et al.	8,678,047	B2	3/2014	Tribble et al.
6,928,452	B2	8/2005	De La Huerga	D715,958	S	10/2014	Bossart et al.
6,950,708	B2	9/2005	Bowman, IV et al.	9,053,218	B2	6/2015	Osborne et al.
6,958,705	B2	10/2005	Lebel et al.	D733,480	S	7/2015	Shao
6,974,437	B2	12/2005	Lebel et al.	D738,152	S	9/2015	Grasselli et al.
6,975,924	B2	12/2005	Kircher et al.	D753,428	S	4/2016	Shao
6,976,628	B2	12/2005	Krupa	9,362,969	B1	6/2016	Burgess et al.
6,979,306	B2	12/2005	Moll	9,382,021	B2	7/2016	Tribble et al.
6,980,958	B1	12/2005	Surwit et al.	9,662,273	B2	5/2017	Ranalletta et al.
6,981,644	B2	1/2006	Cheong et al.	9,930,297	B2	3/2018	Alexander et al.
6,985,870	B2	1/2006	Martucci et al.	9,956,145	B2	5/2018	Thompson et al.
6,991,002	B2	1/2006	Osborne et al.	2001/0001237	A1	5/2001	Stroda et al.
6,995,664	B1	2/2006	Darling	2001/0003177	A1	6/2001	Schena et al.
7,006,893	B2	2/2006	Hart et al.	2001/0007053	A1	7/2001	Bardy
7,015,806	B2	3/2006	Naidoo et al.	2001/0007932	A1	7/2001	Kamen et al.
7,017,622	B2	3/2006	Osborne et al.	2001/0011153	A1	8/2001	Bardy
7,017,623	B2	3/2006	Tribble et al.	2001/0016699	A1	8/2001	Burbank et al.
7,028,723	B1	4/2006	Alouani et al.	2001/0017817	A1	8/2001	De La Huerga
7,096,212	B2	8/2006	Tribble et al.	2001/0021801	A1	9/2001	Bardy
7,117,902	B2	10/2006	Osborne	2001/0025138	A1	9/2001	Bardy
7,151,982	B2	12/2006	Liff et al.	2001/0025156	A1	9/2001	Bui et al.
7,194,336	B2	3/2007	DiGianfilippo et al.	2001/0027634	A1	10/2001	Hebron et al.
7,209,891	B1	4/2007	Addy et al.	2001/0028308	A1	10/2001	De La Huerga
7,240,699	B2	7/2007	Osborne et al.	2001/0030234	A1	10/2001	Wiklof
7,255,680	B1	8/2007	Gharib	2001/0031944	A1	10/2001	Peterson et al.
				2001/0032101	A1	10/2001	Statius Muller
				2001/0034502	A1	10/2001	Moberg et al.
				2001/0034614	A1	10/2001	Fletcher-Haynes et al.
				2001/0034616	A1	10/2001	Giannini

(56)

References Cited

U.S. PATENT DOCUMENTS

2001/0037057	A1	11/2001	Bardy	2002/0198624	A1	12/2002	Greenwald
2001/0037083	A1	11/2001	Hartlaub et al.	2003/0006878	A1	1/2003	Chung
2001/0037217	A1	11/2001	Abensour et al.	2003/0023177	A1	1/2003	Bardy
2001/0037220	A1	11/2001	Merry et al.	2003/0033532	A1	2/2003	Marks
2001/0041920	A1	11/2001	Starkweather et al.	2003/0036783	A1	2/2003	Bauhahn et al.
2001/0044588	A1	11/2001	Mault	2003/0046114	A1	3/2003	Davies et al.
2001/0044731	A1	11/2001	Coffman et al.	2003/0046280	A1	3/2003	Rotter et al.
2001/0047125	A1	11/2001	Quy	2003/0046439	A1	3/2003	Manke et al.
2001/0051764	A1	12/2001	Bardy	2003/0050621	A1	3/2003	Lebel et al.
2001/0053885	A1	12/2001	Gielen et al.	2003/0050731	A1	3/2003	Rosenblum
2002/0002326	A1	1/2002	Causey, III et al.	2003/0052787	A1	3/2003	Zerhusen
2002/0002473	A1	1/2002	Schrier et al.	2003/0060753	A1	3/2003	Starkweather et al.
2002/0004645	A1	1/2002	Carlisle et al.	2003/0060754	A1	3/2003	Reilly
2002/0007285	A1	1/2002	Rappaport	2003/0060765	A1	3/2003	Campbell et al.
2002/0010568	A1	1/2002	Rubbert et al.	2003/0060768	A1	3/2003	Kiyatake
2002/0010679	A1	1/2002	Felsher	2003/0065287	A1	4/2003	Spohn et al.
2002/0013612	A1	1/2002	Whitehurst	2003/0076736	A1	4/2003	Buker et al.
2002/0016567	A1	2/2002	Hochman et al.	2003/0078534	A1	4/2003	Hochman et al.
2002/0016568	A1	2/2002	Lebel et al.	2003/0079746	A1	5/2003	Hickle
2002/0016719	A1	2/2002	Nemeth et al.	2003/0083901	A1	5/2003	Bosch et al.
2002/0016722	A1	2/2002	Kameda	2003/0088238	A1	5/2003	Poulsen et al.
2002/0019606	A1	2/2002	Lebel et al.	2003/0097092	A1	5/2003	Flaherty
2002/0019748	A1	2/2002	Brown	2003/0097368	A1 *	5/2003	Tribble G06F 19/3456
2002/0022776	A1	2/2002	Bardy	2003/0114836	A1	6/2003	Estes et al.
2002/0025796	A1	2/2002	Taylor et al.	2003/0117580	A1	6/2003	Franz et al.
2002/0026104	A1	2/2002	Bardy	2003/0125609	A1	7/2003	Becker
2002/0029157	A1	3/2002	Marchosky	2003/0125611	A1	7/2003	Bardy
2002/0029776	A1	3/2002	Blomquist	2003/0139701	A1	7/2003	White et al.
2002/0032582	A1	3/2002	Feeney et al.	2003/0144878	A1	7/2003	Wilkes et al.
2002/0032602	A1	3/2002	Lanzillo, Jr. et al.	2003/0149599	A1	8/2003	Goodall et al.
2002/0038392	A1	3/2002	De La Huerga	2003/0154108	A1	8/2003	Fletcher-Haynes et al.
2002/0040208	A1	4/2002	Flaherty et al.	2003/0158508	A1	8/2003	DiGianfilippo
2002/0044043	A1	4/2002	Chaco et al.	2003/0160683	A1	8/2003	Blomquist
2002/0046062	A1	4/2002	Kameda	2003/0163088	A1	8/2003	Blomquist
2002/0046185	A1	4/2002	Villart et al.	2003/0163223	A1	8/2003	Blomquist
2002/0046346	A1	4/2002	Evans	2003/0163789	A1	8/2003	Blomquist
2002/0052539	A1	5/2002	Haller et al.	2003/0167035	A1	9/2003	Flaherty et al.
2002/0052542	A1	5/2002	Bardy	2003/0176933	A1	9/2003	Lebel et al.
2002/0052574	A1	5/2002	Hochman et al.	2003/0179287	A1	9/2003	Kozic et al.
2002/0062227	A1	5/2002	Yuyama	2003/0181851	A1	9/2003	Mann et al.
2002/0062229	A1	5/2002	Alban et al.	2003/0182164	A1	9/2003	Blomquist
2002/0065540	A1	5/2002	Lebel et al.	2003/0195397	A1	10/2003	Bardy
2002/0065686	A1	5/2002	Monteleone et al.	2003/0200117	A1	10/2003	Manetta et al.
2002/0067273	A1	6/2002	Jaques et al.	2003/0201697	A1	10/2003	Richardson
2002/0072733	A1	6/2002	Flaherty	2003/0212379	A1	11/2003	Bylund et al.
2002/0073250	A1	6/2002	Ommering	2003/0225596	A1	12/2003	Richardson et al.
2002/0077852	A1	6/2002	Ford et al.	2003/0225728	A1	12/2003	Moura
2002/0077865	A1	6/2002	Sullivan	2003/0231803	A1	12/2003	Huang
2002/0082480	A1	6/2002	Riff et al.	2004/0002874	A1	1/2004	Shaffer et al.
2002/0082865	A1	6/2002	Bianco et al.	2004/0017475	A1	1/2004	Akers et al.
2002/0082868	A1	6/2002	Pories et al.	2004/0019607	A1	1/2004	Moubayed et al.
2002/0084904	A1	7/2002	De La Huerga	2004/0039260	A1	2/2004	Bardy
2002/0087120	A1	7/2002	Rogers et al.	2004/0039264	A1	2/2004	Bardy
2002/0091309	A1	7/2002	Auer	2004/0051368	A1	3/2004	Caputo et al.
2002/0099283	A1	7/2002	Christ et al.	2004/0055611	A1	3/2004	Penny et al.
2002/0099301	A1	7/2002	Bardy	2004/0064343	A1	4/2004	Korpman et al.
2002/0100762	A1	8/2002	Liff et al.	2004/0073329	A1	4/2004	Engleson
2002/0107476	A1	8/2002	Mann et al.	2004/0088187	A1	5/2004	Chudy et al.
2002/0107707	A1	8/2002	Naparstek et al.	2004/0088374	A1	5/2004	Webb et al.
2002/0116226	A1	8/2002	Auer et al.	2004/0111293	A1	6/2004	Firanek et al.
2002/0116509	A1	8/2002	De La Huerga	2004/0115132	A1	6/2004	Young et al.
2002/0128606	A1	9/2002	Cowan et al.	2004/0116862	A1	6/2004	Ray
2002/0128871	A1	9/2002	Adamson et al.	2004/0117215	A1	6/2004	Marchosky
2002/0128880	A1	9/2002	Kunikiyo	2004/0128162	A1	7/2004	Schlotterbeck et al.
2002/0133377	A1	9/2002	Brown	2004/0129616	A1	7/2004	Mori et al.
2002/0140675	A1	10/2002	Ali et al.	2004/0148195	A1	7/2004	Kalies
2002/0143254	A1	10/2002	Maruyama	2004/0158193	A1	8/2004	Bui et al.
2002/0156462	A1	10/2002	Stultz	2004/0172283	A1	9/2004	Vanderveen et al.
2002/0158128	A1	10/2002	Ashiuro	2004/0172289	A1	9/2004	Kozic et al.
2002/0165491	A1	11/2002	Reilly	2004/0172300	A1	9/2004	Mihai et al.
2002/0169636	A1	11/2002	Eggers et al.	2004/0193328	A1	9/2004	Zaitzu et al.
2002/0173875	A1	11/2002	Wallace et al.	2004/0193453	A1	9/2004	Butterfield et al.
2002/0188467	A1	12/2002	Eke	2004/0204673	A1	10/2004	Flaherty
2002/0198473	A1	12/2002	Kumar et al.	2004/0204954	A1	10/2004	Lacko
2002/0198513	A1	12/2002	Lebel et al.	2004/0215490	A1	10/2004	Duchon et al.
				2004/0220829	A1	11/2004	Baharav et al.
				2004/0225528	A1	11/2004	Brock
				2004/0235446	A1	11/2004	Flaherty et al.
				2004/0236630	A1	11/2004	Kost et al.

(56)

References Cited

U.S. PATENT DOCUMENTS

2004/0248295	A1	12/2004	Katsuhiko et al.	
2004/0260233	A1	12/2004	Garibotto et al.	
2004/0260577	A1	12/2004	Dahlin et al.	
2005/0001033	A1	1/2005	Cheong et al.	
2005/0017864	A1	1/2005	Tsoukalis	
2005/0033124	A1	2/2005	Kelly et al.	
2005/0033773	A1	2/2005	Roberge et al.	
2005/0038680	A1	2/2005	McMahon	
2005/0039742	A1	2/2005	Hickle	
2005/0043665	A1	2/2005	Vinci et al.	
2005/0045548	A1	3/2005	Brugger et al.	
2005/0054923	A1	3/2005	Pan	
2005/0060372	A1	3/2005	DeBettencourt et al.	
2005/0080651	A1	4/2005	Morrison et al.	
2005/0187794	A1	8/2005	Kimak	
2005/0209737	A1	9/2005	Kircher	
2005/0228238	A1	10/2005	Monitzer	
2005/0279419	A1	12/2005	Tribble et al.	
2006/0084042	A1	4/2006	Weaver et al.	
2006/0124656	A1	6/2006	Popovic, Jr.	
2006/0136095	A1	6/2006	Rob et al.	
2006/0149416	A1*	7/2006	Mohapatra	G16H 20/10 700/242
2006/0161294	A1	7/2006	DiMaggio	
2006/0173714	A1	8/2006	Grotzinger	
2006/0178578	A1	8/2006	Tribble et al.	
2006/0181391	A1	8/2006	McNeill et al.	
2006/0235881	A1	10/2006	Masarie et al.	
2007/0038915	A1*	2/2007	Alaimo	G11B 20/182
2007/0043767	A1	2/2007	Osborne et al.	
2007/0047980	A1	3/2007	Limer et al.	
2007/0088568	A1	4/2007	Goodall et al.	
2007/0110305	A1	5/2007	Corcoran et al.	
2007/0125442	A1	6/2007	Tribble et al.	
2007/0168228	A1*	7/2007	Lawless	G06F 19/3418 705/2
2007/0179806	A1	8/2007	Knowlton et al.	
2007/0189597	A1	8/2007	Limer et al.	
2007/0192139	A1	8/2007	Cookson et al.	
2007/0216998	A1	9/2007	Sander	
2007/0239482	A1	10/2007	Finn et al.	
2007/0239997	A1	10/2007	Qu et al.	
2008/0046292	A1	2/2008	Myers et al.	
2008/0056556	A1	3/2008	Eller et al.	
2008/0059228	A1	3/2008	Bossi et al.	
2008/0091467	A1	4/2008	Moncrief et al.	
2008/0119958	A1	5/2008	Bear et al.	
2008/0125897	A1	5/2008	DiGianfilippo et al.	
2008/0147554	A1	6/2008	Stevens et al.	
2008/0195246	A1	8/2008	Tribble et al.	
2008/0306926	A1	12/2008	Friedlander et al.	
2009/0024414	A1	1/2009	Mansour et al.	
2009/0080408	A1	3/2009	Natoli et al.	
2009/0097368	A1	4/2009	Vlutters et al.	
2009/0138340	A1	5/2009	Borr et al.	
2009/0188937	A1	7/2009	Kim	
2009/0205877	A1	8/2009	Claypool	
2009/0210252	A1	8/2009	Silver	
2009/0235194	A1	9/2009	Arndt et al.	
2009/0258331	A1	10/2009	Do et al.	
2009/0285762	A1	11/2009	Flower	
2009/0313044	A1	12/2009	Haque et al.	
2009/0323170	A1	12/2009	Lin	
2009/0324032	A1	12/2009	Chen	
2010/0010841	A1*	1/2010	Cooper	G06Q 10/02 705/6
2010/0017031	A1	1/2010	Rob et al.	
2010/0091281	A1	4/2010	Suzuki	
2010/0094653	A1	4/2010	Tribble et al.	
2010/0128165	A1	5/2010	Newcomb et al.	
2010/0157293	A9	6/2010	Rzasa et al.	
2010/0185456	A1	7/2010	Kansal	
2010/0241270	A1	9/2010	Eliuk et al.	
2011/0119088	A1*	5/2011	Gunn	G06Q 50/24 705/3

2011/0191121	A1	8/2011	Fioravanti	
2011/0202366	A1	8/2011	Akers et al.	
2011/0208350	A1	8/2011	Eliuk et al.	
2011/0267465	A1	11/2011	Alexander et al.	
2012/0097290	A1	4/2012	Mikhaeil	
2012/0200596	A1	8/2012	Gotou et al.	
2012/0211565	A1	8/2012	Colavito et al.	
2012/0303388	A1	11/2012	Vishnubhalta et al.	
2013/0079581	A1	3/2013	Agamaite et al.	
2013/0090947	A1	4/2013	Nockley	
2013/0197445	A1	8/2013	Schabbach et al.	
2013/0262138	A1	10/2013	Jaskela et al.	
2013/0279774	A1	10/2013	Helgason et al.	
2013/0304510	A1	11/2013	Chan et al.	
2013/0314535	A1	11/2013	Yuyama et al.	
2013/0342676	A1	12/2013	Amano	
2014/0022569	A1	1/2014	Matsui et al.	
2014/0156064	A1	6/2014	Crawford et al.	
2014/0156294	A1	6/2014	Tribble et al.	
2014/0214436	A1	7/2014	Utech et al.	
2014/0350950	A1	11/2014	Jaskela et al.	
2015/0205932	A1	7/2015	Tribble	
2015/0227719	A1	8/2015	Ranalletta	
2015/0272320	A1*	10/2015	Ranalletta	G06V 20/66 108/23
2015/0278477	A1	10/2015	Tribble	
2015/0286799	A1	10/2015	Padmani	
2016/0072985	A1	3/2016	Sandmann et al.	
2016/0092638	A1	3/2016	Padmani	
2016/0092639	A1	3/2016	Padmani	
2016/0140315	A1	5/2016	Diaz et al.	
2016/0210437	A1	7/2016	Padmani et al.	
2016/0371462	A1	12/2016	Wallen	
2017/0372034	A1	12/2017	Tribble	

FOREIGN PATENT DOCUMENTS

CN	1131076	12/2003
EP	0237588	9/1987
EP	0462466	12/1991
EP	0505627	9/1992
EP	0522527	1/1993
EP	0439355	9/1994
EP	0844581	5/1998
EP	0960627	12/1999
EP	0970655	1/2000
EP	1072994	2/2001
EP	1107158	A1 6/2001
EP	1097671	2/2003
GB	994977	A 6/1965
GB	2210713	2/1987
GB	2279784	1/1995
GB	2285135	6/1995
GB	2379037	2/2003
JP	53137644	12/1978
JP	61066950	4/1986
JP	63068133	3/1988
JP	2111375	4/1990
JP	3423055	B2 1/1994
JP	06327636	11/1994
JP	07204253	A 8/1995
JP	104585	1/1998
JP	10014890	1/1998
JP	10079770	3/1998
JP	2000036032	A 2/2000
JP	03055131	4/2000
JP	2002011095	1/2002
JP	2002092181	A 3/2002
JP	2002520718	7/2002
JP	2003022322	1/2003
JP	2004078970	3/2004
JP	2004326436	11/2004
JP	2004340770	A 12/2004
JP	2005252710	A 9/2005
JP	2006033291	A 2/2006
JP	2006334062	12/2006
JP	2007198934	A 8/2007
JP	2008139201	A 6/2008
JP	4276654	B2 6/2009

(56)

References Cited

FOREIGN PATENT DOCUMENTS

JP	2009265827	A	11/2009
JP	2010056619	A	3/2010
JP	2010170504		8/2010
JP	2010533927		10/2010
JP	2011151430	A	8/2011
JP	2012078265		4/2012
JP	5342197	B2	11/2013
JP	5747150	B2	7/2015
JP	6086813		3/2017
KR	20000036642		7/2000
KR	102000036642		7/2000
KR	20010094703	A	11/2001
KR	1020010094703		11/2001
KR	20050054379		12/2003
KR	20110115927	A	10/2011
KR	1020110115927		10/2011
KR	20130001500		1/2013
WO	WO8400493		2/1984
WO	WO9524010	A1	9/1995
WO	WO9634291	A1	10/1996
WO	WO9741525		11/1997
WO	WO9814275	A1	4/1998
WO	WO9815092	A1	4/1998
WO	WO9824358	A1	6/1998
WO	WO9833433	A1	8/1998
WO	WO9859487		12/1998
WO	WO9904043		1/1999
WO	WO9910029		3/1999
WO	WO9942933		8/1999
WO	WO9944162		9/1999
WO	WO9959472		11/1999
WO	WO0013588		3/2000
WO	WO0029983		5/2000
WO	WO0043941		7/2000
WO	WO0052437		9/2000
WO	WO0052626		9/2000
WO	WO0057339		9/2000
WO	WO0060449		10/2000
WO	WO0069331		11/2000
WO	WO0072181		11/2000
WO	WO0078374		12/2000
WO	WO0101305		1/2001
WO	WO0102979		1/2001
WO	WO0106468		1/2001
WO	WO0145774		6/2001
WO	WO0217777		7/2002
WO	WO02091276	A1	11/2002
WO	WO03025826	A2	3/2003
WO	WO03094073		11/2003
WO	WO2004070557		8/2004
WO	WO2004070994		8/2004

OTHER PUBLICATIONS

Peterson, Charles D. and Anderson, Jr., Howard C.; "The North Dakota Telepharmacy Project: Restoring and Retaining Pharmacy Services in Rural Communities" Feb. 1, 2004.

Cato Reference Manual, Support for Trial Version (Abridged), Vienna, May 2004 Jun. 1, 2004.

Seifert et al.; "The Training of a Telepharmacist: Addressing the Needs of Rural West Texas," American Journal of Pharmaceutical Education 2004; 68 (3) Article 60. Jul. 16, 2004.

Cato Reference Manual, Vienna, May 2005 May 1, 2005.

Phillips, Jon, Associate Director of Telemedicine; "Telepharmacy at Texas Tech," presented Apr. 30, 2003, published at <http://www.ttuhs.edu/telemedicine/publication.htm> at least by Jun. 22, 2003 Jun. 22, 2003.

Singapore Written Opinion for related Singapore Application No. 11201703338P; report dated Apr. 18, 2018; (6 pages).

European Search Report for related European Application No. EP15852627; report dated May 4, 2018; (16 pages).

AHRQ Health Information Technology Program-Update Jun. 2005 Fact Sheet., <http://www.ahrq.gov/research/findings/factsheets/it/hitfact/index.html>—3 pages.

Albert A. Cook, "An integrated nursing-pharmacy approach to a computerized medication dispensing/administration system," Hospital Pharmacy, May 1985, pp. 321-325, vol. 20, JB Lippincott Company, Philadelphia, PA.

Allan T. Pryor, "Current State of Computer-based Patient Record Systems," Aspects of the Computer-based Patient Record, 1992, pp. 67-82, Springer-Verlag, New York, NY.

Anderson, Howard "A Narrative on the History of the Development of Telepharmacy in North Dakota from the Board of Pharmacy's Perspective Recorded by Excerpts from Board Minutes", Feb. 2006.

Angaran, "Telemedicine and telepharmacy: Current status and future implications", Am J Health-Syst Pharm, vol. 56, Jul. 15, 1999, pp. 1404-1405.

Ann Slone Endo, "Using Computers in Newborn Intensive Care Settings," American Journal of Nursing, Jul. 1981, pp. 1336-1337.

Anonymous, "Chains covet customized pharmacy integration" Drug Store New, Aug. 18, 2003, vol. 25, No. 10—p. 73.

Automated Dispensing Technologies: Directory of Vendors, <http://pharmacyautomation.com/vendors.html>, Jun. 5, 2003—3 pages.

Auto Syringe® AS40A Infusion Pump Technical Manual, 1995, 89 pages, Baxter Healthcare Corporation, Deerfield, IL.

Auto Syringe® AS40A: Model AS40A Infusion Pump Operation Manual, undated, 78 pages, Baxter Healthcare Corporation, Deerfield, IL.

Baxa Corporation, DoseEdge The Leading Edge in Dose Management, Brochure, published copyright date 2010—5 pages.

Baxa Corporation, Product Catalog 2010-2011, published at least by Sep. 15, 2012, <https://web.archive.org/web/20120915210739http://www.baxa.com/resources/docs/BaxaCatalog.pdf> (52 pages).

Bell Atlantic Healthcare Systems, Inc., court exhibit, StatLan Functions and Features, Specification, release 3.5, dated Nov. 12, 1992, 49 pages.

Ben Schneiderman, "Designing the User Interface: Strategies for Effective Human-Computer Interaction," 2d Ed., 1992, Chapter 5: Direct Manipulation (56 pages), Addison-Wesley Publishing Company.

"Block Medical: Growing with Home Infusion Therapy," taken from Invivo, the Business and Medicine Report, Apr. 1991, pp. 7-9.

Bynum et al., "The Effect of Telepharmacy Counseling on Metered-Dose Inhaler Technique among Adolescents with Asthma in Rural Arkansas", Telemedicine Journal and e-health, vol. 7, No. 3, 2001, Mary Ann Liebert, Inc., pp. 207-218.

Cabral, Jr. et al., "Multimedia Systems for Telemedicine Systems for Telemedicine and Their Communications Requirements," IEEE Communications Magazine Jul. 1996, pp. 20-27.

Cardinal Health Introduces Rx-source(SM) to Address Pharmacist Labor Shortage and Medication Safety Challenges at Hospitals. PR Newswire, Feb. 25, 2003—5 pages.

Casey, Michelle M. et al., "Pharmacist Staffing and the Use of Technology in Small Rural Hospitals: Implications for Medication Safety" Upper Midwest Rural Health Research Center, Dec. 2005—51 pages.

Charles Safran, M.D. et al., "Computer-Based Support for Clinical Decision Making," Clinical Computin, vol. 7, No. 5 (1990), pp. 319-322.

Clayton M. Curtis, "A Computer-based Patient Record Emerging from the Public Sector: The Decentralized Hospital Computer Program," First Annual Nicholas E. Davies Award Proceedings of the CPR Recognition Symposium, 1995, pp. 75-132, Computer-based Patient Record Institute, Inc., Bethesda, MD.

Clement J. McDonald, M.D. et al., "The Three-Legged Stool: Regenstrief Institute for Health Care," Third Annual Nicholas E. Davies Award Proceedings of the CPR Recognition Symposium, 1997, pp. 131-158, Computer-based Patient Record Institute, Inc., Bethesda, MD.

Clement J. McDonald, M.D. et al., The Regenstrief Medical Record System: 20 Years of Experience in Hospitals, Clinics, and Neighborhood Health Centers, M.D. Computing, 1992 pp. 206-217, vol. 9, No. 4, Springer-Verlag, New York, NY.

(56) **References Cited**

OTHER PUBLICATIONS

- Clifton, G. Dennis et al., "Provision of pharmacy services to underserved populations via remote dispensing and two-way videoconferencing" *Am J Health-Syst Pharm*, vol. 60, Dec. 15, 2003 oe pp. 2577-2582.
- Dan Murphy, "Nuclear Pharmacy Primer", *Radiation Protection Management*, vol. 20, No. 5 (2003), pp. 1-10.
- Dan Scheraga, "Tech firms answer chain pharmacy's call for productivity," *Drug Store News*; Dec. 15, 2003; 25, 17; ProQuest Research Library, p. 31-32.
- Daniel Andresen et al., "Scalability Issues for High Performance Digital Libraries on the World Wide Web," *Proceedings of ADL '96*, 1996, pp. 139-148, IEEE.
- Daniel J. Nigrin et al., "Glucoweb: A Case Study of Secure, Remote Biomonitoring and Communication," *Proceedings of the 2000*, 5 pages, American Medical Informatics Association, Bethesda, MD.
- Darryl V. Wareham et al., "Combination Medication Cart and Computer Terminal in Decentralized Drug Distribution," *American Journal of Hospital Pharmacy*, Jun. 1983, pp. 976-978, vol. 40, American Society of Hospital Pharmacists.
- Dart, Luann, "Digital Doses" *Rural Electric*, Jan. 2005—pp. 28-31.
- Deborah J. Mayhew, "Principles and Guidelines in Software user Interface Designs," 1992, selected portions of Chapter 9, 17 pages, Prentice-Hall, Inc.
- Defendants Initial Invalidity Contentions with Exhibits A and B dated Sep. 8, 2014; Civil Action No. 1:14-cv-00222.
- Dennis D. Cote et al., "Robotic system for i.v. antineoplastic drug preparation: Description and preliminary evaluation under simulated conditions," *American Journal of Hospital Pharmacy*, Nov. 1989, pp. 2286-2293, vol. 46, American Society of Hospital Pharmacists.
- Donna Young, "Loan repayments help pharmacists provide care in medically underserved areas," *American Journal of Health—System Pharmacy*; Nov. 1, 2003, pp. 2186-2188, vol. 60.
- Environmental Scan of Pharmacy Technicians; M. MacInnis; Canadian Pharmacists Association; Sep. 2001.
- Exhibit 1, Publications Manually Reviewed for the Search to U.S. Pat. No. 8,347,887 titled "System and Method for Remotely Supervising and Verifying Pharmacy Functions" As of Jun. 25, 2014.
- Exhibit 1001 U.S. Pat. No. 8,374,887, Alexander issued Feb. 12, 2013.
- Exhibit 1002 Patent File History U.S. Pat. No. 8,374,887.
- Exhibit 1003, Declaration of Mr. Brian T. Hart from U.S. Pat. No. 8,374,887.
- Exhibit 1004, Declaration of Wayne H. Grant from U.S. Pat. No. 8,374,887.
- Exhibit 1005, 22 TAC §§291.20, 291.36, and 291.71—291.74 date issued Mar. 5, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1006 U.S. Pat. No. 6,711,460 Reese issued Mar. 23, 2004 from U.S. Pat. No. 8,374,887.
- Exhibit 1009, Peterson et al., *The North Dakota Telepharmacy Project: Restoring and Retaining Pharmacy Services in Rural Communities; the journal of Pharmacy Technology*, vol. 20, No. 1, Jan./Feb. 2004—pp. 1-39 from U.S. Pat. No. 8,374,887.
- Exhibit 1010, Declaration of Benjamin E. Weed from U.S. Pat. No. 8,374,887.
- Exhibit 1011, Complaint—*Alexander v. Baxter*, (W.D.Texas 2014) filed Mar. 13, 2014 from U.S. Pat. No. 8,374,887.
- Exhibit 1012, Charles F. Seifert et al., "The Training of a Telepharmacist: Addressing the Needs of Rural West Texas," *American Journal of Pharmaceutical Education*, 2004; 68 (3) Article 60—pp. 1-9 from U.S. Pat. No. 8,374,887.
- Exhibit 1013, Assignment Emily H. Alexander to Becton, Dickinson and Company; U.S. Appl. No. 13/747,231; Reel 034110/ Frame 0789 from U.S. Pat. No. 8,374,887.
- Exhibit 1014, Exhibit A—Corrected Parties' Claims Construction Terms, Proposed Construction and cites Civil, 1:14cv-00222-LY—pp. 1-7 from U.S. Pat. No. 8,374,887.
- Exhibit 1015, Information about Telepharmacy presentation 42503 and Presentation Telepharmacy at Texas Tech; Jon Phillips—1-27 from U.S. Pat. No. 8,374,887.
- Exhibit 1017, Declaration of Dr. Roger W. Anderson in Support of Becton, Dickinson & Company's Response to Baxter's Motion for Summary Judgment of Invalidity Based Upon 35 U.S.C. § 101 filed Jan. 15, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1018, Plaintiff's Claim Construction Brief, 1:14-cv-222-LY filed Oct. 17, 2014 from U.S. Pat. No. 8,374,887.
- Exhibit 1019, Plaintiff's Reply Claim Construction Brief, 1:14-cv-222-LY filed Nov. 7, 2014 from U.S. Pat. No. 8,374,887.
- Exhibit 1020, *The United States Pharmacopeia—the Official Compendia of Standards*; 2004 from U.S. Pat. No. 8,374,887.
- Exhibit 1021, Curriculum Vitae of Brian T Hart from U.S. Pat. No. 8,374,887.
- Exhibit 1022, Curriculum Vitae of Wayne H Grant—Expert oversight-Expert Witness-Litigation Support from U.S. Pat. No. 8,374,887.
- Exhibit 1023, Charles D Peterson et al., "The North Dakota Telepharmacy Project: Restoring and Retaining Pharmacy Services in Rural Communities," *J Pharm Technol*, 2004; vol. 20—pp. 028-039 from U.S. Pat. No. 8,374,887.
- Exhibit 1025, Affidavit of Christopher Butler with attached Telemedicine Report Archive dated Mar. 4, 2015—6 pages from U.S. Pat. No. 8,374,887.
- Exhibit 1026, Affidavit of Christopher Butler with attached presentation Telepharmacy at Text Tech—Jon Phillips dated Mar. 4, 2015—31 pages from U.S. Pat. No. 8,374,887.
- Exhibit 1027, Order on Motion for Summary Judgment filed Aug. 3, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1028, Final Judgment filed Aug. 3, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1029 Charles Seifert from U.S. Pat. No. 8,374,887.
- Exhibit 1030 Deposition of Charles Seifert Dec. 4, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1031 Deposition of Diane B. Ginsburg, PhD. Dec. 16, 2015 from U.S. Pat. No. 8,374,887.
- Exhibit 1032 Texas Administrative Code, Title 22, Chapter 291, Subchapter A, Section 291.23 as in effect on Feb. 1, 2004 from U.S. Pat. No. 8,374,887.
- Felkey, Bill G., "Integrating Technology at the Point of Care", *Insight*, Jan. 2004—pp. 8-10.
- Formula for Patient Safety; ScriptPro; Aug. 17, 2003.
- Fred Puckett, "Medication-management component of a point-of-care information system," *Am. J. Health-Syst.Pharm.*, Jun. 15, 1995, pp. 1305-1309, vol. 52, American Society of Health-System Pharmacists, Inc.
- "GE ImageQuant TL 7.0 Image Analysis Software" User Manual , May 2007, <http://nba.uth.tmc.edu/Assets/pdf/other/typhoon-supporting-files/IQTL-UserManual.pdf>, Uppsala, Sweden.
- Gerald E. Meyer et al., "Use of bar codes in inpatient drug distribution," *Am. J. Hosp. Pharm.*, May 1991, pp. 953-966, vol. 48, American Society of Hospital Pharmacists, Inc.
- Ghent, Natale, "Pharmacists go digital to fight shortage", *Pharmacy Practice* 20.11 (Nov. 2004): 47—2 pages.
- Gilad J. Kuperman, M.D. et al., "Innovations and research review: The impact of the HELP computer system on the LDS Hospital paper medical record," *Topics in Health Record Management*, 1991, pp. 76-85, vol. 12, Issue 2, Aspen Publishers, Inc.
- "Global Med Announces First Safetrace TX™ Sale," Apr. 1, 1999, 2 pages.
- Global Med Technologies, Inc. Introduces PeopleMed™.com, inc., A Chronic Disease Management Application Service Provider (ASP) Subsidiary, Jan. 11, 2000, 2 pages, Global med Technologies, Inc., Denver, CO.
- Gretchen A. Barry et al., "Bar-code technology for documenting administration of large-volume intravenous solutions," *American Journal of Hospital Pharmacy*, Feb. 1989, pp. 282-287, vol. 46, American Society of Hospital Pharmacists.
- H. Paul Hammann et al., "A World Wide Web Accessible Multi-Species ECG Database," 1997, pp. 7-12, ISA.
- Halverson, Daniel R. IsoRx: TelePharmacy Software presentation—23 pages.

(56)

References Cited

OTHER PUBLICATIONS

- Henry J. Lowe et al., "WebReport: A World Wide Web Based Clinical Multimedia Reporting System," 1996, pp. 314-318, AMIA, Inc.
- "Hospitals battle errors with bar codes," Mar. 24, 2004, 3 pages, MSNBC.
- Howard L. Bleich et al., "Clinical Computing in a Teaching Hospital," Use and Impact of Computers in Clinical Medicine, 1987, pp. 205-223 and selected p. Springer—Verlag, New York, NY. <http://isorex.com/> Jan. 29, 2004.
- <http://www.scriptpro.com/products/sp-200/main.htm>, Feb. 13, 2004, Product listing for SP 200® Robotic Prescription Dispensing System.
- <http://www.scriptpro.com/products/space/space200.htm>, Feb. 10, 2004, Product listing for SP Automation Center 200TM (Space 200TM) Prescription Dispensing Automation Center.
- Hughes, Shirley, "Bedside Terminals: Clinicom," Clinical Computing, Jan./Feb. 1988, pp. 22-28, vol. 5, No. 1.
- IPR Decision Paper No. 8 Entered Aug. 13, 2015 from U.S. Pat. No. 8,374,887.
- IPR Final Written Decision Paper No. 29 Entered Jul. 11, 2016 from U.S. Pat. No. 8,374,887.
- James Kazmer et al., "The Creation of Virtual Electronic Medical Record," 1996, 17 pages.
- Jennifer Langham, "Taking Automation to New Levels," Insight, the QS/1 Magazine, Oct. 2002; pp. 2-5.
- John Frady, "What's New in RxCare Plus 17.2," Insight, the QS/1 Magazine, Apr. 2002; pp. 2-3, 14.
- Jones, et al., "Use of a remote computerized system for study documentation in clinical trials" Drug Information Journal, Oct.-Dec. 1998, vol. 32, No. 4 oe pp. 1153-1163.
- Karen E. Bradshaw et al., "Physician decision-making—Evaluation of data used in a computerized ICU," International Journal of Clinical Monitoring and Computing, 1984, pp. 81-91, vol. 1, Martinus Nijhoff Publishers, Netherlands.
- Kastango, Eric S. and Bradshaw, Brian D., "USP chapter 797: Establishing a practice standard for compounding sterile preparations in pharmacy" Am J Health-Syst Pharm., Sep. 15, 2004, vol. 61—pp. 1928-1938.
- Kenneth N. Barker et al., "Effect of an automated bedside dispensing machine on medication errors," American Journal of Hospital Pharmacy, Jul. 1984, pp. 1352-1358, vol. 41, No. 7, American Society of Hospital Pharmacists.
- Keelys, Christopher A. et al., "Providing nighttime pharmaceutical services through telepharmacy" Am J Health-Syst Pharm, Apr. 15, 2002, vol. 59—pp. 716-721.
- Khan, Shamima et al., "Is There a Successful Business Case for Telepharmacy?" Telemedicine and e-Health, vol. 14, No. 3, Apr. 2008, pp. 235-245.
- Kimber, Michael B. et al., "Telepharmacy-Enabling Technology to Provide Quality Pharmacy Services in Rural and Remote Communities" Journal of Pharmacy Practice and Research, vol. 36, No. 2, 2006—128-133.
- Kodak DirectView PACS—Rural Hospital Joins the Big Leagues PACS/Enterprise Information management (EIM) Solution—www.kodak.com/go/medical—4 pages.
- Kosub, David, "Device allows pharmacy care in remote areas" Pharmacy Practice, vol. 20, No. 10, Oct. 2004—pp. 12-13.
- Koutnik, Eileen, Assistnat Editor, Pharmacy Times, "The Pharmacy of Tomorrow" Pharmacy Times, Aug. 1, 2003—3 pages.
- Larry B. Grandia, B.S.E et al., "Building a computer-based Patient Record System in an Evolving Integrated Health System," First Annual Nicholas E. Davies Award Proceedings of the CPR Recognition Symposium, 1995, pp. 19-55, Computer-based Patient Record Institute, Inc., Bethesda, MD.
- Leffkowitz, Sheldon et al., "A Trial of the Use of Bar Code Technology to Restructure a Drug Distribution and Administration System," 1991, pp. 239-242, Hospital Pharmacy, vol. 26.
- LP, "ATM-Style Drug Dispensers Taking Hold in Areas With Limited Pharmacist Services" Pharmacy Practice News, Jan. 2004, vol. 31, No. 1—4 pages.
- "The Longitudinal Clinical Record: A View of the Patient," taken from Proceedings of the 1994 Annual HIMSS Conference, Feb. 14, 1994, pp. 239-250, Healthcare Information and Management Systems Society, Chicago, Illinois, USA.
- Lustig, Ahuva, "Medication error prevention by pharmacists—An Israeli solution" Pharmacy World & Science, 2000, vol. 22, No. 1—pp. 21-25.
- Medicaid Memo—Department of Medical Assistance Services (Converting NDCs from 10-digits to 11-digits) May 31, 2007.
- Medcin® Technical Overview, undated, 111 pages, Medicom Systems.
- Michael H. Mackin, "Impact of Technology on Environmental Therapeutic Device Design," Medical Instrumentation, Feb. 1987, pp. 33-35, vol. 21, No. 1, Association for the Advancement of Medical Instrumentation.
- Michelle M. Casey, M.S., Jill Klingner, R.N., M.S., and Ira Moscovice, Ph. D.; "Access to Rural Pharmacy Services in Minnesota, North Dakota, and South Dakota," Working Paper Series, Jul. 2001, #36.
- Monane et al., "Improving Prescribing Patterson for the Elderly Through an Online Drug Utilization Review Intervention", JAMA, Oct. 14, 1998, vol. 280, No. 14—pp. 1249-1252.
- Morris, Aisha M., Schneider, Philip J., Pedersen, Craig A. and Mirtallo, Jay M. "National survey of quality assurance activities for pharmacy-compounded sterile preparations" Am J Health-Syst Pharm, Dec. 15, 2003, vol. 60—pp. 2567-2576.
- Murray, Michael D. et al. "Effects of Computer-based Prescribing on Pharmacist Work Patterns" Journal of the American Medical Informatics Association, Nov./Dec. 1998, vol. 5, No. 6—pp. 546-553.
- Napoli, M et al., "Picture archiving and communication in radiology", Rays, Jan.-Mar. 2003—PubMed—NCBI <http://www.ncbi.nlm.nih.gov/pubmed/14509181>—Abstract.
- Nissen et al., Can telepharmacy provide pharmacy services in the bush, School of Pharmacy, University of Queensland, Brisbane, Australia, Journal of Telemedicine and Telecare 2003, vol. 9 (Suppl. 2): S2:39-41.
- North Dakota Century Code Statute Law—State Board of Pharmacy—219 pages.
- Parks, Liz, "Annual report of retail pharmacy: Using central-fill to maximize dispensing" Drug Store News, Aug. 20, 2001 vol. 24, No. 11—pp. 51, 75.
- Parsons, et al., "Digital Media-Can I Change a Graphic's File Size?," New Perspectives on Computer Concepts—Course Technology, 2011, Cengage Learning, Boston, MA.
- Paul H. Perlstein et al., "Computer-Assisted Newborn Intensive Care," Pediatrics, Apr. 1976, pp. 494-501, vol. 57, No. 4, American Academy of Pediatrics, Inc., Evanston, Illinois.
- Paul H. Perlstein et al., "Future Directions for Device Design and Infant Management," Medical Instrumentation, Feb. 1987, pp. 36-41, vol. 21, No. 1, Association for the Advancement of Medical Instrumentation.
- PCA II Multi-Mode Cartridge Operator's Manual, Sep. 1995, approx. 40 pages, Baxter Healthcare Corporation, Deerfield, IL.
- Pesce, James, "Bedside Terminals: Medtake," Clinical Computing, Jan./Feb. 1988, pp. 16-21, vol. 5, No. 1.
- Peter Lord et al., MiniMed Technologies Programmable Implantable Infusion System, Annals New York Academy of Science, pp. 66-71, describing clinical trials from Nov. 1986.
- Peterson et al., The North Dakota Telepharmacy Project Restoring and Retaining Pharmacy Services in Rural Communities—Presentation North Dakota State University, Fargo, North Dakota. Petition for Inter Partes Review *Baxter International Inc. v. Becton, Dickinson and Company* for U.S. Pat. No. 8,374,887, pp. 1-69.
- Pharmacy Automation Online Vendors Page; Internet Archive Wayback Machine; <http://pharmacyautomation.com/vendors.html>—3 pages.
- Pharmacy Data Management (PDM) Technical Manual/Security Guide Version 1.0, Sep. 1997—55 pages.
- Pharmacy education and practice out of sync? (Roundtable) Chain Drug Review, vol. 25, No. 6, Mar. 17, 2003, RX2 (6).

(56)

References Cited

OTHER PUBLICATIONS

- Prem S. Chopra, Virgil A. Thomason, and Dell M. Stinett; "Voice-Activated Networked Workstation for a Physically Disabled Physician," 10-7803-2050-6/94 1994 IEEE, pp. 478-479.
- Product literature, Baxter Healthcare Corporation, "Flo-Gard® 6201 Volumetric Infusion Pump," 1992, 2 pages.
- Product literature, Baxter Healthcare Corporation, "MultiPlex™ Series 100 Fluid Management System," 1988, 2 pages.
- Product literature, Baxter Healthcare Corporation, "MultiPlex™ Series 100 Fluid Management System," undated, 2 pages.
- Remote Dispensing Regulations, NABPLAW Sep. 2003.
- Woodall, Sandra C., Remote Order Entry and Video Verification; Reducing After-Hours Medication Errors in a Rural Hospital; S. Woodall; Joint Commission on Accreditation of Healthcare Organizations; vol. 30; No. 8; Aug. 2004.
- Rich Muller; "NRx QS/1's Premium Pharmacy Software," Insight, the QS/1 Magazine, Jul. 2003; pp. 2-3, 12-15.
- Rouse, et al., Academy of Managed Care Pharmacy et al., "White paper on pharmacy technicians 2002: Needed changes can no longer wait" Am J Health-Syst Pharm, Jan. 1, 2003, vol. 60—pp. 37-51.
- Rule Section 291.36—Class A Pharmacies Compounding Sterile Pharmaceuticals—1 page.
- Schrenker, Richard and Cooper, Todd, "Building the Foundation for Medical Device Plug-and-Play Interoperability".
- Standard Specification for Transferring Clinical Laboratory Data Messages Between Independent computer Systems, Annual Book of ASTM Standards, Mar. 25, 1988, pp. 1-16, E 1238-88, Global Engineering Documents, Philadelphia, PA.
- Standard Specification for Transferring Clinical Observations Between Independent Computer Systems, Annual Book of ASTM Standards, Jun. Mar. 1994, pp. 132-210, E 1238-94, Philadelphia, PA.
- Standard Specification for Transferring Clinical Observations Between Independent Computer Systems, Aug. 10, 1997, 79 pages, ASTM E 1238-97, West Conshohocken, PA, United States.
- Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems, Annual Book of ASTM Standards, Jun. 1991, 15 pages, E 1394-91, Philadelphia, PA.
- Suzanne Carter, RN, Ed.D et al., "The Computer-based Patient Record: The Jacobi Medical Center Experience," Second Annual Nicholas E. Davies Award Proceedings of the CPR Recognition Symposium, 1996, pp. 71-95, Computer-based Patient Record Institute, Inc., Bethesda, MD.
- T. Allan Pryor et al., "help—A Total Hospital Information System," Proceedings of the Fourth Annual Symposium on Computer Applications in Medical Care, Nov. 2-5, 1980, pp. 3-7, vol. 1, Institute for Electrical and Electronics Engineers, New York, NY.
- T.E. Bozeman et al., "The Development and Implementation of a Computer-Based Patient Record in a Rural Integrated Health System," Third Annual Nicholas E. David Award Proceedings of the CPR Recognition Symposium, 1997, pp. 101-130, Computer-based Patient Record Institute, Inc., Bethesda, MD.
- "Telepharmacy project expands students' practice experience" Telemedicine Report, vol. 6, No. 1, Jan. 2004 oe 4 pages.
- The World's First Fully Integrated Workflow Manager for I.V. Rooms, IntelliFlowRx Brochure, For Health Technologies Inc., United States, May 2008.
- Title 22. Examining Boards, 22 TAC Section 1.161; texinfo.library.unt.edu/Texasregister/html/2001/sep-14/PROPOSED/22.EXAMING BOARDS.html—Sep. 20, 2014, pp. 1-70.
- Ukens, Carol, "Pharmacist shortage boosts telepharmacy" Drug Topics, Jun. 3, 2002; 146, 11—p. 53.
- Valeriy Nenov et al., "Remote Analysis of Physiological Data from Neurosurgical ICU Patients," Journal of the American Medical Informatics Association, Sep./Oct. 1996, pp. 318-327, vol. 3, No. 5.
- Victor J. Perini et al., "Comparison of automated medication-management systems," Am. J. Hosp. Pharm., Aug. 1, 1994, pp. 1883-1891, vol. 51, American Society of Hospital Pharmacists, Inc."
- Vincenzo Della Mae et al., "HTML generation and semantic markup for telepathology," Computer Networks and ISDN Systems, 1996, pp. 1085-1094, vol. 28, Elsevier Science B.V.
- Website information for Cartharsis Medical Technology Products, Dec. 9, 2001, 15 pages.
- Website information for MedPoint™, Mar. 13, 2003, 20 pages, Bridge Medical, Solana Beach, CA.
- William R. Dito et al., "Bar codes and the clinical laboratory: adaptation perspectives," Clinical Laboratory Management Review, Jan./Feb. 1992, pp. 72-85, Clinical Laboratory Management Association, Inc.
- Wills, Robert D., "Drug Images and Drug Imprints" Insight, Apr. 2001—p. 7.
- Yvonne Mari Abdo, "Designing a Patient Care Medication and Recording System that Uses Bar Code Technology," Computers in Nursing, May/Jun. 1992, pp. 116-120, vol. 10, No. 3.
- Jon Phillips, Telepharmacy at Texas Tech, PowerPoint, Jan. 26, 1997, <https://web.archive.org/web/20040509162423/http://www.ttuhsu.edu/telemedicine/Powerpoint/Telepharmacy%20presentation%2042503.ppt>.
- A.H. McMorris et al. "Are Process Control Rooms Obsolete?," Control Engineering, pp. 42-47, Jul. 1971.
- Standard Specification for Transferring Clinical Observations between Independent Computer Systems, Annual Book of ASTM Standards, Nov. 14, 1991, pp. 1-64, ASTM E 1238-91, Philadelphia, PA.
- Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems, Dec. 10, 1997; 15 pages, ASTM E 1394-97, West Conshohocken, PA, United States.
- Web site information, Information Data Management, Inc.'s PCMS: Plasma Center Management System, Dec. 14, 2001, 11 pages.
- Web site Information, Wyndgate Technologies' SafeTrace Tx™, undated, 15 pages.
- Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems, Mar. 11, 1991; 7 pages, ASTM E 1381-91, Philadelphia, PA, United States.
- Atherton, H.D., Dollberg, S., Donnelly, M.M., Perlstein, P. H. Roath, S.B., "Computerized Temperature Control of the Low-Birth-Weight Infant: A 20-Year Retrospective and Future Prospects," Biomedical Instrumentation and Technology, Jul./Aug. 1994, pp. 302-309, vol. 28 No. 4.
- Friesdorf, W., Grob-Alltag, F., Konichezky, S., Schwillk, B., Fatroth, A., Fett, P., "Lessons learned while building an integrated ICU workstation," International Journal of Clinical Monitoring and Computing, 1994, pp. 89-97, vol. 11.
- Gammon, K., Robinson, K., "Bedside Data System Aids Pharmacy," Computers in Healthcare, Dec. 1988, pp. 35-37, vol. 9 No. 12.
- Graseby 3100 Syringe Pump, Graseby Medical Ltd., A Cambridge Electronic Industries Company, England, 2 pages.
- Kampmann, J., Lau, G., Kropp, St., Schwarzer, E., Hernandez Sande, C., "Connection of electronic medical devices in ICU according to the standard 'MIB'," International Journal of Clinical Monitoring and Computing, 1991, pp. 163-166, vol. 8.
- Angaran, "Telemedicine and telepharmacy: Current status and future implications", Am J Health-Syst Pharm, vol. 56, Jul. 15, 1999 (32 pages).
- Carson, Ewart et al., "A Systems Methodology for the Development and Evaluation of a Telematic Home Haemodialysis Service," Proceedings—19th International Conference—IEEE/EMBS Oct. 30-Nov. 2, 1997, Chicago, Illinois, pp. 907-910.
- Japanese Office Action for related Japanese Application No. 2017-522051; action dated Oct. 1, 2019; (4 pages).
- Singapore Written Opinion for related Singapore Application No. 11201703338P; action dated Jun. 24, 2019; (7 pages).
- Extended European Search Report from corresponding European Patent Application No. 20193098.9, mailed Dec. 8, 2020.
- Australian Examination Report for related Australian Application No. 2015335629; report dated Jun. 5, 2020; (5 pages).
- New Zealand Examination Report for related New Zealand Application No. 731172; action dated Aug. 2, 2021; (5 pages).
- Canadian Office Action for related Canadian Application No. 2,965,533; action dated Sep. 13, 2021; (5 pages).

(56)

References Cited**OTHER PUBLICATIONS**

Office Action for related Japanese Application No. 2021-123427; action dated Sep. 13, 2022; (3 pages).

Office Action for related Canadian Application No. 2,965,533; action dated Jan. 28, 2023; (5 pages).

Becton, Dickinson and Company, Appellant v. Baxter Corporation Englewood, Appellee, Case No. 2020-1937 Appeal from the United States Patent and Trademark Office, Patent Trial and Appeal Board in No. IPR2019-00119—Decided: May 28, 2021—Document 30 filed May 28, 2021—21 pages.

Becton, Dickinson and Company, Petitioner, v. Baxter Corporation Englewood, Patent Owner, IPR2019-00212, U.S. Pat. No. 9,474,693 B2 Judgment Final Written Decision Determining All Challenged Claims Unpatentable 35 U.S.C. Section 318(a)—Denying Patent Owner's Revised Motion to Amend to Substitute Claims 20-38 35 U.S.C. Section 318(a)—Denying Petitioner's Motion to Exclude Evidence 37 C.F.R. Section 42.64—Paper No. 63 dated Apr. 29, 2020—102 pages.

Becton, Dickinson and Company, Petitioner, v. Baxter Corporation Englewood, Patent Owner, IPR2019-00119, U.S. Pat. No. 8,554,579 B2 Judgment Final Written Decision Determining No Challenged Claims Unpatentable 35 U.S.C. Section 318(a)—Denying Petitioner's Motion to Exclude Evidence 37 C.F.R. Section 42.64, Paper 51 entered Apr. 29, 2020—63 pages.

Baxter Corporation Englewood, Appellant v. Becton, Dickinson and Company, Appellee, Case Nos. 2020-1806, 2020, 1808, Judgment

Document 40 filed/entered Mar. 8, 2021 IPR2019-00120, IPR2019-00212 *Baxter Corporation Englewood, Appellant v. Becton, Dickinson and Company*, Appellee, Case No. 2020-1806, 2020, 1808, Mandate entered Mar. 8, 2021 Document 43, filed May 17, 2021 IPR2019-00120, IPR2019-00212.

Becton, Dickinson and Company, Petitioner, v. Baxter Corporation Englewood, Patent Owner, IPR2019-00120 U.S. Pat. No. 9,662,273 B2 Judgement Final Written Decision Determining Challenged Claims Unpatentable 35 U.S.C. Section 318 (a)—Denying Patent Owner's Revised Motion to Amend to Substitute Claims 22-42 35 U.S.C. Section 318(a) Denying Petitioner's Motion to Exclude Evidence 37 C.F.R. Section 42.64—Paper 63 dated Apr. 29, 2020, 90 pages.

Baxter Corporation Englewood, Appellant v. Becton, Dickinson and Company, Appellee, Case Nos. 2020-1806, 2020-1808 IPR2019-00120-IPR2019-0012, On Petition for Rehearing En Banc, Order—Document 42, filed May 10, 2021—2 pages.

Baxter Corporation Englewood, Appellant, v. Becton, Dickinson and Company, Appellee, Case Nos. 20-1806, 20-1808 Inter Partes Review Nos. IPR2019-00120, IPR2019-00121—Petition for Rehearing En Banc By Appellant Baxter Corporation Englewood—Document 41, filed Apr. 7, 2021—18 pages.

Inter Partes Review Certificate, Ranalletta et al. U.S. Pat. No. 9,662,273 K1 Certificate issued Dec. 30, 2021—2 pages.

Inter Partes Review Certificate, Tribble et al. U.S. Pat. No. 8,554,579 K1 Certificate issued May 12, 2022—2 pages.

* cited by examiner

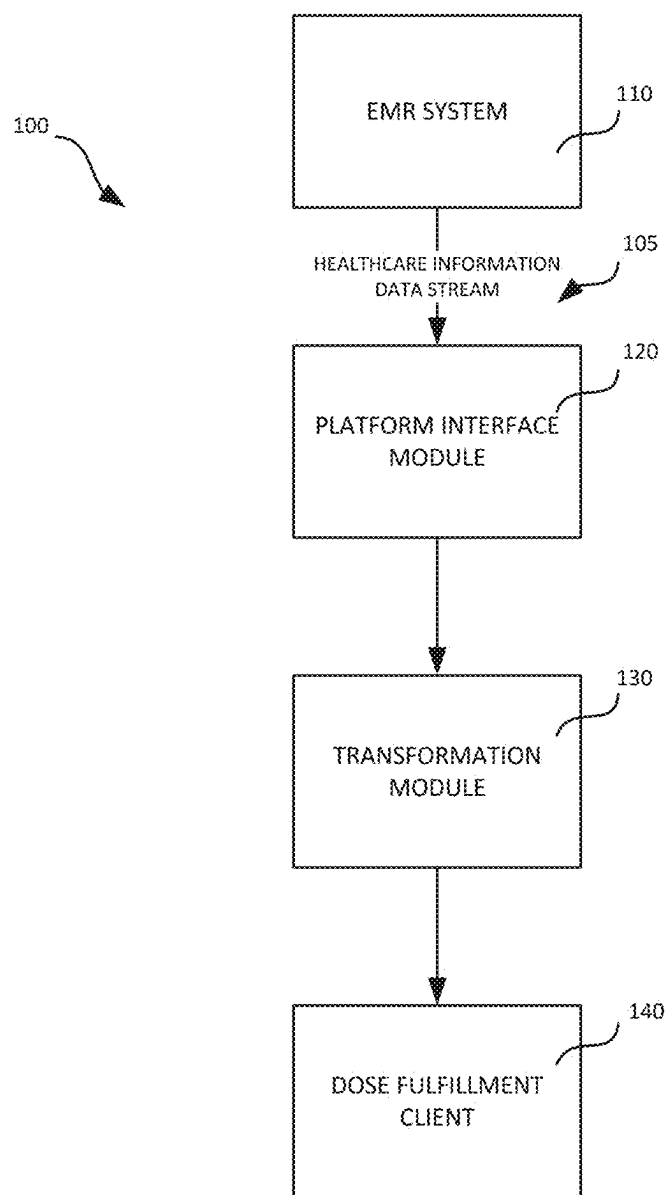
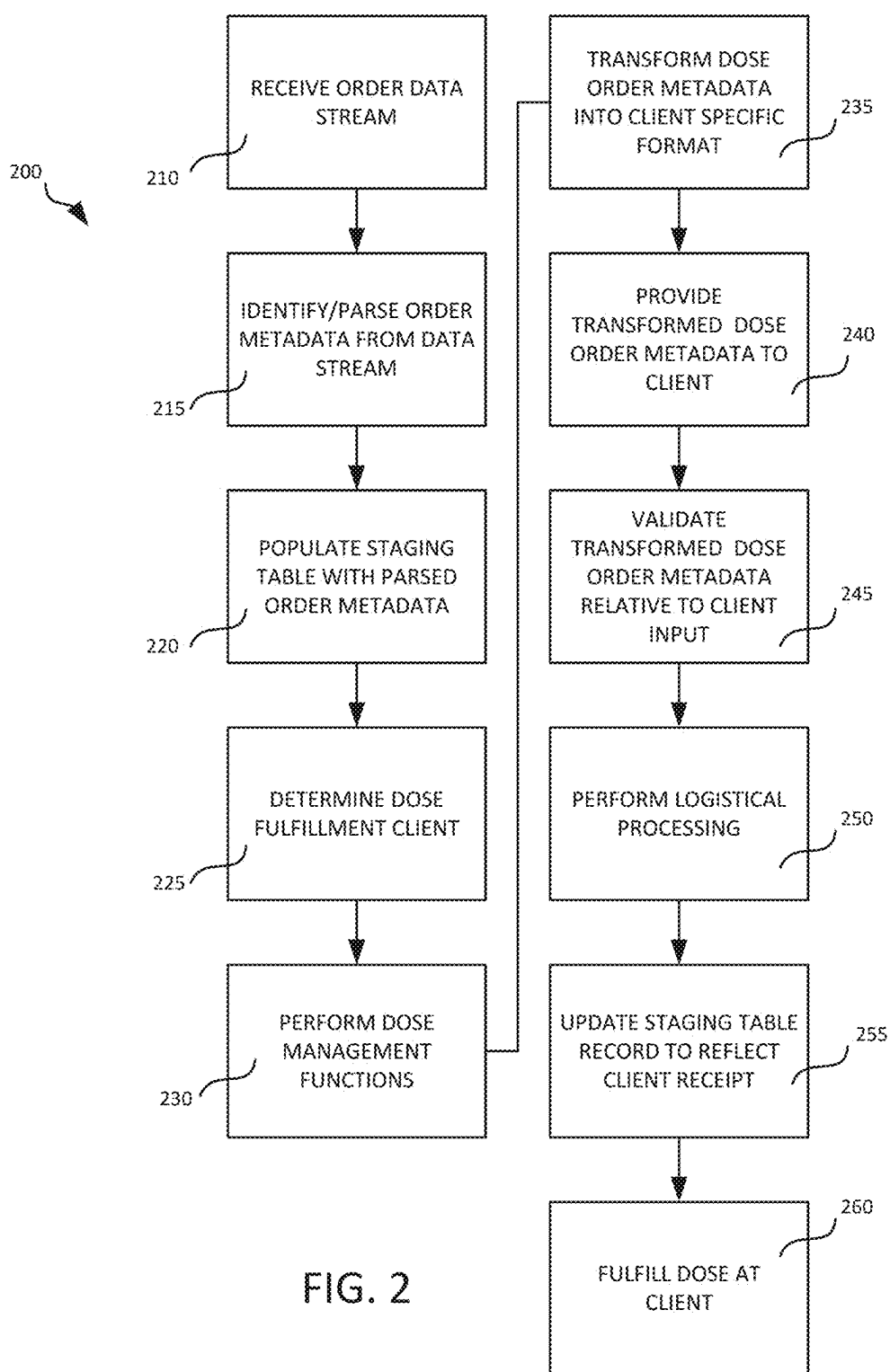


FIG. 1



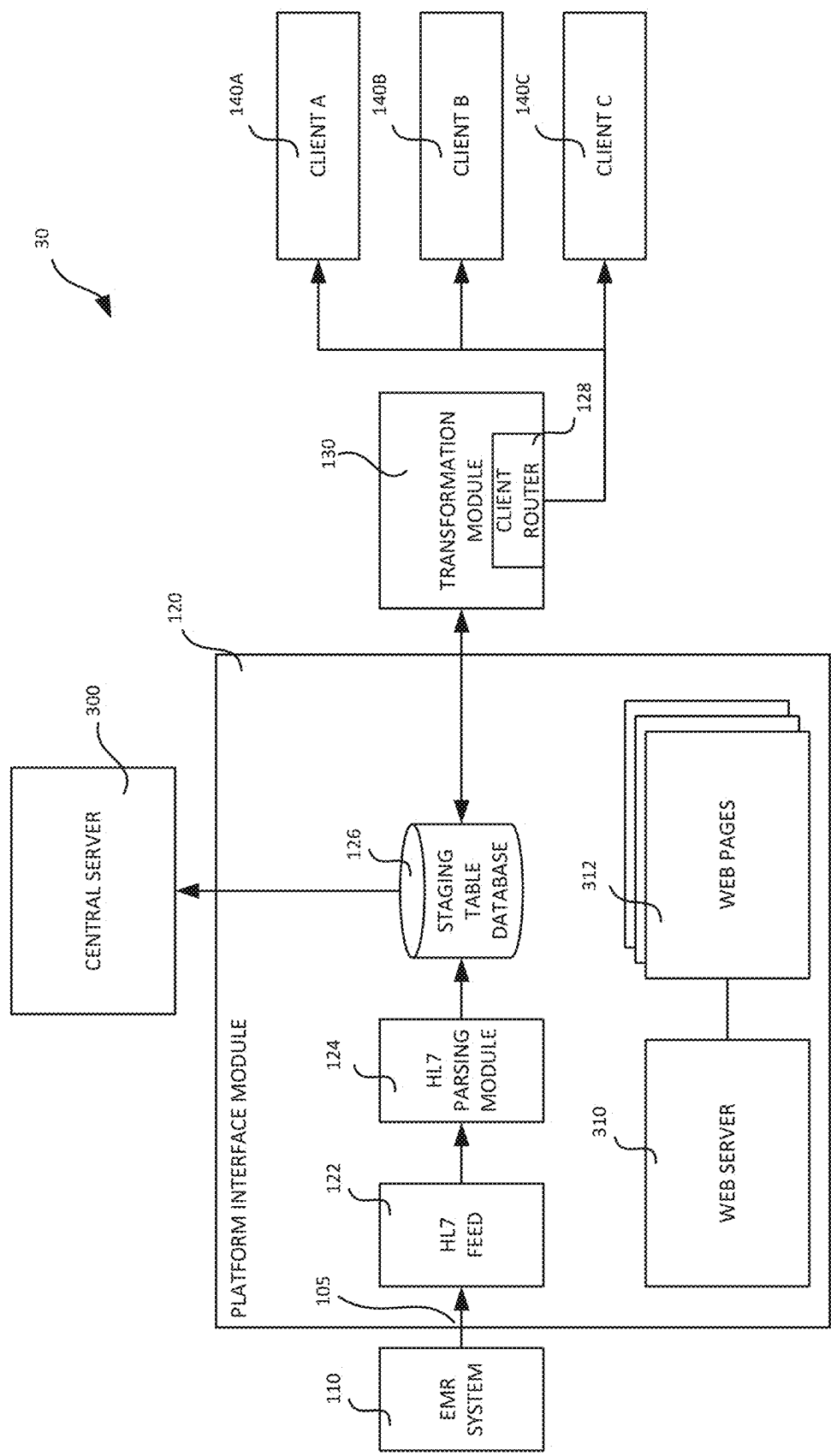


FIG. 3

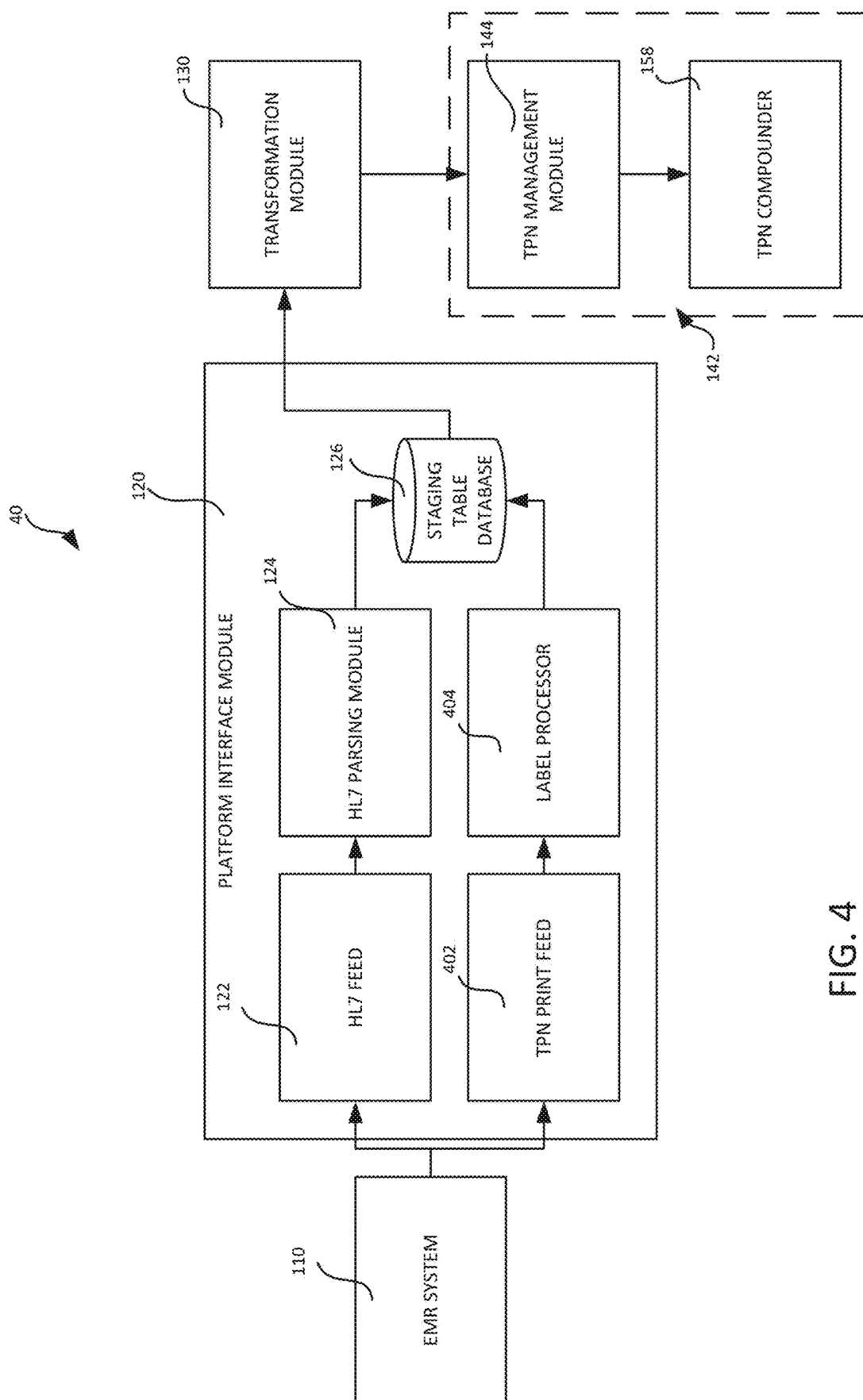


FIG. 4

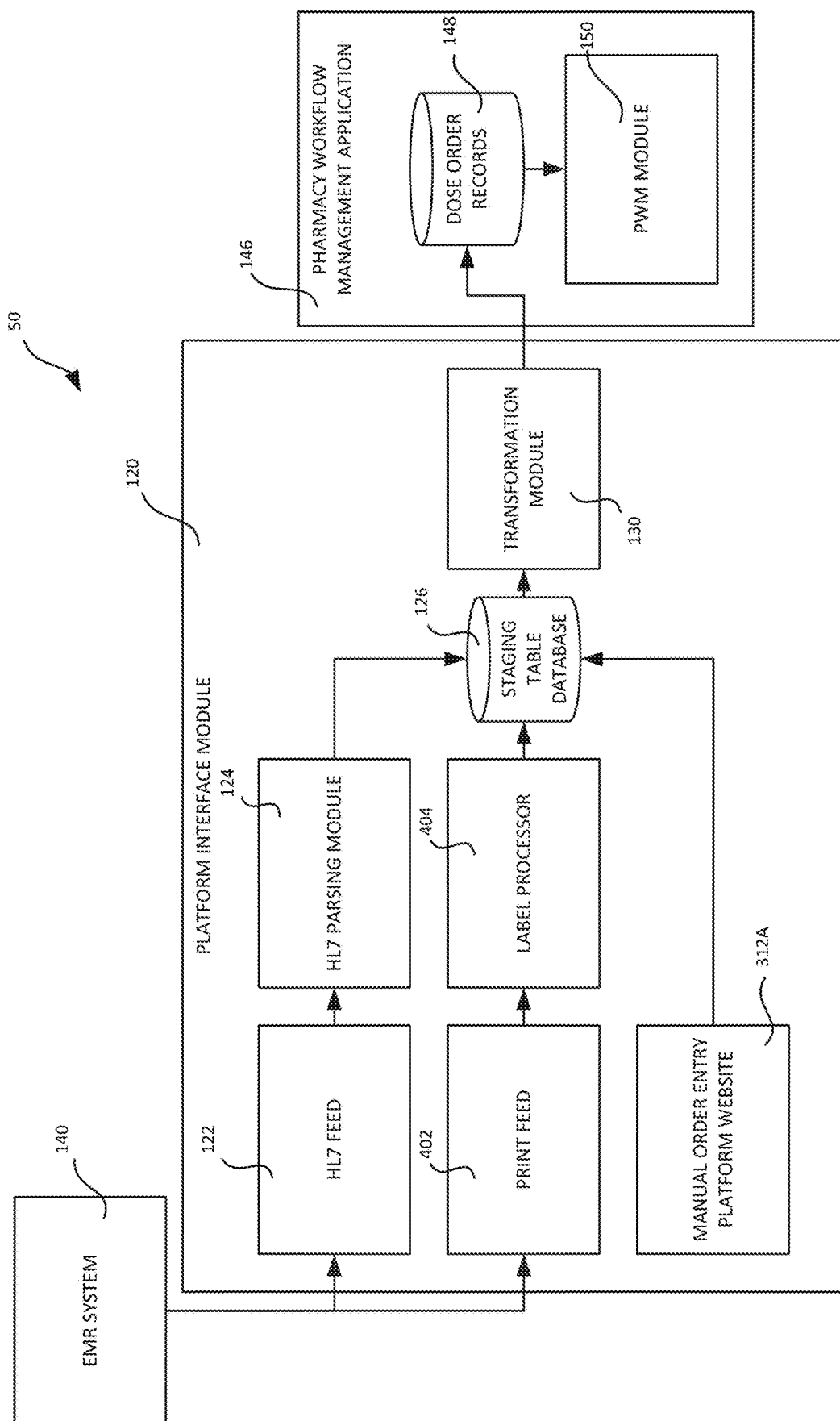
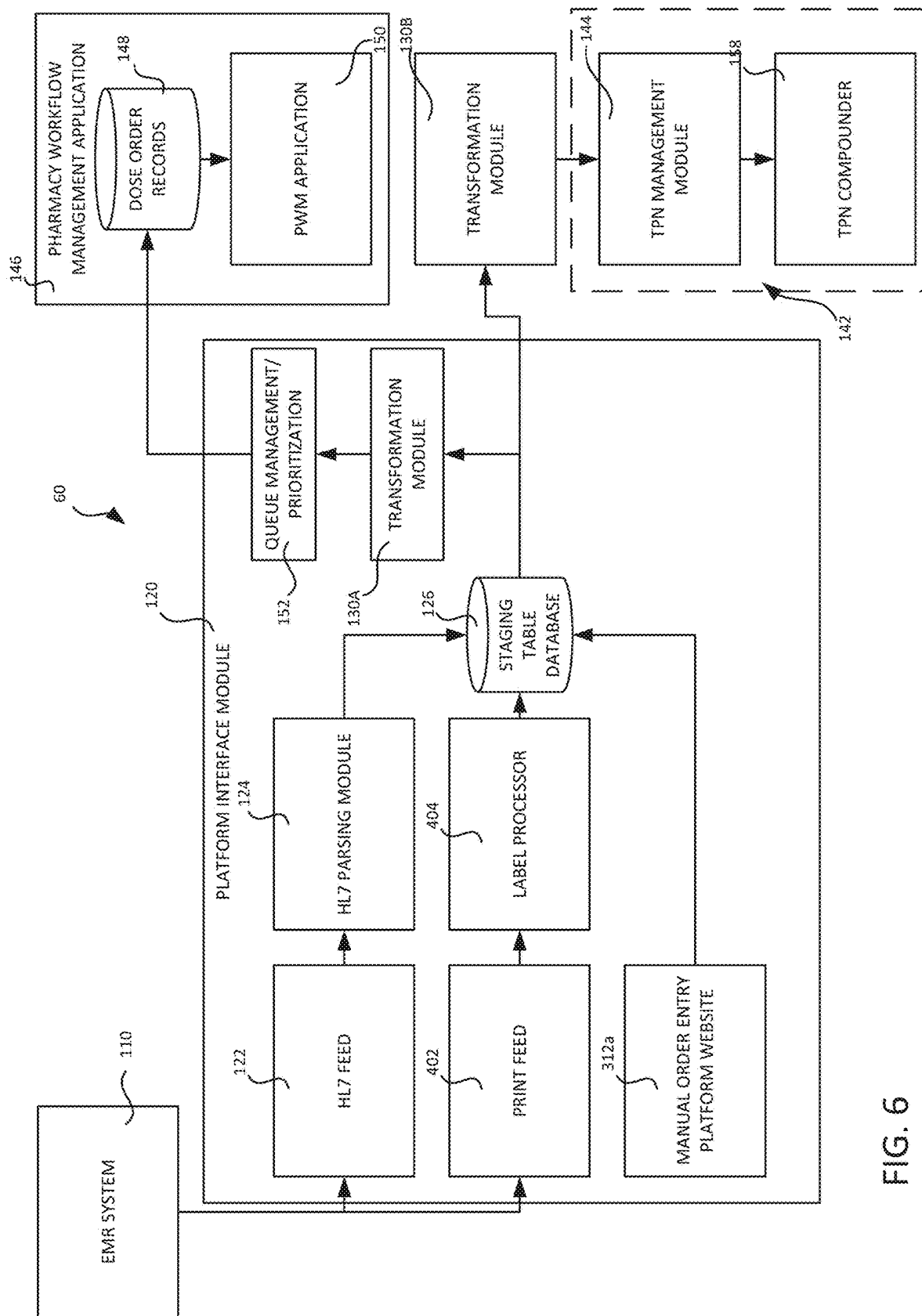
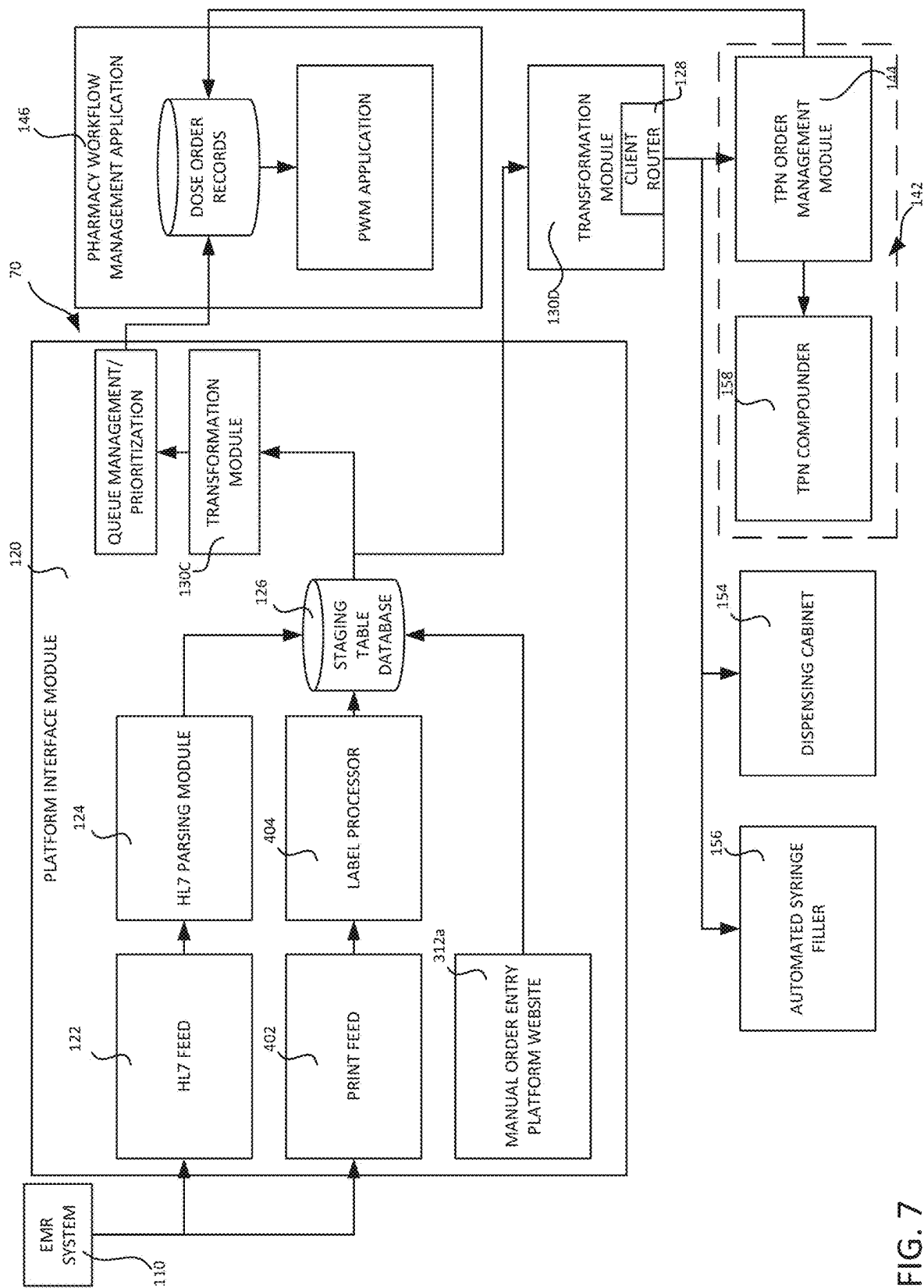


FIG. 5





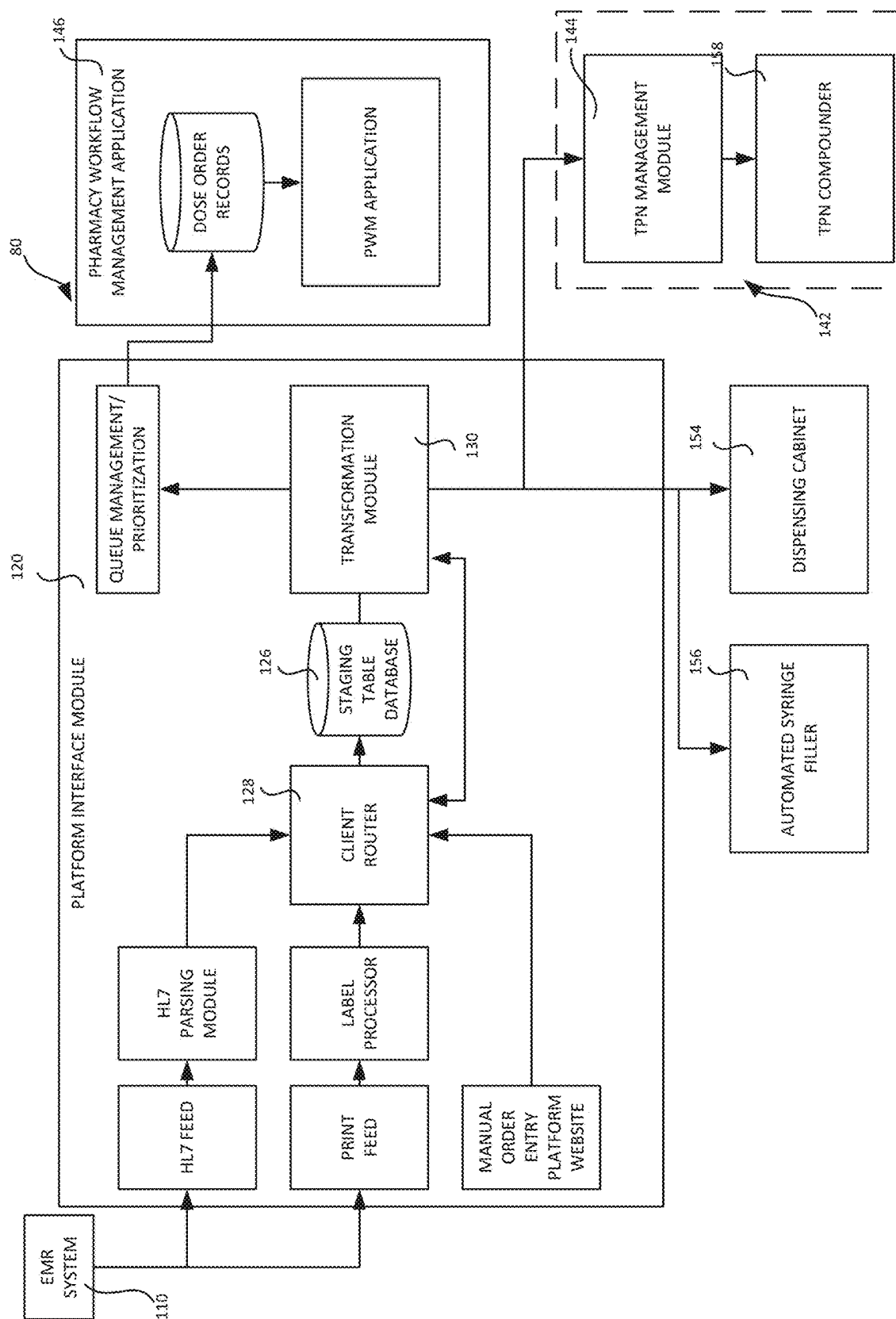


FIG. 8

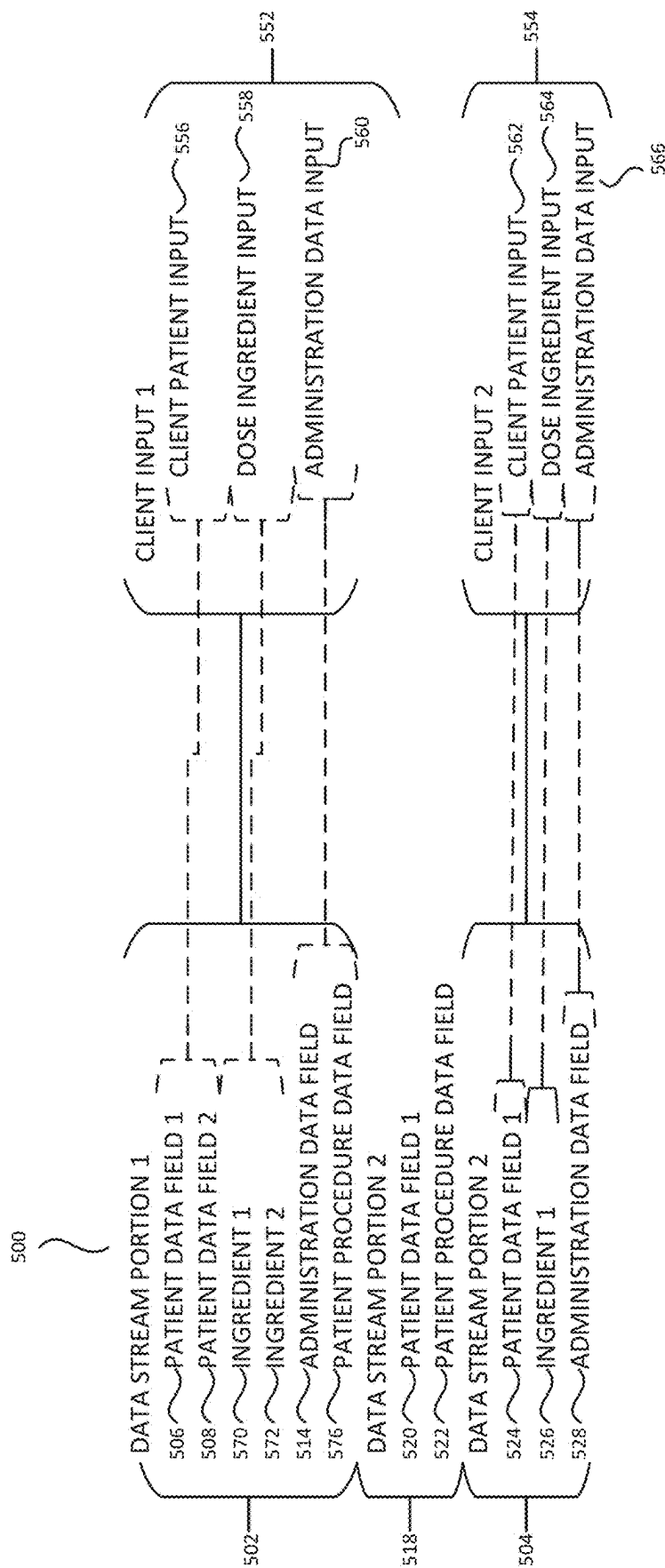


FIG. 9

23456^^^MR-758-49-3029^^^SSN|SMC^TONI||19871229|female

610

600

620

630

640

VID01121346---MR-758-49-3029---SSN||SMC^TONI||19871229|female||---03---||||19801783|758-49-3029|Admitting|This patient will be moved to ICU in three days|P1

ATE12|Dietary|Give vitamin K on Thursdays only|P1

PV1||121000107912|Pool|NONE|10---SMC^10001---|||||||||t|74003474|||||||||120123

ALL12|ABC|PENTICILLIN|||

ALL12|C87|PEANUTS|||

ALL13|C83|Latex|||

DEC|MM|73901182^EPC|||||TBN631102|20130131213000---E-PRESCRIPTION---||20130131154326|STOMACH STOMACHES

AGE||TBN62130---20130131^TPN&2130|ASCORBIC---|PARENTERAL NUTRITION (ADULT STANDARD) RX---|parents

DOB|IV-1971

SUC|1604007-2000 ML FLUX CONT | DEXTROSE 70 & IV SOLN^RX|2512/g/Mg|252/g/Mg

ATE13|Nasolac|Subcutaneous

SUC|1604011-2000 ML FLUX CONT | DEXTROSE 70 & IV SOLN^RX|2512/g/Mg|252/g/Mg

SUC|1604021-100 ML VIAL | TPN ELECTROLYTES IV CONC^RX|33.6 ml/33.6 ml

SUC|1604022-15 ML VIAL | POTASSIUM PHOSPHATE DIHYDRO 3 MEQ/ML IV SOLN^RX|16.8 ml/16.8 ml

SUC|1604024-10 ML SODIUM CHLORIDE 4 MEQ/ML IV SOLN^RX|33.6 MEQ|33.6 MEQ

SUC|1604034-20 ML POTASSIUM ACETATE 2 MEQ/ML IV SOLN^RX|16.8 MEQ|16.8 MEQ

SUC|1604035-10 ML

SUC|1604042-10 ML VIAL | TRACE MINERALS CR-CO-MG-ZN 0.01-1-0.5-5 MG/ML IV SOLN^RX|11 ml/11 ml

SUC|1604043-TPN INSULIN HEPIDAR (CPOM)^RX|25.2 Units/25.2 Units

SUC|158419-1 ML AMPULE | VITAMIN K1 10 MG/ML INJ SOLN^RX|2 MG|2 MG

SUC|1604072-2000 ML FLUX CONT | STERILE WATER FOR INJECTION IV SOLN^RX|500.55 ml/500.55 ml

DEC|13|NN|Must Current|Must current lab value for Na (actual)||1147|mg/L

DEC|13|MM|Must Current|Must current measured weight (actual)||99.5|kg

FIG. 10



810

SMC, TONI
Account #: 15001783
Med Rec #: 123456

Med Rec #: 123456
Patient Type: Adult/Ped: 400g
IV Site: Central
Age: 25 Years
Card Age: 80.5kg
Weight: 80.5kg
Location: 4

Diagnosis:
Allergies:
Patient Notes:
8/16/2013 3:39:40 PM Order: 72496
User: BADA

800

Item	Units	Batch			
Template: Push TPN Aminosyn II					
Qty	Substance	Value	Name	Descript	Units
1	Aminosyn II 15%	100.8	gram	/-	
1	TPN Electrolytes	33.6	mEq	/-	
1	K Phos 30mEq/100ml	16.8	mEq	/-	
1	NaCl 4mEq/ml	33.6	mEq	/-	
1	K Acetate 2mEq/ml	16.8	mEq	/-	

830

840

820

Add Ingredient
Delete Ingredient
Edit Ingredient

Recommendations

FIG. 12

Abacus IPN Order Entry Software [Leach, Doug]

File Patients Order Tools Help

Patients ☐ Hide Patients ☒ Hide Inactive ☐ View Pending Only

S	Status	Patient Type	Last Name	First Name	Account #
		Adult	Doe-Ha	50 mL q	28
		Adult	Doe-Ha	Table 50m.	29
		Adult	Doe-Le	Mary	5893
		Adult	Doe-Le	James	5886
		Adult	Doe-Mu	Patrick	5831
		Adult	Doe-Pa	Indumati	5983
		Adult	Doe-So	Special	1385
		Adult	Doe-Wa	Shella	5839
		Adult	Doe-Wi	Marlin	5828
		Adult	Doe-Zi	Rory	5888

912

910

Orders for Selected Patient ☐ View Pending Only

S	Status	Bag #	Ord.	Order Date/Time	Hang Date/Time	Created By	Approved By
	Received	4	72600	1/29/2014 1:56:30 PM	2/11/2014 8:00:00 PM	Leach, Doug	922
	Received	4	72477	2/9/2014 2:49:15 PM	2/11/2014 8:00:00 PM	Leach, Doug	924
	Complete	3	72132	9/18/2009 6:20:24 AM	9/18/2009 8:00:00 PM	Kelsey Chidress	Kelsey Chidress
	Complete	1	72850	9/16/2009 9:22:24 AM	9/16/2009 8:00:00 PM	Adam Stoeber	Adam Stoeber

926

928

920

FIG. 13

Sig	DC	Substance	Value	Units	Demon	Rule
		Volume	30	ml /		
		Rate	30	ml /hour		
		Pulse	24	hour /		
		Duration	24	hour /		
		Givetail	100	ml /		
		Non-TPN Volume	0	ml /		
		Dextrose 70%	20	% /		
		Invertrol 10%	5	% /		
		KCl 2mEq/ml	30	mEq /L		
		NaCl 4mEq/ml	50	mEq /L		
		MgSO4 4mEq/ml	5	mEq /L		
		Ca Gluconate 10%	5	mEq /L		
		MVI Adult	10	ml /day		
		Adult Trace Conc	1	ml /day		
		Pandoline	1	mg /Kg/Day		

Order Info: Unresolved Items

EMR ID	Substance	Value	Units
5406	Ibrexin	10	unit/L

Administration:
Bag Number: 25065131
Date: 7/15/2014 Time: 9:00 PM

Cancel **Finish Later** **Done**

FIG. 14

Order	Infusion	Energy	Units	Batch	Template	HL7																																																																																																																							
Order Information Cistest Brandon Meisel Account #: 3051627 Med Rec #: MEDIC MFM Patient Type: Adult M Site: Central Age: 33 Years Bed Age: 65 Kg Weight: 65 Kg Location: 8 EAST, D121 Diagnosis: Allergic: Gluten, Peanut, Shellfish Patient Notes:																																																																																																																													
Recommendations																																																																																																																													
Order Information Order Info: Unresolved Items:																																																																																																																													
<table border="1"> <thead> <tr> <th>EMR ID</th> <th>Substance</th> <th>Value</th> <th>Units</th> </tr> </thead> <tbody> <tr> <td>5406</td> <td>Insulin</td> <td>10</td> <td>units/L</td> </tr> </tbody> </table>							EMR ID	Substance	Value	Units	5406	Insulin	10	units/L																																																																																																															
EMR ID	Substance	Value	Units																																																																																																																										
5406	Insulin	10	units/L																																																																																																																										
<table border="1"> <thead> <tr> <th>Std</th> <th>QC</th> <th>Substance</th> <th>Value</th> <th>Number</th> <th>Denom</th> <th>Rate</th> </tr> </thead> <tbody> <tr> <td></td> <td></td> <td>Volume</td> <td>2160</td> <td>ml</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Rate</td> <td>90</td> <td>ml</td> <td>/hour</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Period</td> <td>24</td> <td>hour</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Duration</td> <td>24</td> <td>hour</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Overfill</td> <td>100</td> <td>ml</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Non-IPN Volume</td> <td>0</td> <td>ml</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Dextrose 70%</td> <td>20</td> <td>S</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Treosol 10%</td> <td>5</td> <td>S</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>KCl 2mEq/ml</td> <td>30</td> <td>mEq</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>NaCl 4mEq/ml</td> <td>50</td> <td>mEq</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>MgSO4 4mEq/ml</td> <td>5</td> <td>mEq</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Ca Gluconate 10%</td> <td>5</td> <td>mEq</td> <td>L</td> <td></td> </tr> <tr> <td></td> <td></td> <td>MVI Adult</td> <td>10</td> <td>ml</td> <td>/day</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Adult Trace Conc</td> <td>1</td> <td>ml</td> <td>/day</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Insulin</td> <td>1</td> <td>mg</td> <td>Kg/Day</td> <td></td> </tr> <tr> <td></td> <td></td> <td>Regulate Insulin</td> <td></td> <td></td> <td>units</td> <td>L</td> </tr> </tbody> </table>							Std	QC	Substance	Value	Number	Denom	Rate			Volume	2160	ml	L				Rate	90	ml	/hour				Period	24	hour	L				Duration	24	hour	L				Overfill	100	ml	L				Non-IPN Volume	0	ml	L				Dextrose 70%	20	S	L				Treosol 10%	5	S	L				KCl 2mEq/ml	30	mEq	L				NaCl 4mEq/ml	50	mEq	L				MgSO4 4mEq/ml	5	mEq	L				Ca Gluconate 10%	5	mEq	L				MVI Adult	10	ml	/day				Adult Trace Conc	1	ml	/day				Insulin	1	mg	Kg/Day				Regulate Insulin			units	L
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Administration Bag Number: 1 Prescription: 25308813 1 Date: 7/15/2014 Time: 8:00 PM																																																																																																																													
Order Information Add Ingredient Replace Ingredient Delete Ingredient																																																																																																																													
Order Information Label Notes: Refrigerate until 1 hour before administration																																																																																																																													
Order Information Summary Ca/PO4 Formula Limits																																																																																																																													

FIG. 15

1000

Formulary Editor

Ingredients

Groups

Attributes

Intrinsics

Options

☒ View Available Only

Ingredient

Calcium Gluconate 453mEq

Dextrose 30%

Dextrose 70%

Heparin 100 units/mL

Intralipid 20%

K Acetate 2mEq/mL

K Chloride 2mEq/mL

K Phosphate 3mMol/mL PO4

Magnesium Cl

Magnesium Sulfate 4.06mEq

Manganese 0.1mg/mL

Mannitol 25%

Multitrac-4 Pediatric

MVI Adult

MVI-Pediatric

Na Bicarbonate

Na Chloride 2.5mEq/mL

Na Chloride 4mEq/mL

PremaSol 10%

Ranitidine Inj 35mg/mL

Selenium 40mcg/mL

Sodium Lactate 3mEq

Sterile Water for Inj

Thiamine 120mg/mL

TPN Electrolyte

Travasol 10%

Trehalmine 10%

Trehalmine 1mg/mL

Zinc Sulfate 5mg/mL

Ingredient Detail

Name

Regular Insulin 100units

Label Name

Regular Insulin

Abbreviation

Regins

Concentration Units

100 units/mL

NDC #

0159-2449-10

EMR ID

INSULIN

Type

☐ Base

☐ Lipid

☐ Electrolyte

☒ Other

☐ Pre-Mix PN Solution

Billing

Cost Per Package

\$0.00

Package Size

10 mL

Charge Code

Options

☐ Do Not Scale

☐ Phantom

☐ Unavailable

☐ Do Not Print On Label

☒ OSHA

Attribute

Value

Units

BCAA

Calcium

Calories

KCal/ml

Carbohydrates

Chloride

Lipid

Magnesium

Non-protein Calories

KCal/ml

Phosphate

Potassium

Protein

Intrinsic

Value

Units

Ascorbic Acid Vit C

Carnitine

Chromium

Cimetidine

Copper

Cyanocobalamin VitB12

Deoxypanthol Vit B5

Dextrose

Famotidine

Folic Acid

Heparin

Insulin

100 units/ml

Save

Cancel

Close

FIG. 16

1710 Facilities And Nursing Units 1730 Map EMR

Facilities | Nursing Units | Workstations | Mixing Locations | Printers | Field Na8780

Substance Type	Abacus Name	EMR ID
1712	Acetate	1722
Attributes	Calcium	1732
Attributes	Chloride	
Attributes	Magnesium	
Attributes	Phosphate	
Attributes	Potassium	
Attributes	Sodium	
1714	Adult Trace Coric	1724
Ingredients	Amnodyn 10%	ADULT TRACE 1734
Ingredients	Ca Gluconate 10%	AMNODYN
Ingredients	Dextrose 70%	CAGLUC10
Ingredients	KCl 2mEq/ml	Dextrose
Ingredients	M V.I. Adult	KCL2
Ingredients	MgSD4 4mEq/ml	MV1
Ingredients	NaCl 4mEq/ml	MGSD4
Ingredients	Ranitidine	NaCL4
Ingredients	Tiavasol 10%	RANTON
Ingredients	Water for Inj	TRA10
Special Substances	Duration	Water
Special Substances	Non-TPN Volume	DURATION
Special Substances	Overfill	NONTPN
Special Substances	Overfill	OVERFILL
Special Substances	Period/Cycle	OF
Special Substances	Period/Cycle	PERIOD
Special Substances	Period/Cycle	DATE

1702

Substance Type

Ingredients 1740

Abacus Substance

Regular Insulin 1750

EMR Identifier

Insulin 1760

Save Cancel Close

New

FIG. 17

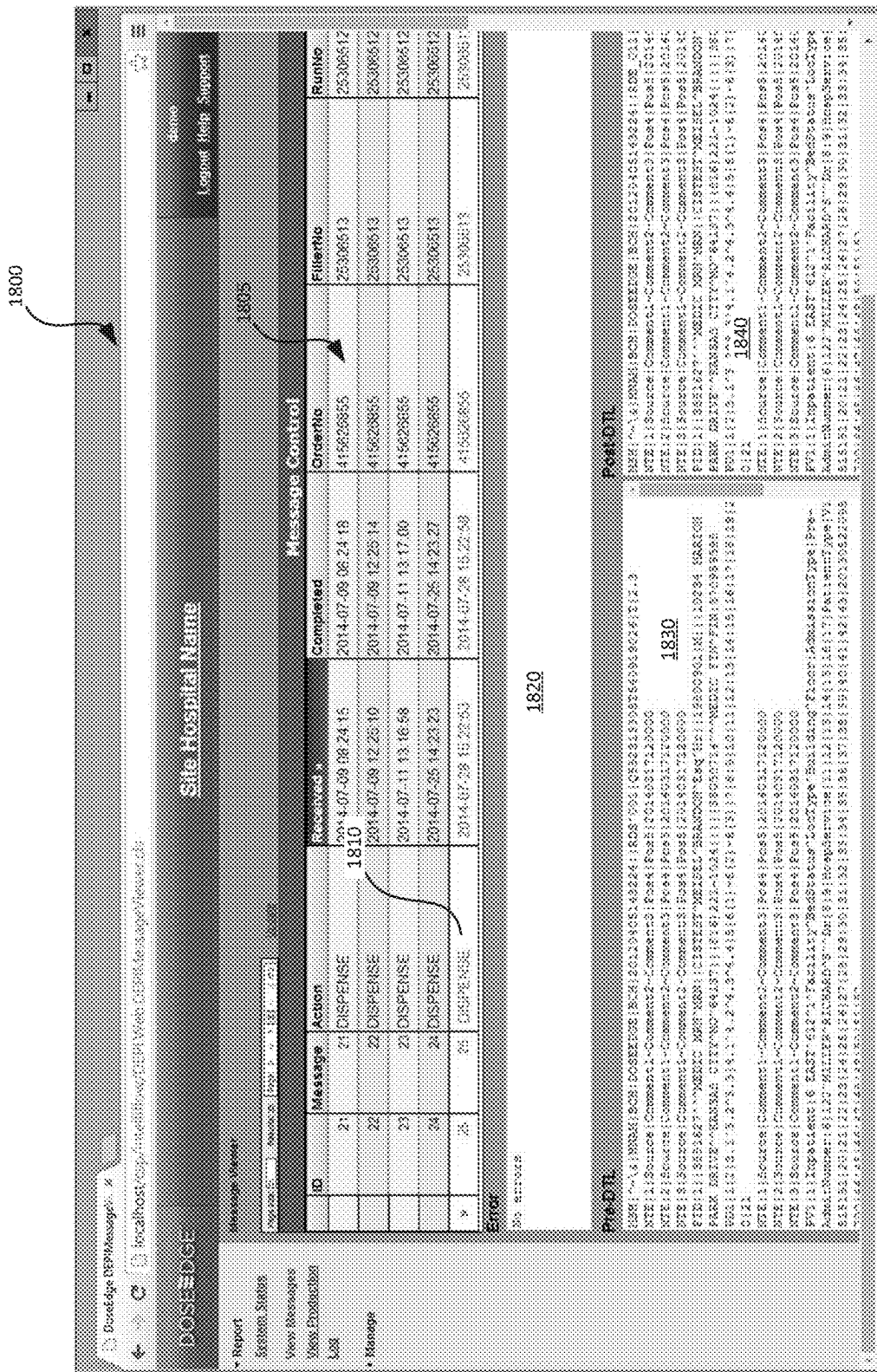


FIG. 18

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AUTOMATED EXCHANGE OF HEALTHCARE INFORMATION FOR FULFILLMENT OF MEDICATION DOSES

CROSS-REFERENCE TO RELATED APPLICATION

This application claims priority from U.S. Provisional Application No. 62/068,301 filed on Oct. 24, 2014, entitled "AUTOMATED EXCHANGE OF HEALTHCARE INFORMATION FOR FULFILLMENT OF MEDICATION DOSES," the contents of which are incorporated by reference herein as if set forth in full.

BACKGROUND

Despite advances in electronic healthcare information systems, processes related to the exchange of information may still rely upon manual intervention by a human user. While independent systems may be operative to maintain an electronic medical record (EMR), the exchange of information between independent systems may be complicated by a lack of a standardized data format or data exchange protocol. Furthermore, to the extent data may be exchanged between independent systems, use of unique semantics at each independent system may result in complications when exchanging data therebetween.

For example, in the context of dose order processing, a physician or the like may rely upon a hardcopy order form to request a medication order to be administered to a patient. In turn, the hardcopy of the order form may be provided to a pharmacy by way of, for example, physical transport, facsimile, relay by telephone, relay by email, etc. In turn, upon receipt of the information related to the dose order at the pharmacy, a technician may be required to manually input the information (e.g., by transcribing information from the hardcopy or an electronic copy thereof) regarding the requested dose order. As such, a new instance of an EMR related to the dose order requested based on the transcribed dose order may be generated in the pharmacy. In turn, the dose order may be fulfilled by the pharmacy (e.g., including calculation, review, compounding, and delivery of the dose order).

As such, at one or more instances during the dose ordering process, information related to the requested dose order may be required to be manually processed by a human user. For example, a human user may be required to relay information from a physical hardcopy order form to a pharmacy system. Furthermore, a human user may be required to retrieve information from a first system and transcribe information into a second system. For example, a human user may be required to input received order information into a pharmacy information system (PIS). In this regard, despite advances in EMR systems for dose order generation and pharmacy fulfillment, the exchange of healthcare information may still be subject to human user handling.

In turn, the potential for delays and errors may be compounded because of the requirement for human intervention. For example, it is been found that 60% of clinicians working with parenteral nutrition orders report one to five ordering errors per month. Significantly, near fatal complication or death may result in 4.8% of parenteral nutrition order errors in one two year span that was reviewed. Further still, transcription may provide delay in order fulfillment as it has been found that up to 10% of parenteral nutrition orders require clarification. The most common causes of the need for clarification include illegible writing, missing essential

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ingredients, or unstable macronutrient content. In this regard, human errors may come in the form of transcription errors, EMR order entry errors, errors resulting in information asymmetry, errors resulting in order form legibility, errors related to reentry of orders in the pharmacy system, or other points within a data flow that require human intervention. Accordingly, improvements in exchange of healthcare information between independent healthcare systems may reduce the reliance upon human intervention, thereby reducing error rates in improving patient outcomes.

SUMMARY

In view the foregoing, the present disclosure is generally related to improved exchange of healthcare information. Specifically, the present disclosure presents systems and methods that may at least in part help reduce reliance upon manual human interaction in the exchange of healthcare information between independent healthcare systems. In turn, the reliance upon human interaction to facilitate information exchange may be reduced, and the speed and accuracy at which healthcare information may be exchanged between independent healthcare systems may be improved. In turn, an at least partially automated approach to healthcare information exchange may be facilitated.

Accordingly, use of an automated health information exchange as described herein may improve patient outcomes by assisting in reduction of errors associated with human interaction in the exchange of healthcare information. Furthermore, the speed at which information may be exchanged may be increased for more efficient pharmacy operation. For instance, one context in which automated healthcare information exchange may be utilized is in the context of dose order fulfillment. Specifically, automation of the exchange of healthcare information from dose order entry to dose order fulfillment may be facilitated to assist in reduction of (and potentially elimination of) the need for human intervention in connection with the exchange of healthcare information between an EMR system that facilitates dose order entry, pharmacy information systems (PIS) for management of pharmacy operations, and dose fulfillment clients for use in fulfillment of medication doses that are to be administered to a patient. As discussed herein, dose orders may relate to requested medication doses in any number of contexts including, for example, IV doses, parenteral nutrition (PN) doses, total parenteral nutrition (TPN) doses, or the like.

Specifically, the disclosure presented herein may facilitate receipt and processing of a healthcare information data stream by a platform interface module. The platform interface module may perform identification and parsing of the healthcare information data stream to identify individual dose orders from the healthcare information data stream. In this regard, the healthcare information data stream may include a multitude of data including, for example, a plurality of dose orders or other data related to EMR data or hospital information system (HIS) data. Upon identification of a dose order from the healthcare information data stream, the platform interface module may be operative to parse the dose order such that dose order data fields may be identified and populated with dose order metadata related to the dose order. The dose order data fields may correspond to dose order characteristics such as an ingredient list, product requirements, administration characteristics, patient information, or other appropriate data types related to the dose order. In turn, corresponding dose order metadata for each respective dose order may be parsed by the platform inter-

face module. A staging table may be generated and stored at the platform interface module in relation to a dose order. The staging table may include a standardized intermediate format for storage of dose order metadata in corresponding relation to dose order data fields for a given dose order.

In this regard, it may be appreciated that different various EMR systems (e.g., one or more HISs, one or more physician order entry (POE) systems, or the like) may be used to generate and/or transmit dose orders to a pharmacy. In turn, receipt and processing of the health information data stream to generate staging tables may facilitate generation of a standardized intermediate format for the dose order. The standardized intermediate format may be beneficial as the dose order may be routed to one of a plurality of different dose order fulfillment clients for use in fulfillment of the medication dose associated with the dose order. In turn, rather than having to facilitate specific respective correlations between different EMR system formats and specific dose fulfillment client input formats, the standardized intermediate form (e.g., the staging table) may be utilized such that a plurality of different EMR system formats may be standardized into the standardized intermediate form and the standardized intermediate form may be in turn transformed into one of the plurality of different input formats associated with different respective ones of dose fulfillment clients. In this regard, upon addition of different EMR systems, the data from the EMR system may be standardized into the standardized intermediate format such that the system may process dose orders received from the new EMR system without having to generate application functionality specific to each dose fulfillment client for the new EMR system. Furthermore, as new dose fulfillment clients are added the system, transformation of the data in the standardized intermediate format may be accomplished without having to generate application functionality for the new dose fulfillment client specific to each EMR system. Accordingly, a robust and modular data exchange system may be realized that allows for any one of a plurality of EMR systems to be utilized in conjunction with any one of a plurality of dose fulfillment clients.

In at least some embodiments, the dose order records maintained in a staging table may have some dose order management functionality applied thereto. For example, functions in connection with viewing, modifying, prioritizing, or otherwise organizing the dose orders may be applied to the dose order information stored in the staging tables. In turn, improved pharmacy management may be provided in that personnel associated with the operation of the pharmacy may be able to perform pharmacy management functions collectively on the dose orders stored in the staging tables. For instance, such pharmacy management may be performed on the dose orders prior to provision of the dose orders to a plurality of dose fulfillment clients. Further still, dose order management may be performed locally relative to one or more dose fulfillment clients. In turn, pharmacy resources may be efficiently managed in connection with dose orders having different priorities, urgencies, or the like.

Dose order information from the staging table may be retrieved by and/or provided to a dose fulfillment client for use in preparation of a dose associated with the dose order. In this regard, the dose order information in the staging table may be transformed from the standardized intermediate form of the staging table into an input format associated with a dose fulfillment client. For instance, a unique input format for a given dose fulfillment client may be provided. As such, identification of a specific dose fulfillment client for use in preparation of the dose order may allow for the correspond-

ing unique input format associated with the identified dose fulfillment client to be used to transform the data from the staging table into the unique input format for a given client.

In connection with the transformation of information from the staging table into the dose from a client specific format, a mapping and/or translation processes may occur. For example, the transformation of the dose order information from the standardized intermediate format to the unique input format for a given dose fulfillment client may include mapping dose order data fields from the staging table to corresponding respective dose fulfillment client input fields. In addition to the specific mapping, translations of the data may be performed. For example, the unique input format for a dose fulfillment client may specify a form in which the data is expected. As such, data may be translated from the form in which it is provided in the staging table such that the data complies with the form of the unique input format.

Additionally, dose order metadata related to the corresponding respective dose fulfillment client input fields may be validated to determine whether a valid input is provided for the dose order from the staging table. In the event the data may not be validated (e.g., due to a parsing error, a mapping error, a translation error, or some other error), exception processing may occur. That is, exception processing may allow for a user to resolve any errors identified during validation, mapping, or translation. The resolution of the errors by a user may be used to modify processing rules (e.g., validation rules, mapping rules, or translation rules) or the like for subsequent dose orders. The receipt of an input corresponding to a resolution of an error by a user may be incorporated into the transformation and/or validation of subsequent orders. Thus, in connection with the transformation process, exceptions related to the mapping, translation, and/or validation of dose order metadata from the staging table may be processed as will be discussed herein. Furthermore, logistical processing may be performed such that changes, cancellations, or other modifications to a dose order received during the processing of the dose order may be facilitated. The exemption and/or logistical processing may be performed at the same location or different locations. Furthermore, the exemption and/or logistical processing may occur at the platform interface module, transformation module, and/or dose fulfillment client.

In addition, logging may occur throughout the exchange of healthcare data between independent systems described herein. In turn, auditing, troubleshooting, or the like may be facilitated for any or all of the steps of the exchange between systems described herein. The log information generated during the healthcare information exchange may be maintained at each independent system or aggregated into a single repository related to a portion of or the entire path of the data exchange. In turn, improve record-keeping for the purposes of auditing or the like may be provided in connection with the exchange of healthcare information between systems described herein.

As contemplated herein, a plurality of dose fulfillment clients may be employed in connection with the healthcare information exchange facilitated by the disclosure presented herein. For instance, dose fulfillment clients may include a pharmacy workflow management application for use in manual preparation of dose orders, automated syringe filling platforms, TPN management modules, automated compounding (e.g., for use in fulfilling PN and TPN orders), automated dose dispensing cabinets, or other dose fulfillment clients that may be utilized to realize a physical dose associated with the dose order. Accordingly, at least some of the dose fulfillment clients may comprise automated dose

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preparation systems for fulfillment of dose orders without the need for human intervention. In turn, in at least some embodiments facilitated herein, the processing performed on the dose order may be completely automated such that there is no human intervention between the entry of the dose order by a physician or the like and the fulfillment of the dose order by an automated dose fulfillment client.

A first aspect of the disclosure presented herein includes a method for automated transformation of healthcare information by a transformation module between a first form from an electronic medical record (EMR) system and a second form associated with a dose fulfillment client for preparation of a dose in accord with a dose order of the health information data. The method includes retrieving dose order metadata, at a transformation module, for a dose order from a staging table corresponding to the dose order. The staging table is populated with dose order metadata received from a healthcare information data stream in a first form from the EMR system. Additionally, the staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The method further includes transforming the dose order metadata into a predefined second form corresponding with a dose fulfillment client and providing the dose order metadata in the second form to the dose fulfillment client for fulfillment of a dose associated with the dose order based on the dose order metadata.

A number of feature refinements and additional features are applicable to the first aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the first aspect.

For example, the method may further include receiving the healthcare information data stream at a platform interface module from the EMR system. The healthcare information data stream may include information related to the dose order (e.g., among other information related to different dose orders and/or unrelated data). The method may also include identifying the dose order from the healthcare information data stream and parsing the dose order metadata for the dose order from the healthcare information data stream. The dose order metadata may include one or more dose characteristics associated with the dose order. The method may further include populating the staging table, stored in a staging table database at the platform interface module, with the dose order metadata for the dose order.

The staging table may store the dose order metadata a standardized intermediate format. For example, the method may further include receiving a first dose order in a first EMR form and receiving a second dose order in a second EMR form. The first dose order and the second dose order may correspond to a dose order with identical constituent ingredients. The form of the constituent ingredients in the first EMR form may differ from the second EMR form (e.g., because of different EMR sources or different EMR formats utilized). The form of the constituent ingredients for the first dose order may, in turn, be identical to the form of the constituent ingredients for the second dose order in the respective staging tables corresponding to the first dose order and the second dose order. Furthermore, other dose data may be standardized such as, for example, patient data, dose administration data, or the like.

The method may also include identifying, at least in part based on the dose order metadata, the dose fulfillment client from a plurality of dose fulfillment clients for fulfillment of

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the dose order. As such, the plurality of dose fulfillment clients each have respective predefined second forms. In turn, the transforming may at least in part be based on the respective predefined second form of the identified dose fulfillment client. The staging table may be independent of any of the plurality of dose fulfillment clients.

The transforming may include mapping the plurality of dose order data fields of the staging table to corresponding respective ones of a plurality of dose fulfillment client input fields. In an application, the dose fulfillment client input fields may be defined by a unique input format associated with the dose fulfillment client. The transforming may additionally include validating the dose order metadata of the plurality of dose order data fields with respect to the unique input format for corresponding ones of the dose fulfillment client input fields. The validating may include correlating the dose order metadata of the plurality of dose order data fields to formulary records of the dose fulfillment client with respect to the corresponding respective ones of the dose fulfillment client input fields.

In an embodiment, the method may include exception processing (e.g., in response to the validating, mapping, or translating of dose order data). In this regard, the method may include generating an exception in response to an error associated with at least one of the mapping or validating. Additionally, the exception may include prompting a human user to resolve the error. As such, the method may include receiving, from the human user, an input associated with the resolution of the error. In turn, the input may include at least one of a correct mapping between a dose order data field of the staging table and a corresponding respective one of a dose fulfillment client input fields or a correct correlation between a portion of dose order metadata and a formulary record of the dose fulfillment client. As such, the exception processing may further include updating at least one of a mapping logic or a correlation logic for use in the mapping and correlating, respectively, based on the input received from the human user.

In an embodiment, the method may include updating the staging table to indicate the dose order metadata for the dose order has been retrieved by the dose fulfillment client. As such, the processing status of a dose may be maintained at the staging table to provide an indication of dose orders that have been processed and those that have not been processed. The method may also include managing the dose order (e.g., by organizing, prioritizing, etc.) prior to providing the dose order metadata in the second form to the dose fulfillment client. For instance, the managing may include providing a user interface to allow a user to perform a management function relative to the dose order. Specifically, the managing may include at least one of modification of dose order metadata, cancellation of the dose order, organization of the dose order relative to other dose orders, or prioritization of the dose order relative to other dose orders.

Additionally, the method may include performing logistical processing on the dose order. The logistical processing may include performing an action relative to the dose order in response to receipt of an EMR system message subsequent to the receipt of the dose order. The EMR system message may include at least one of a dose order change message or a dose order discontinuation message. As such, the logistical processing may include determining if the dose order metadata has been provided to the dose fulfillment client, and wherein the logistical processing is at least in part based on whether the determining. The logistical processing may include performing an action relative to the dose order record corresponding to the EMR system message for a dose

order record that has not yet been provided to the dose fulfillment client. For instance, the logistical processing may include providing data to the dose fulfillment client related to the EMR system message for a dose order record that has been provided to the dose fulfillment client. The data provided to the dose fulfillment client related to the EMR system message may also be transformed into a form corresponding to the dose fulfillment client.

In an embodiment, the dose fulfillment client may include at least one of a pharmacy workflow management application for manual preparation of the dose order, an automated dose preparation device, an automated total parenteral nutrition (TPN) compounder, or a dose dispensing cabinet. Additionally, the EMR system may include a hospital information system (HIS). As such, the method may include exchange of a message from independent healthcare systems comprising the EMR system and the dose fulfillment client. The data exchanged between the EMR system and the dose fulfillment client may be otherwise incompatible without the use of the healthcare information exchange system. For instance, the healthcare information data stream may be a Health Level 7 (HL7) format.

The method may also include logging activity taken with respect to the dose order to generate a log regarding activities relative to the dose order. The logging may include aggregating a plurality of logs generated relative to different respective actions taken with respect to the dose order.

A second aspect includes a method for automated exchange of healthcare information between an electronic medical record (EMR) system and a dose fulfillment client for fulfillment of a dose order. The method includes receiving a healthcare information data stream at a platform interface module. The healthcare information data stream includes information related to a dose order. The method also includes identifying the dose order from the healthcare information data stream and parsing dose order metadata for the dose order from the healthcare information data stream. The dose order metadata includes one or more dose characteristics associated with the dose order. The method further includes populating a staging table with the dose order metadata for the dose order. The staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The method further includes transforming the dose order metadata into a predefined format corresponding to a particular dose fulfillment client to generate transformed dose order metadata. Also, the method includes providing the particular dose fulfillment client access to the transformed dose order metadata, wherein the dose fulfillment client is operable to retrieve the transformed dose order metadata from the staging table to fulfill the dose order based on the transformed dose order metadata.

A number of feature refinements and additional features are applicable to the second aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the second aspect. For instance, any of the foregoing features described in relation to the first aspect may be applicable to the second aspect.

A third aspect includes a healthcare information exchange system for exchange of healthcare information between an electronic medical record (EMR) system and a dose fulfillment client for fulfillment of a dose order. The system includes a stream processing module in operative communication with an EMR system to receive a healthcare infor-

mation data stream from the EMR system. The system further includes a staging table database in operative communication with the stream processing module for storage of a staging table populated with dose order metadata included in the healthcare information data stream. The staging table includes a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata. The system further includes a transformation module operatively disposed between the staging table data base and a dose fulfillment client. The transformation module is operative to identify the dose fulfillment client for use in fulfillment of the dose order and transform the dose order metadata into a predefined format corresponding to the dose fulfillment client. Specifically, the predefined format defines dose order data fields and corresponding dose order metadata formats associated with the dose fulfillment client.

A number of feature refinements and additional features are applicable to the third aspect. These feature refinements and additional features may be used individually or in any combination. As such, each of the following features that will be discussed may be, but are not required to be, used with any other feature or combination of features of the third aspect. For instance, any of the foregoing features described in relation to the first aspect may be applicable to the third aspect.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic view of an embodiment of a system for facilitating exchange of healthcare information.

FIG. 2 is a flowchart depicting an embodiment of a method for exchange of healthcare information.

FIG. 3 is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a plurality of dose fulfillment clients.

FIG. 4 is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a dose fulfillment client for fulfillment of total parenteral nutrition dose.

FIG. 5 is a schematic view of an embodiment of a system for facilitating exchange of healthcare information between an EMR system and a pharmacy workflow management application for fulfillment of a dose order.

FIGS. 6-8 are schematic views of embodiments of a system for facilitating exchange of healthcare information between an EMR system and a plurality of dose fulfillment clients.

FIG. 9 is an embodiment of a textual representation of a mapping between a healthcare information data stream and an input format for a dose fulfillment client.

FIG. 10 is an embodiment of a human readable representation of an EMR system message comprising a dose order.

FIG. 11 is an embodiment of a human readable representation of the operation of a transformation module executing relative to the EMR system message of FIG. 10.

FIG. 12 is an embodiment of a dose fulfillment client interface corresponding to the dose order of the EMR system message of FIG. 10.

FIG. 13 is an embodiment of an interface for display of dose orders processed by a healthcare information exchange system.

FIGS. 14-16 depict a sequence of interfaces related to an embodiment of exception handling of a dose order with processing errors in relation to the exchange of data related to the dose order.

FIG. 17 is an embodiment of an interface for management of a transformation of healthcare information in a healthcare information exchange system.

FIG. 18 is an embodiment of an interface for a platform interface module depicting standardization of an EMR message into an intermediate standardized format.

DETAILED DESCRIPTION

While the invention is susceptible to various modifications and alternative forms, specific embodiments thereof have been shown by way of example in the drawings and are herein described in detail. It should be understood, however, that it is not intended to limit the invention to the particular form disclosed, but rather, the invention is to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the claims.

FIG. 1 schematically depicts an embodiment of a system 100 for automated exchange of healthcare information. The system 100 may include an EMR system 110 that is in operative communication with a platform interface module 120. In this regard, the EMR system 110 may provide a healthcare information data stream 105 to the platform interface module 120. While a single EMR system 110 is shown in FIG. 1, it may be appreciated that a plurality of EMR systems 110 may be in operative communication with the platform interface module 120. The EMR systems 110 may correspond to different EMR systems 110 at a given facility or EMR systems 110 from a plurality of facilities. In any regard, the platform interface module 120 may be in operative communication with a transformation module 130. The transformation module 130 may in turn be in operative communication with a dose fulfillment client 140. As discussed in greater detail below, the transformation module 130 may be in operative communication with a plurality of dose fulfillment clients 140. As such, the transformation module 130 may include a dose routing module and/or perform dose routing functions in relation to dose orders received at the transformation module 130.

In this regard, the system 100 in FIG. 1 may be operative to facilitate automated data exchange between independent healthcare information systems. For example, the EMR system 110 and the dose fulfillment client 140 may each correspond to independent healthcare information systems and/or products. As described above, the EMR system 110 and the fulfillment client 140 may use unrelated or incompatible data formats. In turn, the platform interface module 120 and transformation module 130 may act upon the healthcare information data stream 105 to transform the data from the form received from the EMR system 110 such that the healthcare information (e.g., comprising dose order data) may be exchanged between the EMR system 110 in the dose fulfillment client 140. Specifically, the data may be exchanged between the EMR system 110 and the dose fulfillment client 140 in an automated manner that avoids the requirement for human intervention with respect to the exchange of the healthcare information. In this regard and as described above, the automation of the exchange of information between the EMR system 110 and the dose fulfillment client 140 provided according to the disclosure presented herein may assist in reduction of errors and/or delays associated with human involvement with the healthcare information data stream 105 as it passes from the EMR system 110 to the dose fulfillment client 140.

The EMR system 110 may comprise a hospital information system (HIS). Accordingly, the healthcare information data stream 105 may comprise information regarding dose

orders to be prepared by the dose of fulfillment client 140. However, the healthcare information data stream 105 may also include other healthcare information that is unrelated to dose orders or preparation of dose orders. For example, in the case of an HIS, the healthcare information data stream 105 may include patient related data (e.g., admission/discharge data), hospital management data (e.g., resource availability, scheduling, etc.), billing and accounting data, or other ancillary data unrelated to dose orders or the preparation of dose orders.

The healthcare information data stream 105 may be provided in a number of different formats. Furthermore, a given one platform interface module 120 may interface with a plurality of different ERM systems 110 each potentially providing healthcare information data streams 105 in differing formats. In an embodiment, the healthcare information data stream 105 may comprise a Health Level 7 (HL7) format. The HL7 format refers to set of loosely standardized conventions for the transfer of clinical and administration data between HISs. Despite the loose standardization of conventions provided by HL7 messages, the exchange of healthcare information between various independent healthcare systems employing HL7 formatting is not seamless. In this regard, different systems may utilize different lexicons, conventions, semantics, or other different standards within the HL7 format such that the exchange of information between two systems utilizing an HL7 format may still require intervention to allow for data exchange between systems.

Accordingly, the platform interface module 120 may receive the healthcare information data stream 105 from the EMR system 110 regardless of the nature of the format used (e.g., regardless of variations in HL7 formats). For instance, different specific EMR systems 110 (e.g., different HISs) may utilized varying HL7 formats, all of which may be received by the platform interface module 120. As will be discussed in greater detail below, the platform interface module 120 may be operative to identify and/or parse dose orders from the healthcare information data stream 105. The platform interface module 120 may in turn store the dose order records identified and parsed from the healthcare information data stream 105 in an a standardized intermediate format. For example, the standardized intermediate format may comprise a staging table for each respective dose order for which dose order metadata is parsed from the healthcare information data stream 105. In turn, the parsed dose order metadata may be used to populate fields of the staging table that correspond to dose order data fields from the healthcare information data stream 105.

It should be noted that the standardized intermediate format (i.e., the staging table) may be standardized relative to different HL7 formats used by different HL7 providers. That is, if first HL7 provider provides a dose order with identical constituent ingredients as a second HL7 provider, the portions of the HL7 messages corresponding to the identical constituent ingredients for the doses may still differ based on differing usage of the HL7 format by the first and second providers to describe the identical constituent ingredients. However, the platform interface module 120 may standardize the differing portions of the HL7 messages regarding the constituent ingredients for the identical orders such that the constituent ingredients for the orders would appear identical in the standardized intermediate format. Furthermore, patient data, dose administration data, or other data related to the dose may be provided in different formats corresponding to different uses of the HL7 format. However, this data may also be standardized in the standardized intermediate

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format. In this regard, the staging table comprising the standardized intermediate format may standardize HL7 messages received in different healthcare data streams **105**.

The generation of a standardized intermediate format is illustrated in FIG. **18** which depicts a user interface **1800** and may display a listing **1805** of receive dose orders. A user may be able to select a selected one **1810** of the dose orders from the listing **1805** of dose orders. Upon selection of the selected one **1810**, corresponding information regarding the selected dose order **1810** may be displayed in an error pane **1820**, a pre-standardization pane **1830**, and a post-standardization pane **1840**. In this regard, the error pane **1820** may provide information to a user regarding errors (e.g., parsing errors, identification errors, or other errors that may occur with respect to receipt of the dose order from an EMR system **110**). Furthermore, the pre-standardization pane **1830** may display the dose order record as received from the EMR system **110**. In this regard, the pre-standardization pane **1830** may display a textual representation of an HL7 message received from the EMR system **110** prior to undergoing standardization for population of a staging table. The post-standardization pane **1840** may display the corresponding textual representation of the dose order after having been standardized into the standardized intermediate format. That is, the post standardization pane **1840** may provide a standardized version of the pre-standardized HL7 format displayed in the pre-standardization pane **1830**. Accordingly, the user interface **1800** may allow a user to review errors related to a selected dose order **1810**. Furthermore, a user may be operative to review pre-standardization and post-standardization formats side-by-side for a given dose order to facilitate review and or troubleshooting related to the dose order.

The use of a standardized intermediate format may be particularly beneficial in the context of multiple healthcare data stream providers (e.g., multiple EMR systems **110** that may include, for example, HISSs, PISs, POEs, or other healthcare information data stream **105** sources) and multiple dose fulfillment clients **140**. For instance, rather than having to develop specific transformation logic for all possible combinations of EMR systems **110** and all dose fulfillment clients **140**, the system **100** may provide a more modular approach. That is, as new EMR systems **110** are added to the system **100**, a parsing module (described in greater detail below) may be provided to format the healthcare information data stream **105** into the standardized intermediate format for storage in a staging table. In turn, all existing dose fulfillment clients **140** may receive information as a transformation module **130** may be capable of transforming the standardized intermediate format into each dose fulfillment client **140** specific format. Further still, as different dose fulfillment clients **140** are added to the system that require unique input formats, rather than having to develop transformation logic for each potential EMR system **110**, transformation logic relative to the standardized intermediate format may be generated. Accordingly, the standardized intermediate format allows for improved, modular development as additional components to which data is to be exchanged are added to the system **100**.

The transformation module **130** may be operative to access the data regarding dose orders from the staging table maintained at the platform interface module **120**. The transformation module **130** may be operative to transform the accessed data in the standardized intermediate format regarding the dose orders from the standardized intermediate format of the staging table into a specific format association with a dose fulfillment client. In this regard, the client-

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specific format may correspond to a unique input format associated with and/or required by a dose fulfillment client **140**. In this regard, the unique input format for a given dose fulfillment client **140** may have input fields corresponding to dose order data fields of the staging table. In this regard, the transformation module **130** may map dose order data fields to corresponding input fields for the unique input format for a given dose fulfillment client **140**. Furthermore, the transformation client **130** may translate dose order metadata from a first format stored in the staging table into a target format associated with the unique input format for a given dose fulfillment client **140**. In this regard, the transformation module **130** may be operative to apply a transformation that is specific to a dose fulfillment client **140** to dose order data stored in the staging table based on a dose fulfillment client **140** that is identified to be used to fulfill the corresponding dose order.

As such, data regarding a dose order may be provided to a dose fulfillment client **140** in a format associated with a unique input format associated with the dose fulfillment client **140** such that the dose order data may be automatically provided to the dose fulfillment client **140** for use by the dose fulfillment client **140** in fulfillment of the dose corresponding to the dose order data. As will be appreciated from the further discussion below, the dose fulfillment client **140** may comprise any one or more of a number of different types of dose fulfillment clients that may be utilized in the fulfillment of a dose associated with a dose order. For example, the dose fulfillment client **140** may comprise any one or more of a pharmacy workflow management application for management of manual preparation of the dose, a total parenteral nutrition (TPN) client for processing in connection with and/or preparation (e.g., automated preparation) of a TPN dose order, an automated syringe filler, a dose dispensation cabinet with pre-stocked doses that may be allocated based on received dose order data to fulfill a dose, etc.

As will also be appreciated from the discussion below, the platform interface module **120** and transformation module **130** may each include hardware and/or software that comprise each respective module. Furthermore, the platform interface module **120** and the transformation client **130** may be collectively provided as a single module. That is, each module may comprise hardware and/or software for execution of functionality described below in relation to each respective one of the platform interface module **120** and/or transformation module **130**. For example, the respective modules may individually or collectively comprise one or more processors in operative communication with a memory device that stores non-transitory machine-readable data that may be used to specifically configure the one or more processor for execution of functionality related to the modules as described below. In this regard, the respective modules may also comprise the memory device that stores non-transitory machine-readable data structure (e.g., such as a hard drive, flash drive, memory module, or other appropriate physical electronic memory) for storage of the non-transitory machine-readable data used for specific configuration of the one or more processors.

With further reference to FIG. **2**, an embodiment of a method **200** for automated exchange of healthcare information between an EMR system **110** and a dose fulfillment client **140** is shown as a flowchart. While the method **200** is described herein in connection with the system **100** of FIG. **1**, it will further be appreciated that the method **200** may be performed in connection with any other embodiment presented below. Furthermore, while the method **200** is

depicted in the form of a sequential flow chart with specific discrete operations in a particular order, it is contemplated that the steps of the method discussed below may be performed in any order (or concurrently) unless otherwise specified. Also, further embodiments of the method are contemplated wherein some steps may be omitted. That is, all steps of the process need not be performed in each embodiment.

The method 200 may include receiving 210 a dose order data stream 105 (e.g., such as a health information data stream 105 comprising one or more dose orders therein). As may be appreciated, the receiving 210 may include transmission of the dose order data stream from the EMR system 110. In this regard, the dose order data stream may be sent upon initiation from the EMR system 110 (i.e., “pushed” from the EMR system 110) and/or be sent upon request by the platform interface module 120 (i.e., “pulled” from the EMR system 110). The method 200 may also include identifying/parsing 215 dose order metadata from the dose order data stream. In this regard, the identifying/parsing 215 may include analyzing the dose order data stream 105 to identify individual dose orders from the data stream 105. For example and as described above, the data stream 105 may include a plurality of dose orders and/or data that is unrelated to dose orders at all. In turn, the identifying may include delineating individual dose orders from the data stream 105. As such, the identifying may include recognition of data in a specific format indicative of a dose order. Additionally or alternatively, the data stream 105 may include express delineators to assist in identifying individual ones of the dose orders in the data stream 105. Additionally, dose order metadata may be parsed from the data stream 105 in regard an individual dose order. In this regard, the parsing may include locating one or more specific dose order data fields and corresponding dose order metadata related to the data fields. In turn, the dose order data fields and dose order metadata related thereto may be used in populating a dose order data field. In addition, standardization of the dose order metadata may be performed such that the staging table comprising the dose order data fields may include standardized dose order metadata (i.e., in a standardized intermediate format).

Any one or more of a plurality of dose order data fields may be included in the data stream 105 and, in turn, in a corresponding staging table related to the identified/parsed dose order may be provided and/or generated. As an example, a predefined staging table may be provided that includes predefined data fields that relate to patient specific data, ingredients of the dose order, administration details for the dose order, or other appropriate data fields that relate to the dose order, the administration of the dose order, or a patient which the dose orders to be administered. That is, any data field relevant to the preparation and/or administration of the dose order may be contained in the data stream 105. The data provided in relation to such data fields may in turn be standardized into a corresponding data field in a staging table that may include predefined data fields to which the dose orders in the data stream 105 are standardized. Additionally or alternatively, a staging table with data fields corresponding to the fields of the dose order in the data stream 105 may be generated upon receipt of the data corresponding to the dose order.

In this later regard, the method may include populating 220 a staging table with the identified/parsed dose order data from the data stream 105. For example, the staging table may include data fields stored as a specific data structure corresponding to the standardized intermediate format. For

instance, the standardized intermediate format may be defined by a database schema associated with dose orders. In this regard, to the extent dose order metadata for a given dose order data field is parsed with respect to a given one of the dose orders in the data stream 105, a corresponding staging table instance (e.g., a row in the table and/or a unique staging table entry or record) may be generated and appropriate respective data fields according to the standardized intermediate format may be populated 220 the corresponding dose order metadata identified/parsed 215 from the data stream 105. In turn, a staging table corresponding to each identified individual dose order may be generated and populated 220 with parsed dose order data. It may be appreciated that the individual staging tables regarding corresponding individual dose orders may be collectively stored in a database. In this regard, individual database files need not be generated for each individual dose order, but rather an identified staging table or portion of a larger staging table (e.g., a unique row or record) may be specifically related to a given dose order for specific identification relative thereto.

The method 200 may, as an option in at least some embodiments, include determining 225 a dose fulfillment client for use in preparation of a given dose order. For example, the dose order data contained in the data stream 105 for a given dose order or plurality of doses may include an express identification of a dose fulfillment client 140 for use in fulfillment of the corresponding dose order(s) (e.g., the healthcare information data system 105 may include an indication of the dose fulfillment client 140 for given a given dose order that originates as data provided by the EMR system 110). That is, the EMR system 110 may send data indicative of the desired dose fulfillment client 140 to be used to fulfill a particular dose order (e.g., by indicating dose fulfillment client type or by indicating a specific identity of a dose fulfillment client) that is contained in the healthcare data stream 105. In another example, a source of the data stream 105 may be identified, and the identity of the source of the data stream 105 may at least in part be used to determine the dose fulfillment client 140 to which the dose order is to be provided for fulfillment. For example, all dose orders received from a specific given source may be defined as being dose orders corresponding to a specific dose fulfillment client 140 for use in fulfillment of those specific dose orders from the specific source. Further still, the determination 225 of the dose fulfillment client 140 for a dose order may be based upon one or more portions of dose order data. For example, the dose order data may define a dose characteristic (e.g., an ingredient, ingredient amounts, ingredient types, etc.) that may be used to determine 225 dose fulfillment client 140 for use in fulfillment of the dose order. In this regard, logic may be established in accord with any of the foregoing examples to facilitate determination 225 of the dose fulfillment client 140 for a given dose order. For instance, the logic may be executed by a client router 128 (e.g., shown in FIG. 3 et seq.) that may be provided at the transformation module 130. In other contemplated embodiments, the client router 128 may be provided remotely from the transformation module 130 (e.g., at the platform interface module 120 and/or as a completely separate module).

Furthermore, the method 200 may, as an option in at least some embodiments, include performing 230 dose management functions with respect to the dose orders received and stored in corresponding staging tables. For example, the dose orders stored in corresponding staging tables may be reviewed, modified, prioritized, grouped, organized, or otherwise managed. It should be noted that the dose manage-

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ment functions that may be performed **230** may occur prior to receipt of dose order data at a dose fulfillment client **140** and/or upon receipt of dose order data at the dose fulfillment client **140**. That is, the dose management functions may be performed by the platform interface module **120**, the transformation module **130**, and/or the dose fulfillment client **140**. For instance, the dose management functions may be performed by a queue management and prioritization module **152**, which may be located at the platform interface module **120** (e.g., as a standalone module and/or as a component of the client router **128** or transformation module **130**), at a dose fulfillment client **140**, and/or disposed between the platform interface module **120** and a dose fulfillment client **140**.

The method **200** may further include transforming **235** dose order data from the standardized intermediate format (e.g., the staging table) into a specific format associated with a dose fulfillment client **140** (e.g., a unique input format for a given dose fulfillment client **140**). As will be described in greater detail below, the transforming **235** may include mapping dose order data fields of the staging table to input fields for a unique input format of a given dose fulfillment client **140**. Furthermore, dose order metadata for a given dose order stored in a dose order data field may be translated into a format associated with the unique input format of the given dose fulfillment client **140**. In this regard, the transformation **235** may be at least in part depending upon the determination **225** of the dose fulfillment client for a given dose order as described above.

The method **200** may further include providing **240** the transformed dose order data to a dose fulfillment client **140**. In this regard, the dose fulfillment client may request the transformed dose order data (e.g., pull the data from the platform interface **120** and/or transformation module **130**) and/or the dose fulfillment client **140** may have the transformed dose order metadata transmitted thereto (e.g., the data may be pushed from the platform interface **120** and/or transformation module **130**).

In any regard, the method **200** may include validating **245** the transformed dose order metadata relative to the unique input format for the dose fulfillment client **140** that receives or is to receive the dose order data. For instance, the validating **245** may include determining whether required dose fulfillment client input fields are present in the transformed dose order data. Furthermore, the validating **245** may include analyzing a portion of transformed dose order data to determine if a valid corresponding portion of data is available in the unique input format for the dose fulfillment client **140**. In this regard, in the event the validating **245** is unsuccessful, an exception may be generated with respect to a dose order. In turn, the method **200** may include exemption processing to resolve an error associated with the validation **245**. Examples of exception processing are provided below in greater detail.

The method **200** may also, as an option in at least some embodiments, include performing **250** logistical processing with respect to the dose orders. For example, the dose order data stream **105** may include order messages of differing types. For instance, dose order message types may relate to new dose orders, dose order change requests, dose order cancellation requests, or other appropriate dose order requests. That is, at least some of the dose order message types processed may require subsequent actions to be taken on a previously received dose order. Accordingly, in the event dose order data is processed and provided to a dose fulfillment client **140** and a subsequent dose order message is received that requires action be taken with respect to the

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dose order data provided to the dose fulfillment client **140**, additional logistic processing may be performed **250**. This may include analysis of prior dose orders to determine the state of the previous dose order (e.g. whether the dose order has yet been sent to the dose fulfillment client, whether the dose has been fulfilled, etc.). Additionally, the logistical processing may also include performing actions (e.g., modification, cancellation, etc.) of a prior dose order as required by a subsequently received dose order message. Furthermore, it may be appreciated that a dose order message may apply to a dose order that has been routed to a first dose fulfillment client **140**, but upon the execution of the dose order message is to be recalled from the first dose fulfillment client **140** and provided to a second dose fulfillment client **140**.

The method **200** may also include updating **255** a staging table to reflect receipt of a dose order at the dose fulfillment client **140**. For instance, the updating **235** may include modifying a data field of the staging table associated with an indication of the dose order status to indicate the dose order record has been provided to and/or received by the dose fulfillment client **140**.

The method **200** may further include fulfilling **260** a dose at the dose fulfillment client **140**. The fulfilling **260** may include preparation of a dose corresponding to a dose order at the dose fulfillment client **140**. The preparation may be a manual operation to prepare the dose order, an automated operation to fulfill the dose order, or a combination thereof. Further still, the fulfilling **260** may include making available a dose (e.g., a pre-prepared) to satisfy a dose order (e.g., such as providing instructions to a dose dispensation cabinet to make a prepared dose available for fulfillment of the dose order).

With further reference to FIG. 3, a schematic depiction of an embodiment of a system **300** for automated healthcare exchange is depicted. The system **30** may include an EMR system **110** that provides a healthcare information data stream **105** in the form of an HL7 data feed to a HL7 data feed port **122** of the platform interface module **120**. In turn, the HL7 data received at the HL7 data feed port **122** may be provided to an HL7 parsing module **124**. In this regard, the HL7 parsing module **124** may identify/parse dose orders from the HL7 data feed received at the HL7 data feed port **122**. As such, the HL7 parsing module **127** may comprise a stream processing module. The HL7 parsing module **124** may also determine if the dose order identified from the data stream **105** is to be processed by the exchange system **30**. For instance, a dose may be processed by the system as described herein or may be diverted to a different dose handling system by the parsing module **124** based on the dose order identified.

In any regard and as described above, a staging table may be populated for a dose order identified/parsed from the data stream **105**. The staging table may be a predefined data structure that is populated with identified/parsed data from the data stream **105**. In turn, the dose order data may be used to populate fields in the staging table corresponding to dose order data fields with dose order metadata from the data stream **105**. In turn, the staging table regarding the dose order may be stored in a staging table database **126**.

The staging table database **126** may be in operative communication with a central server **300** that is disposed remotely from the platform interface module **120**. In this regard, data regarding the dose orders stored as staging tables in the staging table database **126** may be provided to the central server **300**. The central server **300** may receive dose order data from a plurality of platform interface mod-

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ules 120 (e.g., from different healthcare facilities implementing platform interface module 120). The central server 300 may facilitate backup and/or data aggregation services in relation to the dose order data.

The staging table database 126 may be in operative communication with a transformation module 130. In this regard, the transformation module 130 may be operative to retrieve a staging table from the staging table database for transformation of the dose order data contained therein. Additionally, the transformation module 130 may include a client router 128. In turn, the client router 128 be operative to provide transformed dose order data (e.g., in a unique input format for a given dose fulfillment client 140) to an appropriate dose fulfillment client 140. While shown as a common module, the client router 128 may also be provided separately from the transformation module 130 (e.g., as a standalone module or as a module provided in connection with the platform interface module 120).

As described briefly above, various methods for determining an appropriate dose fulfillment client 140 to which the dose order is to be directed may be applied by the client router 128. In this regard, the client router 128 may employ any one or more of the approaches described above to determine an appropriate dose fulfillment client 140 to which the transformed dose order data may be provided. In this regard, shown in FIG. 3, a plurality of clients 140A, 140B, and 140C may be provided in operative communication with the client router 128. In turn, the client router 128 may provide an appropriate one of the fulfillment clients 140A, 140B, 140C with the transformed dose order data such that the appropriate dose fulfillment client 140 may be used to fulfill the dose corresponding to the dose order.

As may be appreciated, the determination of the appropriate dose fulfillment client 140 to which the dose order will be provided may affect the transformation of the dose order by the transformation module 130. In this regard, the transformation module 130 may comprise the client router 128 as depicted or may be in bidirectional communication with the client router 128 such that the transformation module 130 and client router 128 may collectively act upon the data in a staging table to determine the dose fulfillment client 140 to which the dose order will be prepared such that the identity of the determined dose fulfillment client 140 may at least in part affect the transformation applied to the dose order data by the transformation module 130. For instance, upon identification of an appropriate dose fulfillment client 140 by the client router 128, a unique input format for the identified dose fulfillment client 140 may be used to transform the dose order data from the staging table.

In other embodiments (e.g., such as that depicted in FIG. 5 discussed in greater detail below), the transformation module 130 of the embodiment of the system 30 may be provided at the platform interface module 120. Furthermore and as addressed in greater detail below in relation to other embodiments, the client router 128 may be located the platform interface module 120. In this regard, the fulfillment clients 140A, 140B, and 140C may each be operative to receive transformed dose order data for use in fulfillment of the dose corresponding to the dose order. That is, in at least some embodiments, all processing regarding the dose order data for exchange with the dose fulfillment client 140 may occur at the platform interface module 120. However, in other embodiments, such as those described below, different functionality with respect to the transformation module 130 and/or client router 128 may be provided in a distributed manner remotely from the platform interface module 120.

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The platform interface module 120 may also include a web server 310 that provides for remote access to the functionality provided by the platform interface module 120. For example, the web server 310 may maintain and/or execute one or more web pages 312 corresponding to the various modules and/or functionality provided by the platform interface module 120. In turn, the web pages 312 may facilitate functionality in connection with the platform interface module 120 and/or modules provided comprising the platform interface module. For instance, various settings, parameters, or other administrative functions may be performed with respect to platform interface module 120. Such functionality may be facilitated by way of the web pages 312 that are accessed by way the web server 310.

With further reference to FIG. 4, another embodiment of a system 40 for automated exchange of healthcare information is depicted. The system 40 may be particularly useful in the context of fulfillment of total parenteral nutrition (TPN) dose orders received from the EMR system 110. The system of FIG. 4 may be configured to receive TPN dose orders from the EMR system 110 in a plurality of formats. For example, the platform interface module 120 may include an HL7 data feed port 122 to which HL7 format messages from the EMR system 110 may be routed and parsed as described above in connection with FIG. 3. The platform interface module 120 may also include a TPN print data feed 402 to which print feed data messages from the EMR system 110 may be routed. In this regard, the platform interface module 120 may facilitate both the capability to process HL7 format messages as well as print feed messages received from the EMR system 110. Examples of HL7 message formats that may be processed may include, but are not limited to, HL7 messages originating from EMR systems provided by Cerner Corporation, Epic Systems Corporation, Meditech, Omnicell, Inc., or others. Furthermore, examples of supported print feed formats may include, but are not limited to, Zebra programming language provide by Zebra Technologies, DataMax format provided by DataMax-O'Neil, Intermec format provided by Intermec, Inc., portable document format (PDF) provided by Adobe Systems, plain text format, etc.

In this regard, HL7 messages may be provided from the HL7 data feed port 122 to an HL7 parsing module 124 as described above in relation to FIG. 3. Additionally, the TPN print data feed 402 may provide print feed messages to a label processor 404 of the platform interface module 120. In any regard, the HL7 parsing module 124 and/or the label processor 404 may provide individual dose order metadata in relation to corresponding dose order data fields for storage as a staging table in a staging table database 126.

In the example provided in FIG. 4, the platform interface module 126 may be in direct communication with a TPN fulfillment client 142. In this regard, the TPN fulfillment client 142 may be a dose fulfillment client 140 capable of fulfilling TPN dose orders. For instance, the TPN fulfillment client 142 may include a TPN management module 144. The TPN management module 144 may facilitate dose order calculation and management. For instance, the TPN management module 144 may comprise ABACUS Calculation Software provided by Baxter Healthcare Corporation of Deerfield, IL. In any regard, the TPN fulfillment client 142 may provide further processing in relation TPN dose orders in connection with the fulfillment of TPN dose orders (e.g., calculations related to ingredient interactions, etc.). The TPN management module 144 may be in further communication with a TPN compounder 158. The TPN compounder 158 may be operative to autonomously prepare a

dose corresponding to a TPN order. In this regard, in an embodiment, the dose fulfillment client **140** may comprise the TPN fulfillment client **142** that collectively includes the TPN management module **144** and/or the TPN compounder **158**. That is, the transformation module **130** may communicate directly to the TPN management module **144** or the TPN compounder **158**. As such, the unique input format utilized by the transformation module **130** may correspond to the TPN management module **144** or the TPN compounder **158**.

In this regard, and as depicted in FIG. 4, a transformation module **130** may be provided in operative communication with the staging table database **126** and the TPN client **142**. In this regard, the transformation module **130** may access the staging table database **126** to retrieve dose order data in the standardized intermediate format of the staging table that is transformed by the transformation module **130** into a unique input format for use with the TPN client **142**. In this regard, the system **40** of FIG. 4 may not include a client router **128**. For instance, the platform interface module **120** may be utilized in a dedicated manner to receive only TPN dose orders from the EMR system **110**. In this regard, the TPN client **142** may access the staging table database **126** to retrieve all staging tables contained therein for transformation by the transformation module **130** into the unique input format associated with the TPN management module **144**. That is, the platform interface module **120** may be used in a dose context-specific application wherein only a single type of dose orders to be fulfilled by a given dose fulfillment client **140** (e.g., the TPN fulfillment client **142**) are processed by the platform interface module **120**.

FIG. 5 depicts another embodiment of a system **50** for exchange of healthcare information. The system **50** may also receive HL7 format messages as well as print feed messages from an EMR system **110** that are processed by an HL7 data feed port **122** and a print feed port **402**, respectively, as described above in relation to FIG. 4. As described above, data messages may be routed to an appropriate one of an HL7 parsing module **124** or a label processor **404** for generation of a staging table stored in the staging table database **126** for corresponding receive dose orders. FIG. 5 further includes a manual order entry website **312A** (e.g., as provided in websites **312** access by web server **310** of FIG. 3). In this regard, user may access the platform interface module **120** directly using the manual order entry website **312A** to generate a dose order directly at the platform interface module **120** that are stored in a staging table in the staging table database **126**.

In the embodiment of the system **50** depicted in FIG. 5, platform interface module **120** may be in operative communication with a pharmacy workflow management application **146**. In this regard, the pharmacy workflow management application **146** may be the only dose fulfillment client **140** with which the platform interface module **120** is in operative communication. In this regard, as depicted in FIG. 5, the transformation module **130** may be provided with platform interface module **120** for transformation of dose order data stored in staging tables of the staging table database **126** into a unique input format associated with the pharmacy workflow management application **146**. Upon receipt of the transformed dose order data, the data may be stored in the dose order record database **148** of the pharmacy workflow management application **146**. In turn, the dose orders may be provided to a pharmacy workflow management module **150**. The pharmacy workflow management module **150** may allow for manual preparation of a dose order by a pharmacy technician or the like. For example, embodiments of a

pharmacy workflow management application **146** is described in the commonly assigned U.S. provisional Application No. 62/057,906 filed on Sep. 30, 2014 entitled "MANAGEMENT OF MEDICATION PREPARATION WITH FORMULARY MANAGEMENT", the entirety of which is incorporated by reference herein, may be utilized to fulfill dose orders at the pharmacy workflow management application **146**.

With further reference to FIG. 6 another embodiment of a system **60** for automated healthcare information exchange is depicted. The system **60** may receive dose order information from an EMR system **110** by way of an HL7 data feed port **122**, a print data feed port **402**, or by way of manual order entry from a manual order entry website **312A** as described above in connection to FIG. 5. In this regard, dose order data may be received and used to populate corresponding staging tables that are stored in the staging table database **126**.

The system **60** of FIG. 6 further includes a plurality of dose fulfillment clients **140**. For instance, the platform interface module **120** is in operative communication with a pharmacy workflow management application **146** and a TPN fulfillment client **142**. The system **60** of FIG. 6 may have a local transformation module **130A** provided at the platform interface module **120**. The local transformation module **130A** may perform data transformation on dose order data that are provided to the pharmacy workflow management application **146**. Additionally, a queue management and prioritization module **152** may be provided for management of dose orders corresponding to dose order data that are to be provided or that are provided to the pharmacy workflow management application **146**.

Additionally, the TPN fulfillment client **142** may have associated therewith a remote transformation module **130B** located remotely from the platform interface module **120** for transformation of dose order data locally with respect to the TPN fulfillment client **142** for dose order data corresponding to dose orders directed to the TPN fulfillment client **142**. In this regard, the system **60** of FIG. 6 may have different transformation modules **130A** and **130B** for transformation of dose order data provided to the pharmacy workflow management application **146** and the TPN fulfillment client **142**, respectively. Furthermore, a queue management and prioritization module **152** may be provided for at least a portion of the dose orders being directed to a given one of the dose fulfillment clients (e.g., the pharmacy workflow management application **146**). The queue management and prioritization module **152** may be operative to manage dose orders prior to or after data transformation has occurred with respect thereto. Thus, while shown as residing between the transformation module **130** and the pharmacy workflow management application **146** in FIG. 6, one or more queue management and prioritization modules **152** may be provided elsewhere for management of dose order queues (e.g., at the staging table database **126**, at the dose fulfillment client **140**, etc.).

As described above briefly, the determination of the fulfillment client **140** to which the dose order data may be provided may occur in a number of manners. For example, each respective client may access the staging table database **126** and request dose order data that are flagged (e.g., include a data field indicative of) as being designated for the given dose fulfillment client **140** requesting the information. Furthermore, logic may be applied to the dose order data based on a number of different potential parameters to identify a dose fulfillment client **140** for use in fulfillment of a given dose order. In this regard, the fulfillment clients **140**

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may be provided dose order data for use in fulfillment of the dose order based on a data field in which the designated dose fulfillment client **140** is indicated as determined by application of the logic.

With further reference to FIG. 7, an embodiment of a system **70** for automated exchange of healthcare information is depicted. The system **70** may receive dose order data from an EMR system **110** as described above. The system **70** may also include a transformation module **130D** that facilitates transformed data being provided to one or more of a plurality of dose fulfillment clients **140** (e.g., comprising the TPN client **142**, a dispensing cabinet **154**, or an automated syringe filler **156** in the depicted embodiment). The transformation module **130D** may include a client router **128**.

In this regard, the dose fulfillment clients **140** of the system **70** may include a pharmacy workflow management application **146**, a TPN client **142**, a dispensing cabinet **154** (e.g., an automated dose dispensing cabinet), and/or an automated syringe filler **156**. A portion of the dose orders from the staging table database **126** may be provided to the pharmacy workflow management application **146** by way of transformation module **130**. Another portion of dose orders may be provided to the transformation module **130D**. In turn, the client router **128** of the transformation module **130D** may be operative to identify a dose fulfillment client **140** to which the dose order should be provided. As such, the client router **128** may apply any appropriate logic with respect to the dose order to determine the appropriate dose fulfillment client **140**. For example, the client router **128** may identify and utilize a dose fulfillment client flag in the dose order has received from the EMR system to direct the dose order to an appropriate dose fulfillment client **140**. Furthermore, the client router **128** may dynamically route the dose order to an appropriate fulfillment client **140** based on the dose order metadata contained by the dose order. Specifically, the client router **128** may be operative to determine whether the dose order is appropriate for automated preparation (e.g., by an automated syringe filler **156**). In this regard, if the dose order is appropriate for automated preparation, the client router **128** may provide the dose order to an appropriate automated dose fulfillment client **140** (e.g., such as automated syringe filler **156**). Further still, the client router **128** may be operative to identify if the dose order is appropriate for fulfillment by way of an automated dispensing cabinet **154**.

As may be appreciated, it may be depended upon the specific dose fulfillment client **140** to which a dose order is provided as to the processing of the specific dose order data. For example, upon provision of dose orders to one of the pharmacy workflow management application **146** transformation module **130C** that is provided locally to the platform interface module **120** may allow for transformation of the dose order data in the staging table database **126** into a unique input format corresponding to pharmacy workflow management application **146**. Furthermore, the transformation module **130D** may be provided (e.g., remotely from the platform interface module **120**) for transformation of dose order data with respect to the TPN fulfillment client **142**, dispensing cabinet **154**, or automated syringe filler **156**. In this regard, the TPN fulfillment client **142** may include a TPN compounder **158** for compounding of the TPN dose order according to the dose order data received at the TPN client **142** and transformed by the transformation module **130D**. Furthermore, the TPN order management module **144** may provide information to the pharmacy workflow management application **146**. While shown as a direct connection, it is anticipated that the dose order data may be

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provided by way of the platform interface module **120** and/or transformation module **130C**). Further still, the dose order data may be provided to both the pharmacy workflow management application **146** and the TPN fulfillment client **142** and an indication of a correlation between the dose order data provided to each respective dose fulfillment client **140**, respectively.

FIG. 8 further depicts a system **80** that, like the system **70** in FIG. 7 is in operative communication with a pharmacy workflow management application **146**, a TPN client **142**, a dispensing cabinet **154**, and/or an automated syringe filler **156**. However, unlike the system **70** in FIG. 7, the system **80** in FIG. 8 may include a single transformation module **130** provided at the platform interface module **120** for transformation of dose order data from the standardized intermediate format of the staging table stored in the staging table database **126** into a corresponding unique input format for a given dose fulfillment client **140**. In this regard, the platform interface module **120** may include a client router **128** that is disposed between the staging table database **126** and the EMR system **110** or other means of order entry. In this regard, the client router **128** may be operative to apply logic to determine an appropriate one of the dose fulfillment clients **140** for use with each respective one of the dose orders. In this regard, the client router **128** may append information to the data in the staging table for a given dose order indicative of a dose fulfillment client **140** to which the dose order is to be provided. Further still, the client router **128** may be in operative communication with transformation module **130** to provide information regarding the dose fulfillment client **140** to which a dose order for a given staging table may be provided.

In any of the foregoing embodiments, the respective systems for automated transformation healthcare information may also facilitate logging with respect to actions taken relative to a dose order. The logging may include generating logs that reflect the actions taken on the dose order. The logs may be maintained for later review by human user (e.g., in the performance of an audit, troubleshooting, or the like). The logging may occur, and the logs may be stored, at different respective portions within the system. For example, the platform interface module **120** may perform logging to generate corresponding logs. Furthermore, the transformation module **130** (e.g., regardless of their specific location within the systems) may also perform logging to generate corresponding logs. In this regard, the logs may be maintained at different respective portions within the system. Additionally or alternatively, the logs may be transmitted within the system to a common location. For example, the platform interface module **120** may be in operative communication with transformation modules **130**, or dose fulfillment clients **140** to receive logs there from regarding processing in relation to dose orders. In this regard, the logs for one or more portions of the system may be collectively stored at a given point in the system such as, for example, the platform interface module **120**.

With further reference to FIG. 9, a representation of a data transformation performed by a transformation module **130** is shown. Specifically, FIG. 9 includes a textual representation of the data transformation in human perceivable form. FIG. 9 includes a representation of an embodiment of a data feed **500**. As may be appreciated, the data feed **500** may be analyzed (e.g., by a parsing module of the platform interface module **120**). The parsing may result in a dose order **502** and a dose order **504** being identified. As may be appreciated, a portion **518** of the data stream **500** may not correspond to a dose order.

Dose order **502** may be parsed such that a number of dose order data fields are recognized. Specifically, the dose order **502** may include a first patient data field **506**, a second patient data field **508**, a first ingredient field **510**, a second ingredient field **512**, an administration data field **514**, and a patient procedure data field **516**. Additional or fewer data fields may be present such that those presented are for demonstrative purposes only. The second dose order **504** may be parsed such that a first data patient field **524**, a first ingredient field **526**, and an administration data field **528** are identified. In this regard, the data for the first dose order **502** and the second dose order **504** may be stored in respective staging tables. As such, the textual representation of the first dose order **502** may correspond to the staging table for the first dose order **502** and the textual representation of the second dose order **504** may correspond to the staging table for the second dose order **504**. While depicted textually in FIG. 9, it may be appreciated that the actual staging tables may be maintained in a different format such as an XML format, a SQL database format, or other appropriate format.

In any regard, the transformation module **130** may be operative to map data fields from the staging tables **502** and **504** to corresponding fields in unique input formats for one or more dose fulfillment clients **140**. For instance, a first input **552** may be generated that corresponds to the staging table for the first order **502**. As may be appreciated, the first input **552** may have defined fields according to a unique input format for a specific dose fulfillment client **140**. As such, appropriate ones of the data fields in the staging table **502** may be mapped to fields in the first input **552**. Specifically, the first patient data field **506** and the second patient data field **508** may be mapped to the client patient info field **556** of the first input **552**. In this regard, it may be appreciated the multiple data fields from the staging table **502** may be mapped to a single field in the first input **552**. In this regard, the data from the multiple fields (e.g., the first patient data field **506** and the second patient data field **508**) of the staging table **502** may be aggregated or reformatted for inclusion in the client patient input **556**. Furthermore, a portion of the data contained in one of the first patient data field **506** or second patient data field **508** may be omitted.

Further examples of the matching between data fields in the staging tables **502**, **504** and the first and second input **552**, **554** provided in FIG. 9. Specifically, the first ingredient field **510** and the second ingredient field **512** may be mapped to a dose ingredient input **558** of the first input **552**. Furthermore, the administration data field **514** of the first staging table **502** may be mapped to the administration data input **560** of the first input **552**. Of note, not all data field of the first staging table **502** may be mapped to the first input **552**. For example patient procedure data field **516** may be irrelevant for the dose order, and therefore not have a corresponding input field of the first input **552** to which the patient procedure data field **516** is mapped. Furthermore, while the first staging table **502** demonstrates that multiple data fields from the staging table **502** may be mapped to a single info field in the first input **552**, it may also be appreciated that one to one correspondence may be provided such as in the case of the second dose order **504**. In this regard, the first patient data field **524** of the second dose order **504** may be mapped to the client patient input **562** of the second input **554**. Furthermore, the first ingredient field **526** may be mapped to the dose ingredient input **564**, and the administration data field **528** may be mapped to the administration data input **566**. Furthermore, it should be noted that portion **518** of the data stream not corresponding to a dose order may not be mapped to any client input. That is, at least

one platform interface module **120** or transformation module **130** may recognize the portion **518** is not corresponding to the dose order such that no corresponding input is generated or mapped to the portion **518**.

In addition to the mapping of data fields corresponding to dose orders to client input fields, the transformation module **130** may also perform translation with respect to the data contained in each respective field such that the data within each respective field may be translated into a format or form expected by the client in a given client info field. For instance, with further reference to FIG. 17, a user interface **1700** that includes representations of potential translations to be performed by the transformation module **130** is provided. The user interface **1700** may include a type listing **1710**, a client form listing **1720**, and an EMR form listing **1730** provided in a table **1702**. In this regard, the table **1702** may include a plurality of listings (e.g., arranged by type **1710**) for translation from a EMR form **1730** to a client form **1720**. As may be appreciated, some listings may provide identical forms for the EMR form **1730** and the client form **1720**. For example, the form corresponding to listing **1712** for "Acetate" may be the same for the EMR form **1732** and the client form **1722**. However, as may be appreciated from the listing **1702**, other forms may differ between the EMR form **1730** and the client form **1720**. For example, listing **1714** may have a EMR form of "ADULTTRACE" and a client form **1724** of "Adult Trace Conc." Accordingly, should the EMR form **1734** for listing **1714** appear in a data field for a staging table, the data may be translated to the client form **1724** by the transformation module **130**. In this regard, the user interface **1700** may also include a type filter **1740** and a client form filter **1750** that may be used to narrow the results presented in the table **1702**. Furthermore, upon selection of a given listing within the table **1702**, the EMR form **1730** for a given listing may be modified using the EMR form input **1760**. For instance, upon selection of a listing from the table **1702** for a given portion of data, the EMR form **1730** may be modified for the listing using the EMR form input **1760**. Furthermore, listings may be added or removed from the listing **1702** by the user. In this regard, it may be appreciated that the user may be able to maintain the definitions provided in the listing **1702** regarding translations to be applied by the transformation module **103** using the user interface **1700**. In this regard, the user interface **1700** may be provided by a webpage **312** of the platform interface module **120** and/or may be provided by a dose fulfillment client **140** that comprises a transformation module **130** as described above.

With further reference to FIGS. 10-12, an example of the processing of a dose order from an EMR system **110** to a dose fulfillment client **140** is depicted. Specifically, FIG. 10 depicts a textual representation of the data stream **600** from which the dose order is identified. In turn, FIG. 11 presents a textual representation of a log maintained by a transformation module **130** when acting upon the identified dose order from the data stream **600**. FIG. 12 represents a user interface **800** of a dose fulfillment client **140** that is received the dose order. In this regard, the textual representation of the data stream **600** may include a plurality of data fields. For example, highlighted data fields corresponding to a patient data field **610**, a dose order type field **620**, a macro ingredients detail field **630**, and a micro ingredients detail field **640** that are identified for a given dose order from the data stream **600**. In this regard, the metadata for each corresponding field may be stored in a staging table. For example, the metadata found within the patient data field **610** may be used to populate one or more fields in a staging

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table related to the patient. Specifically, the name “TONI SMC,” a gender “female,” a date of birth “19871229,” and an EMR number “123456” may be identified in the patient data field 610. Furthermore, an identifier indicating the dose order is a TPN order may be identified from the dose order type field 620. Furthermore, a macro ingredient 630 “AMINOSYN II 15% IV SOLN RX|100.8|g|100.8|g” for the dose order 630 may be contained within the macro ingredient data field 630. Furthermore, to micro ingredients “SODIUM CHLORIDE 4 MEQ/ML IV SOLN RX|33.6| MEQ| 33.6|MEQ” and “POTASSIUM ACETATE 2 MEQ/ML IV SOLN RX|16.8|MEQ|16.8|MEQ” may be identified in the micro ingredients detail field 640.

With further reference to the log 700 generated by the transformation module 130, it may be appreciated that the transformation module 130 may convert the patient data field 610 into an input for creation of a patient in the dose fulfillment client 140 (e.g., in this case an ABACUS Calculation Software application) at line 710. Furthermore, the dose order type field 620 may be used to create an input for the creation of an order at the dose fulfillment client 140 at line 720. In turn, the order for the dose fulfillment client 140 may be populated with inputs corresponding to the macro ingredient detail 630 and the micro ingredient detail 640 such that those data fields are represented in the order input data 734 the dose fulfillment client 140. As may be appreciated, the macro ingredient detail field in the micro ingredient detail field 630 and 640 may be transformed into a format associated with unique input format associated with the dose fulfillment client 140. In this regard, the input fields 630' and 640' corresponding to the macro ingredient detail field 630 and the micro ingredient detail field 640 may contain data in a translated format. In turn, the inputs to the dose fulfillment client 140 reflected in the log 700 may be used to provide inputs to the dose fulfillment client 140.

In turn, FIG. 12 displays a user interface 800 of the dose fulfillment client 140 after receiving the commands reflected in the log 700. As may be appreciated, patient information file 10 may be presented with information corresponding to the data contained in the patient data field 610. Specifically, the patient name, the EMR number, an age based upon the date of birth provided in the patient data field 610, or other information may be provided in the patient information field 810. Furthermore, in ingredient listing 820 of the dose fulfillment client 140 may be provided. As may be appreciated, the macro ingredient 830 listed in the ingredient listing 820 corresponds to the data contained in the macro ingredient data field 630. The micro ingredients 840 listed in the ingredient listing 820 correspond to the data contained in the micro ingredient data field 840. As may be appreciated, the form of the ingredients 830 and 840 may be different from that contained in the data fields 630 and 640, thus reflecting the translation performed by the transformation module 130.

With further reference to FIG. 13, the user interface 900 for a dose fulfillment client 140 is depicted. The user interface 900 may include a patient listing 910. The patient listing 910 may list patients for which dose orders are pending. Upon receipt of a dose order from a transformation module 130, the dose fulfillment client 140 may update the patient listing 910 with the status indicator 912 to indicate that the dose order has been received. In this regard, the user interface 900 may further include an order listing field 920. The order listing field 920 may list complete and pending dose orders for a given patient selected from the patient listing 910. As may be appreciated, a first pending order 922 and second pending order 924 may be indicated as having

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been received by the dose fulfillment client 140. Status indicators for the received pending doses 922 and 924 may be provided. Specifically, a first status indicator 926 may be provided to indicate that no exceptions were generated in connection with the processing of the first pending order 922 by the platform interface module 120 and/or transformation module 130. In this regard, the first status indicator 926 may indicate the first pending dose order 922 is ready to be fulfilled by the dose fulfillment client 140.

The second status indicator 928 provided with the second pending dose order 924 may indicate that an exception was generated during the processing of the second pending dose order 924. Upon selection of the second pending dose 924, a dose detail screen 950 depicted in FIG. 14 may be provided to the user. The dose detail screen 950 may include a patient information field 810 is described above in connection with FIG. 12. Furthermore, the dose detail screen 950 may include an ingredient listing 820 as described above in connection with FIG. 12. The dose detail screen 950 may also include an exception field 960 that may list exceptions identified during the processing of the dose order by the platform interface module 120 and/or transformation module 130. For example, exceptions may be identified with respect to an unrecognized product, an unrecognized value, and unrecognized unit, and unrecognized data field, or other error associated and the processing of the dose order. For instance in FIG. 14, an exception 962 may be provided corresponding to an error that occurred with respect to the mapping of an ingredient “Insulin” to the dose fulfillment client 140. In this regard, the exception listing 960 may include an EMR form, a substance description, a value, and a unit associated with the exception 962. Accordingly, the user may select the exception 962 and resolve the error associated therewith. For example, in FIG. 15, the dose detail screen 950 is shown such that the user has added a further ingredient 964 to the ingredient listing 820 associated with the ingredient that is the subject of the exception 962. In this regard, the user may flag the exception 962 as having been processed and made proceed with preparation of the dose by the dose fulfillment client 140.

Furthermore, in FIG. 16 a user interface 1000 may be provided for use in resolution of exceptions, such as the exception 962 identified in FIGS. 14 and 15. For example, the exception 962 may be caused by an improper mapping between the EMR form for the drug “Insulin” and the client input form associated with the dose fulfillment client 140. That is, the mapping between the EMR form and the client input form may have caused an error when the platform interface module 120 and/or the transformation module 130A to process the data field containing the data related to insulin for this dose order. In this regard, the user interface 1000 may allow user to select the product 1002 for modification by the user interface 1000. In turn, the user may be operative to modify the formulary record for the selected product 1002.

Specifically, the user interface 1000 may also include a EMR correspondence field 1004 that may be used to provide the corresponding EMR form for the product 1002 at the dose fulfillment client 140. In this regard, the user may use the user interface 1000 to rectify the error in mapping between the EMR form and the client form for the drug “insulin”. In turn, upon further processing of the subsequent dose order containing the ingredient “insulin”, the appropriate translation may be provided such that the exception 962 may not occur in a subsequent order having the same product 1002 provided therewith. While the user interface

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1000 provides for exception resolution at the dose fulfillment client 140, the exception may also be flagged at the platform interface module 120 and/or the transformation module 130 in appropriate user interfaces that may be provided to facilitate provision of a correct mapping and/or translation for a product in a dose order in a manner similar to that discussed with respect to FIG. 16. That is, the exception processing described in relation to FIG. 16 from the perspective of the dose fulfillment client 140 may also be provided in connection with the platform interface module 120 and/or transformation module 130. Furthermore, while ingredient specific processing may be fulfilled using the user interface 1000, bulk changes to the mapping between EMR forms 1730 and client forms 1720 may be facilitated by the user interface 1700 described above in connection with FIG. 17.

While the invention has been illustrated and described in detail in the drawings and foregoing description, such illustration and description is to be considered as exemplary and not restrictive in character. For example, certain embodiments described hereinabove may be combinable with other described embodiments and/or arranged in other ways (e.g., process elements may be performed in other sequences). Accordingly, it should be understood that only the preferred embodiment and variants thereof have been shown and described and that all changes and modifications that come within the spirit of the invention are desired to be protected.

What is claimed is:

1. A method for an automated exchange of healthcare information between an electronic medical record (EMR) system and a selected dose fulfillment client for fulfillment of a dose order, the method comprising:

executing non-transitory machine-readable data for specifically configuring processors of a platform interface module, a transformation module, and a client router, execution of the non-transitory machine-readable data causing the processors to:

receive a healthcare information data stream at the platform interface module via a network from the EMR system, wherein the healthcare information data stream comprises information including dose order metadata related to dose orders that is provided in a Health Level 7 ("HL7") format;

identify, via the platform interface module, individual dose orders from the healthcare information data stream by identifying express delineators within the healthcare information data stream and identifying the dose order metadata corresponding to dose order data fields, wherein the express delineators mark separations between the dose orders in the healthcare information data stream;

parse, via the platform interface module, the dose order metadata for each of the dose orders from the healthcare information data stream, wherein the dose order metadata comprises one or more dose characteristics associated with the respective dose order;

populate, via the platform interface module, a staging table with the dose order metadata for each of the dose orders, wherein the staging table comprises a plurality of dose order data fields populated with corresponding respective portions of the dose order metadata parsed from the healthcare information data stream, and wherein the dose order metadata is populated in a standardized intermediate format;

determine, via the transformation module from the dose order metadata, a dose fulfillment client type, among a plurality of different dose fulfillment client types

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for fulfilling each of the dose orders, the dose fulfillment client types including an automated dose preparation device, an automated total parenteral nutrition (TPN) compounder, and an automated dose dispensing cabinet;

select, via the transformation module, a dose fulfillment client that matches the dose fulfillment client type for fulfillment of each of the dose orders;

transform, via the transformation module, the dose order metadata from the standardized intermediate format into a predefined format dictated by the selected dose fulfillment client to generate transformed dose order metadata,

wherein the transforming is at least in part based on the respective predefined format of the selected dose fulfillment client, and

wherein the transforming includes mapping the plurality of dose order data fields of the staging table to corresponding respective ones of a plurality of dose fulfillment client input fields for the selected dose fulfillment client, including at least:

mapping one data field of the plurality of dose order data fields of the staging table to one data field of the plurality of dose fulfillment client input fields, and

mapping at least two data fields of the plurality of dose order data fields of the staging table to one data field of the plurality of dose fulfillment client input fields;

route, via the client router of the transformation module, the transformed dose order metadata to the selected dose fulfillment client via the network; and cause each of the dose orders to be fulfilled automatically using the selected dose fulfillment client based on the transformed dose order metadata to provide a corresponding medication dose.

2. The method of claim 1, further comprising:

receiving a first dose order in a first EMR form; and receiving a second dose order in a second EMR form, wherein the first dose order and the second dose order correspond to a dose order with identical constituent ingredients,

wherein a form of the constituent ingredients in the first EMR form differs from the second EMR form, and wherein a form of the constituent ingredients for the first dose order is identical to a form of the constituent ingredients for the second dose order in the respective staging tables corresponding to the first dose order and the second dose order.

3. The method of claim 1, wherein the staging table is independent of any of the plurality of dose fulfillment clients.

4. The method of claim 1, wherein the dose fulfillment client input fields are defined by a unique input format associated with the dose fulfillment client.

5. The method of claim 4, wherein the transforming comprises:

validating the dose order metadata of the plurality of dose order data fields with respect to the unique input format for corresponding ones of the dose fulfillment client input fields.

6. The method of claim 5, wherein the validating comprises:

correlating the dose order metadata of the plurality of dose order data fields to formulary records of the dose

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fulfillment client with respect to the corresponding respective ones of the dose fulfillment client input fields.

7. The method of claim 6, further comprising:

generating an exception in response to an error associated with at least one of the mapping or validating.

8. The method of claim 7, wherein the exception comprises prompting a human user to resolve the error.

9. The method of claim 8, further comprising:

receiving, from the human user, an input associated with the resolution of the error, wherein the input comprises at least one of a correct mapping between a dose order data field of the staging table and a corresponding respective one of a dose fulfillment client input fields or a correct correlation between a portion of dose order metadata and a formulary record of the dose fulfillment client.

10. The method of claim 9, further comprising:

updating at least one of a mapping logic or a correlation logic for use in the mapping and correlating, respectively, based on the input received from the human user.

11. The method of claim 1, further comprising:

updating the staging table to indicate the dose order metadata for each of the dose orders that have been routed to the dose fulfillment client.

12. The method of claim 1, further comprising:

managing each of the dose orders prior to providing the dose order metadata in a second form to the dose fulfillment client.

13. The method of claim 12, wherein the managing includes providing a user interface to allow a user to perform a management function relative to each of the dose orders.

14. The method of claim 13, wherein the management function comprises at least one of modification of dose order metadata, cancellation of the dose order, organization of the dose order relative to other dose orders, or prioritization of the dose order relative to other dose orders.

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15. The method of claim 1, further comprising: performing logistical processing on each of the dose orders.

16. The method of claim 15, wherein the logistical processing comprises performing an action relative to each of the dose orders in response to receipt of an EMR system message subsequent to the receipt of the respective dose order.

17. The method of claim 16, wherein the EMR system message comprises at least one of a dose order change message or a dose order discontinuation message.

18. The method of claim 17, wherein the logistical processing comprises determining if the dose order metadata has been provided to the dose fulfillment client.

19. The method of claim 18, wherein the logistical processing comprises performing an action relative to each of the dose orders corresponding to the EMR system message for a dose order that has not yet been provided to the dose fulfillment client.

20. The method of claim 19, wherein the logistical processing comprises providing data to the dose fulfillment client related to the EMR system message for a dose order that has been provided to the dose fulfillment client.

21. The method of claim 20, wherein the data provided to the dose fulfillment client related to the EMR system message is transformed into a form corresponding to the dose fulfillment client.

22. The method of claim 1, further comprising:

logging activity taken with respect to each of the dose orders to generate a log regarding activities relative to the respective dose order.

23. The method of claim 22, wherein the logging comprises aggregating a plurality of logs generated relative to different respective actions taken with respect to the respective dose order.

24. The method of claim 1, wherein the healthcare information data stream is received from the EMR system.

25. The method of claim 24, wherein the EMR system comprises a hospital information system (HIS).

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